

Implementation of Generative AI in the Air Navigation Test Station of a FAL

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Abstract

This work addresses the transformation, through the use of an AI assistant, of the functional testing process for aircraft in an Aeronautical Assembly Line.

The assistant's objective will be to help operators during the testing process by answering various technical questions about procedures, test equipment, fault investigation, safety, etc.

The project idea stems from recent advances in generative AI (LLMs, AI agents, embodied AI, etc.) that enable the shift from the current Industry 4.0 (data automation and efficiency) to Industry 5.0, which is more focused on human-machine collaboration and where resilience and sustainability are key.

Because the set of aircraft tests is broad, we will limit ourselves to testing the air-navigation systems, which are associated with complex technologies (electronics, buses, algorithms…) that require operators with a high technical level. Although the approach can be easily extended to other types of functional tests.

First, a description of the different types of aircraft assembly lines and their work stations is provided.

Next, the standard navigation systems in an aircraft are briefly explained, and the various problems of the testing process are analyzed, such as defects in installations, equipment, human factors, documentation, etc.

From this analysis, the set of functional, ergonomic, and architectural (hard-ware and software) requirements that the assistant must satisfy is determined.

Subsequently, a study of the different solutions currently available on the market regarding the orchestration platform, language model, and databases is carried out, selecting the most appropriate ones based on technical and eco-nomic criteria.

Finally, a methodology and evaluation metrics are proposed with the aim of creating a test suite that can assess how good the assistant is according to various criteria.

The project is completed with an economic analysis that determines development costs, annual operating costs, and savings, following the usual financial profitability metrics.

TABLE OF CONTENTS

1 OBJECTIVE....11
2 SCOPE....13
3 BACKGROUND....15
3.1 General Overview of an Aircraft Assembly Line....15
3.1.1 Types of Aircraft Assembly Lines....16
3.1.2 Main Phases of an Aircraft Assembly Line....19
3.2 Aerospace Industrial Model. Industry 4.0 and Industry 5.0....26
3.2.1 Introduction....26
3.2.2 Industry 4.0....27
3.2.3 Industry 5.0....29
3.3 Artificial Intelligence and Industry 5.0....32
3.3.1 Classic and Generative Artificial Intelligence....32
3.3.2 Impact of AI on Current I4.0 Industry....33
3.3.3 Role of Artificial Intelligence in Industry 5.0....34
4 STANDARDS AND REFERENCES....36
4.1 Legal provisions and applied standards....36
4.1.1 Artificial Intelligence Regulation (EU) 2024/1689....36
4.1.2 Data Protection (GDPR and LOPDGDD)....36
4.1.3 Occupational Risk Prevention (ORP)....37
4.1.4 Digital Accessibility (Law 11/2023)....37
4.2 Base Software....38
4.2.1 N8N....38
5 DEFINITIONS AND ABBREVIATIONS....40
5.1 Definitions....40
5.1.1 Fundamental AI and Data Concepts....40
5.1.2 Generative Artificial Intelligence....40
5.1.3 Manufacturing and Implementation Environment....40
5.2 Abbreviations....41
5.2.1 Reference and Flight Systems....41
5.2.2 Air Navigation....41
5.2.3 Air Communications....42
5.2.4 Engine Control and Avionics Systems....42

6 FIELD OF APPLICATION.....43
6.1 Case study: King Air 360 Aircraft.....43
6.2 King Air 360 Final Assembly Line (Wichita, USA).....45
6.3 Integrated Digital Avionics.....46
6.4 Air Navigation Systems.....49
6.4.1 Air Data System (ADS).....50
6.4.2 Attitude and Heading Reference System (AHRS).....53
6.4.3 Weather Radar System (WRS).....56
6.4.4 Radio Altimeter System (RALT).....59
6.4.5 Traffic Alert and Collision Avoidance System (TCAS).....62
6.4.6 Global Navigation Satellite System (GNSS).....65
6.4.7 VHF Radio Navigation System (RNAV VOR/ILS/MKR/ADF).....68
6.4.8 DME System (DME).....71
6.4.9 Transponder System (XPDR).....74
6.5 Air Navigation Test Execution.....77
6.5.1 Overview.....77
6.5.2 Work Team.....78
6.5.3 Planning.....79
7 DESIGN REQUIREMENTS.....81
7.1 Introduction.....81
7.2 Avionics Functional Testing Issues.....81
7.2.1 Electrical Installation Defects.....82
7.2.2 Equipment Defects.....85
7.2.3 Human Factors.....86
7.2.4 Training and Knowledge.....88
7.2.5 Documentation.....89
7.2.6 Test Equipment.....90
7.3 Functional Requirements.....92
7.3.1 Functional Test Assistance.....92
7.3.2 Diagnostic Assistance in Case of Failure.....92
7.3.3 Repair Assistance.....92
7.3.4 Assistance in the Use of Test Equipment.....93
7.3.5 Training Assistance.....93
7.3.6 Human Factors.....93
7.4 Ergonomic Requirements.....94

7.5 Architecture Requirements....95
7.5.1 Hardware Requirements (Processing)....95
7.5.2 Hardware Requirements (Interface)....95
7.5.3 Software Requirements....95
7.5.4 Software Requirements (RAG Retrieval Augmented Generation)....96
7.6 Requirements Verification....97
8 ANALYSIS OF SOLUTIONS....98
8.1 Introduction....98
8.2 Expert Systems....98
8.3 Document Search Engines....99
8.4 Traditional Chatbots....99
8.5 Classic Machine Learning Systems....100
8.6 Large Language Models....100
8.7 Justification of the choice....101
8.8 Intelligent Assistant....102
8.9 General Architecture....103
8.9.1 Orchestration Platforms....103
8.9.2 Language Models....104
8.9.3 Vector Databases....105
8.10 N8N Orchestration Platform....106
8.10.1 Key Components....106
8.10.2 Main Features....106
8.10.3 Agent Architecture....107
8.10.4 Orchestrator Agent (version 1)....114
8.10.5 Orchestrator Agent (version 2)....116
8.11 Mistral 14B Language Model....119
8.12 Pinecone Database....120
9 FINAL RESULTS....123
9.1 Evaluation Methodology....123
9.2 Evaluation Metrics....124
9.3 Survey....125
9.4 Validity Criteria....128
9.5 Conclusions....129
10 ECONOMIC ANALYSIS....130
10.1 Development Costs....130

10.2 Annual Operating Costs....132
10.3 Quantification of Savings....133
10.4 Financial Profitability Metrics....134
11 BIBLIOGRAPHY....135
12 ANNEX 1 - SPECIFICATIONS....138
12.1 General Conditions....138
12.1.1 Purpose and Scope....138
12.1.2 Intellectual Property and Software Licenses....140
12.1.3 Responsibilities of the Parties....141
12.1.4 Validity and revision of the document....142
12.2 Technical Conditions Materials/Software....142
12.2.1 Server hardware requirements....142
12.2.2 Network and communications requirements....142
12.2.3 Server software requirements....143
12.2.4 Documentary Knowledge Base (RAG)....144
12.2.5 Operator Interface Devices....145
12.3 Execution and Installation Conditions....146
12.3.1 Preconditions before start....146
12.3.2 Installation Procedure....146
12.3.3 Phased Commissioning Plan....147
12.3.4 Acceptance tests....148
12.4 Operation and Maintenance Conditions....149
12.4.1 Normal operating conditions....149
12.4.2 Access Control and Role Management....150
12.4.3 System maintenance....150
12.4.4 RAG knowledge base management....151
12.4.5 Data protection and GDPR/LOPDGDD compliance....151
12.4.6 Incident procedure....152
13 ANNEX 2 - FRONTEND....153
13.1 Frontend Implementation....153
13.1.1 HTML interface structure and organization....154
13.1.2 Operating logic and conversation management....155
13.1.3 Visual design and CSS styles....157
13.2 Docker as a Deployment Platform....158
13.2.1 Docker fundamentals....158

13.2.2 Docker Images....158
13.2.3 Docker Containers and Deployment Architecture....159
13.2.4 Advantages of using Docker....161
13.3 Server Architecture....162

LIST OF FIGURES

Figure 1-1 AI in Industry 5.0....12
Figure 2-1 King Air Final Assembly Line (Kansas)....14
Figure 3-1 F-35 Final Assembly Line Fort Worth (Texas)....15
Figure 3-2 X-59 Prototype Stationary Line (California)....16
Figure 3-3 Airbus C-295 Pulse Line (Seville)....17
Figure 3-4 Boeing 777 Moving Line (Washington)....18
Figure 3-5 Origin of Airbus A320 Major Components (Toulouse-France)....19
Figure 3-6 A400M Wing Transport from Filton to Seville....20
Figure 3-7 A321 Fuselage Transport Saint-Nazaire/Hamburg to Mobile....20
Figure 3-8 Fuselage Sections at Reception (Mobile-Alabama)....20
Figure 3-9 Boeing B777 Fuselage Join Robots (Everett – Washington)....21
Figure 3-10 Airbus A400M Harness Installation (San Pablo-Seville)....22
Figure 3-11 Airbus Cockpit Units Installation (Toulouse)....23
Figure 3-12 Computer-Assisted Functional Tests (Airbus)....25
Figure 3-13 A321 Final Close-up Mobile (Alabama)....25
Figure 3-14 Aerospace Industrial Model....26
Figure 3-15 Industry 4.0 Technologies....28
Figure 3-16 Industry 5.0 Values....30
Figure 3-17 Humanoid robots building an aircraft....31
Figure 3-18 Evolution of Artificial Intelligence....32
Figure 4-1 AI Risk Levels....37
Figure 4-2 Automation with n8n....39
Figure 5-1 ICAO....42
Figure 6-1 King Air 360 Image....43
Figure 6-2 PT6A-60A Powerplant and McCauley C780 Propellers....44
Figure 6-3 Cockpit and Cargo Cabin....44
Figure 6-4 King Air 360 Assembly Line (Wichita – Kansas)....45
Figure 6-5 Cockpit Avionics “Collins Proline Fusion”....47
Figure 6-6 Avionics Block Diagram....48
Figure 6-7 King Air 360 Air Navigation Systems....49
Figure 6-8 Air Navigation Aid Systems....49
Figure 6-9 ADS Operation....50
Figure 6-10 Dual Pitot Tube System & Temperature Sensor....51
Figure 6-11 ADS Block Diagram....51
Figure 6-12 Air Data Test Set....52
Figure 6-13 Cap Kit and Aircraft Connection....52
Figure 6-14 AHRS Operation....53
Figure 6-15 AHRS Controls....54

Figure 6-16 AHRS Block Diagram....54
Figure 6-17 AHRS System Test Equipment....55
Figure 6-18 AHRS Test System Software....55
Figure 6-19 Weather Radar Operation....56
Figure 6-20 Weather Radar System....57
Figure 6-21 WRS Block Diagram....57
Figure 6-22 Digital Inclinometer....58
Figure 6-23 Radio Altimeter Operation....59
Figure 6-24 Radio Altimeter System....60
Figure 6-25 Radio Altimeter Block Diagram....60
Figure 6-26 Radio Altitude Simulator Delay Unit....61
Figure 6-27 ARINC Source....61
Figure 6-28 TCAS System Operation....62
Figure 6-29 TCAS System....63
Figure 6-30 TCAS Test Equipment....64
Figure 6-31 GNSS System Operation....65
Figure 6-32 GNSS System....66
Figure 6-33 GNSS Hangar Test Equipment....67
Figure 6-34 Radio Navigation System Operation....68
Figure 6-35 Radio Navigation System....69
Figure 6-36 VHF Radio Navigation System Block Diagram....69
Figure 6-37 Radio Navigation Test Equipment....70
Figure 6-38 DME Operation....71
Figure 6-39 DME System....72
Figure 6-40 DME Block Diagram....72
Figure 6-41 DME Test Equipment....73
Figure 6-42 XPDR System Operation....74
Figure 6-43 XPDR System....75
Figure 6-44 XPDR Block Diagram....75
Figure 6-45 XPDR Test Equipment....76
Figure 6-46 Technician Performing Avionics Tests (King Air 360)....77
Figure 6-47 Avionics Specialists Testing Systems....78
Figure 6-48 King Air Systems Checklists....79
Figure 6-49 Technicians Troubleshooting a Fault on a King Air 360....80
Figure 7-1 Installation of Electrical Harnesses in an Aircraft....82
Figure 7-2 Electrical Harness Defects....83
Figure 7-3 Amphenol Connector, Inserter/Extractor and Crimper....84
Figure 7-4 Incorrect Software Configuration....85
Figure 7-5 Evolution of Aviation Accident Causes....87
Figure 7-6 Work on Aircraft with PPE....87
Figure 7-7 Assembly Line Training....88
Figure 7-8 Typical Functional Test with Warning....89
Figure 7-9 Navigation Test Equipment Manual....91
Figure 7-10 App Ergonomics....94
Figure 7-11 Typical Virtual Assistant Architecture....96
Figure 8-1 Chatbot with LLM....102

Figure 8-2 N8N Platform.....103
Figure 8-3 AI Language Models.....104
Figure 8-4 Vector Databases.....105
Figure 8-5 Agent Architecture.....108
Figure 8-6 ATA Chapters.....109
Figure 8-7 N8N Orchestrator Agent Version 1.....115
Figure 8-8 N8N Orchestrator Agent Version 2.....118
Figure 8-9 LLM Comparison.....119
Figure 8-10 Database Comparison.....121
Figure 8-11 Synchronization from Drive to Pinecone.....122
Figure 9-1 Virtual Assistant Evaluation Criteria.....123
Figure 9-2 Technical Evaluation of the AI Assistant.....124
Figure 13-1 THOT Structure.....153
Figure 13-2 Data Flow.....156
Figure 13-3 THOT Interface.....157
Figure 13-4 Deployment Architecture.....161
Figure 13-5 Server Architecture.....163

1 OBJECTIVE

The main objective of the project is the transformation, through the use of AI Assistants, of an industrial process under Industry 5.0.

An industrial AI Assistant is a software tool based on generative artificial intelligence specifically designed to support complex operations in manufacturing, assembly and supply chain environments.

Unlike a generic AI assistant, the industrial one is trained with technical data on the product, machinery and processes to help operators and engineers make faster and more accurate decisions.

The project idea stems from recent advances in generative AI (LLMs, AI Agents, Embodied AI, etc.) that enable the transformation of current I4.0 industry (automation and data efficiency) towards an I5.0 industry, more focused on human-machine collaboration and where resilience and sustainability are key.

Within this human-machine cooperation model, applying AI to industrial processes can transform daily operations by empowering the operator and optimizing production [1].

Some of the possible applications would be:

- *Work Instructions Assistant:* provides step-by-step guides and instant answers on complex assembly or test processes, eliminating dependence on paper manuals.
- *Troubleshooting Assistant:* helps detect the causes of failures during the product's functional tests.
- *Predictive Maintenance:* warns of possible machine failures before they occur by analyzing sensor data (IoT) and guides the technician through the repair process with rapid diagnostics.
- *Real-Time Inventory Management:* provides information on stock levels or order status through natural language integrated with corporate ERP systems.
- *Continuous Improvement:* identifies inefficiency in industrial processes and helps improve it based on execution data.
- *Training:* facilitates learning for new operators by answering frequent questions and offering interactive training directly at the workstation. It is also very useful when new products or machines are introduced into the industrial process.
- *Occupational Safety:* monitors and warns of hazardous conditions, protecting the team through real-time analysis of safety data.

All of this produces benefits in:

- *Increased Productivity:* allows employees to focus on higher value-added tasks by automating activities such as repetitive queries, searching for materials, equipment, etc.
- *Reduced Downtime:* by providing immediate solutions and predicting failures, unplanned stoppages are minimized.
- *Reduced Technical Support Staff:* in many cases it allows technical support staff (engineers, specialized technicians, etc.) to be reduced by providing operators with precise technical solutions. In addition, with 24/7 availability.
- *Improved Quality:* it can act as a checkpoint, ensuring quality protocols are followed and reducing human error.
- *Reduced Training Time:* by providing technical answers to any question from novice operators.

In short, thanks to generative AI, the set objectives can be achieved at a lower cost by eliminating waste (searching for information, time lost due to lack of knowledge of the systems...), resulting in an increase in the overall efficiency of the assembly line.

In addition, the use of natural language and personalization allows operators to feel they are in a less aggressive technological environment, in line with the principles of Industry 5.0.



Figure 1-1 AI in Industry 5.0 (Source: Cimatron)

2 SCOPE

Although there is a wide variety of industrial processes that can be improved by introducing AI, this work will focus on the field of aeronautical manufacturing and, in particular, on Aircraft Final Assembly Lines.

Here, AI actions can range from logistics, engineering and quality to production areas (assembly, rigging, testing, etc.).

Due to its difficulty, criticality and relevance to aircraft safety, the functional testing process will be taken as the main focus of the project.

Functional Tests

The testing process is the final phase of aircraft assembly and the most critical, given the complexity of the systems and the high quality and safety standards this type of product requires.

These demanding standards mean that highly qualified staff, up-to-date documentation, test machines in perfect condition, etc. are needed.

Generative AI can help improve this process by acting on all production aspects (operation, safety, quality…) through virtual technical agents that assist in the established tasks.

These agents act as an intelligent bridge between the operator and the complex process information, optimizing execution through real-time support.

For example, some of the functions a virtual assistant for functional tests could have would be:

- *Real-Time Assistance and Personalized Training:* the assistant acts as a mentor providing step-by-step instructions, diagrams or videos for complex functional tests or new product variants, eliminating the need for physical manuals and accelerating on-the-job learning.
- *Problem Solving:* the assistant analyzes historical data and suggests specific solutions to the operator, enhancing their decision-making ability and reducing stress.
- *Knowledge Capture:* allows experienced operators to record voice or text notes on operational "tricks" that the AI processes and shares with the rest of the team, preserving human talent within the organization.
- *Sustainability and Well-being:* it can monitor the line's energy consumption and suggest adjustments to the operator to reduce waste, thus integrating Industry 5.0 values into daily operations.

In short, in this work, we will focus on developing an industrial virtual technical assistant that operates in the avionics functional testing phase (air navigation), evaluating applicable technologies and the advantages and disadvantages of their use.

We will do so for a typical medium-sized Assembly Line such as that of the King Air 360 aircraft.

This line is located in Wichita (Kansas) and the aircraft model belongs to the Beechcraft King Air family, with more than 3,000 units built.

This line was chosen because of how easy it is to obtain information on the internet about the aircraft (King Air 360), its avionics (Rockwell Collins Proline Fusion) and its processes (medium-sized turboprop aircraft).



Figure 2-1 King Air Final Assembly Line (Kansas) (Source: Kingairnation)

3 BACKGROUND

3.1 General Overview of an Aircraft Assembly Line

In the FAL (Final Assembly Line) the final manufacturing process of an aircraft is carried out. This is where all the components defined in the design phase are joined, ranging from large structural components to harnesses, pipes, equipment, control units, etc. [2]

When the aircraft leaves the FAL it is a fully functional aircraft (although it cannot fly yet), certified and ready for the next phases of the process (Flight Line, Production Flights and Customer Delivery) [3].

It differs notably from other lines, such as the automotive one, in aspects such as the high technical complexity requiring multidisciplinary integration (electrical, electronic, hydraulic.).

Also in the high quality and safety requirements, as well as exhaustive process tracking systems, certifications...

All of the above is essential to guarantee a safe product where any failure could cost many human lives.



Figure 3-1 F-35 Final Assembly Line Fort Worth (Texas) (Source: Airandspaceforces)

3.1.1 Types of Aircraft Assembly Lines

Depending on the construction flow of the assembly line, they can be classified as:

Stationary Line [4]**:** the aircraft remains in a fixed position while the work teams, materials and industrial resources (tools, test equipment, …) move to it.

All construction activities are carried out in the same place (inside and around the aircraft) and once assembly is finished, the aircraft leaves the FAL and goes to the flight line for final tests and production flights.

This assembly model is ideal for short series (<10 aircraft/year), prototypes and highly customized aircraft where the work is not very repetitive.

Due to the small size of the aircraft, it has the drawback of greater interaction between work teams, causing interference and waiting if preceding operations have not been completed.

In addition, there is a continuous flow of materials, equipment, etc. since not all the necessary elements can be placed around the aircraft due to lack of surrounding space.

All this leads to low efficiency, increases in non-value-added activity and waste due to travel, …



Figure 3-2 X-59 Prototype Stationary Line (California) (Source: Aviationweek)

Pulse Line [5]: in this case the aircraft moves between several stations (usually fewer than 10 stations) and at each of them an activity is carried out (assembly of

major components, equipment installation and rigging, functional tests...) according to the construction progress status.

The move between stations is done synchronously for all stations and takes place at regular intervals (“takt time”). The move is usually carried out at night when the impact on production is minimal. This workflow is ideal for medium production series (<25 aircraft/year) with medium or low customization.

In this model the different work groups, together with the necessary materials and industrial equipment, are fixed at the different stations. This allows greater operator specialization, less time lost to interference and waiting, and less travel. As a drawback, there is usually duplication of basic industrial facilities, common tools… because each station must have them.

It is less flexible because operators are more specialized and, in addition, when work is not completed at one station, having to do it at another causes performance losses due to travel and interaction with scheduled activities.

All this makes efficiency higher than in stationary assembly, although it does not reach the levels of moving lines. As a drawback, it requires a certain technical complexity of the movement systems and an increase in costs for kit preparation (done outside the moving line).



Figure 3-3 Airbus C-295 Pulse Line (Seville) (Source: Airbus)

Moving Line [6]**:** based on the automotive manufacturing model and adapted to the specific case of an aircraft (larger than a car both outside and inside).

In this flow, the aircraft, together with assembly stands, moves along the hangar pulled by an electric tug at a speed of 1 or 2 meters per hour depending on the

desired production rate. The movement goes almost unnoticed and allows the operators to move along with the aircraft and stands, so they can complete their operations.

In normal operation, when mechanics arrive at work, following the lean point-of-use (“PoU”) philosophy, they receive a pre-prepared kit according to the assigned operation, consisting of the materials, tools, consumables (sealants, adhesives...) and instructions, either on paper or on an electronic tablet.

With this way of working, staff do not have to move around looking for the production elements they need, and value added is maximized. On the other hand, by towing the facilities (stands, electrical and pneumatic power,) the cost of duplicating industrial resources that occurred in pulse lines is reduced.

In addition, the movement of the aircraft, although small, creates a stress very similar to that of automotive lines, which increases productivity. This model is ideal for large series (< 250 aircraft/year) and has very high efficiency.



Figure 3-4 Boeing 777 Moving Line (Washington) (Source: Aerospace ARNI)

3.1.2 Main Phases of an Aircraft Assembly Line

Although they may vary depending on the aircraft model and the industrial strategy (internal or external manufacturing, “Make or Buy”), in general there are the following phases [7], [8]:

Station 0: Reception and Preparation of Major Subassemblies

Aircraft, due to their design, can be built from their fundamental structures, also called MC “Major Components”.

These subassemblies are normally manufactured outside the FAL and sometimes by different suppliers specialized in structures. The elements may come from different plants located in different countries, so they arrive at the FAL by various means of transport such as trucks, trains, ships or even other aircraft.

This phase includes the reception of elements such as the fuselage (forward, center and rear sections), the wings (complete or semi-assembled), the stabilizers (horizontal and vertical), rudders and flaps, engines, landing gear, ...

The key activities in this phase are inspections (visual and dimensional), documentary verification (certificates and traceability) and their installation in positioning tools ("jigs").

In some cases (wings), functional checks are carried out on the included systems (fuel tanks, flap and aileron actuators...) that complement those already carried out by the structure manufacturers (leak tightness).

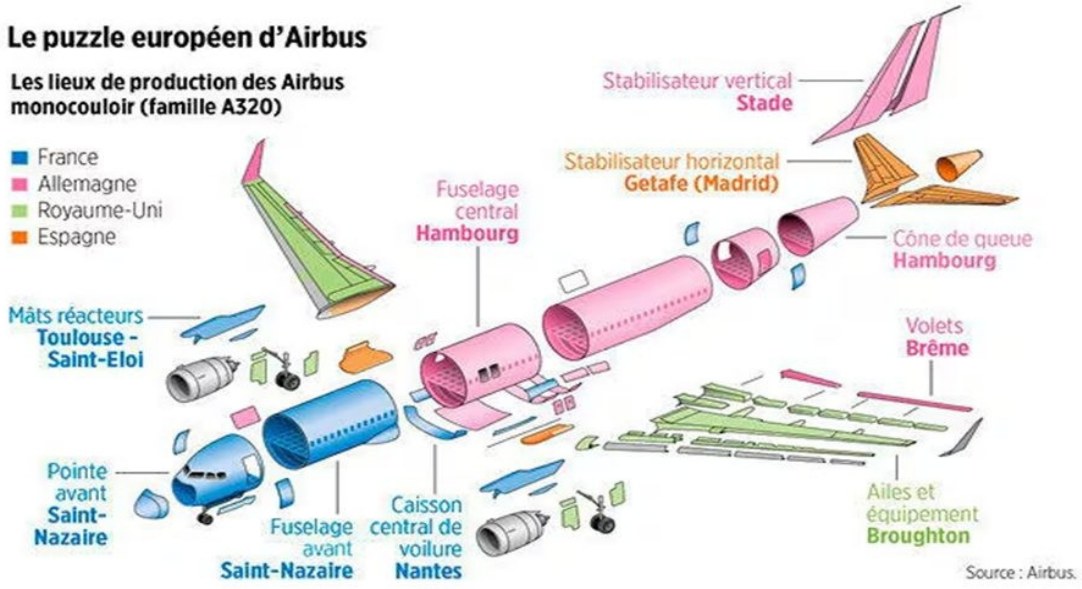


Figure 3-5 Origin of Airbus A320 Major Components (Toulouse-France) (Source: Airbus)



Figure 3-6 A400M Wing Transport from Filton to Seville (Source: Hispaviación)



Figure 3-7 A321 Fuselage Transport Saint-Nazaire/Hamburg to Mobile (Source: Airbus)



Figure 3-8 Fuselage Sections at Reception (Mobile-Alabama) (Source: Airbus)

Station 1: Primary Structure Assembly

In this phase the aircraft begins to take shape, turning the different parts into a complete structural airframe.

The activities carried out range from joining the different parts of the fuselage (nose, center fuselage and tail fuselage) to integrating the wings with the fuselage, installing horizontal and vertical stabilizers, etc...

It is done using rivets and bolts, in a manual or semi-automatic process with robotic machines.

Here precision is critical, as it will define the aerodynamic shape of the aircraft and therefore its flight performance.

These activities usually make up the first station of the FAL.

They normally require operators qualified for mechanical tasks.



Figure 3-9 Boeing B777 Fuselage Join Robots (Everett – Washington) (Source: Seattletimes)

Station 2: Basic Systems Installation

Once the main structure of the aircraft is available, the next phase is the installation of minor structures such as doors or ramps and of the key elements for system operation (electrical harnesses and hydraulic, pneumatic, fuel, air conditioning lines, flight controls...).

The electrical harnesses come from external facilities and in the FAL they are laid out along the interior of the aircraft and fixed to the different structural supports.

The basic elements of the electrical system (electrical distribution boxes, batteries...) are also installed at this point.

Likewise, the different hydraulic pipes are installed and cleaned of dirt and metal debris (swarf) through the process called “flushing”, where pressurized oil is repeatedly passed through the pipes until all impurities are removed.

It is completed with the installation of hydraulic pumps, filters, reducers...

The engines are also installed and interconnected to the power distribution, pressurization, air conditioning, anti-ice systems...

Similarly, the installation of flight controls (control wheels and columns, control cables, control surfaces...) is completed.

All these activities are usually included in a second station.



Figure 3-10 Airbus A400M Harness Installation (San Pablo-Seville) (Source: Miracle)

Station 3: Complete Equipment Installation and System Rigging

Here the installation of the systems is completed by fitting the equipment and control units (cockpit) and interconnecting them with the previously installed harnesses and pipes.

This installation is not done at the previous station to avoid mechanical damage to the units and system parts during the assembly work of the previous station.

Special attention must be paid to the installation of all avionics systems (computers, displays, sensors) since, as they contain software, it must be in the correct version according to the aircraft/customer specifications.

Likewise, the systems (engine, flight controls, doors...) are rigged to leave them in accordance with the design specifications.

Partial leak, electrical and structural continuity tests are also carried out...

Highly qualified staff in installation, rigging and testing are required.



Figure 3-11 Airbus Cockpit Units Installation (Toulouse) (Source: Airbus)

Station 4: Functional Tests

Once all system components are installed and rigged, each complete system must be tested to check that it works as designed [9].

The set of tests will depend on the systems installed in the aircraft, but they can basically be grouped as follows:

- *Electrical Tests*: they begin with powering up the aircraft via an external GPU or batteries, checking that the electrical distribution and protection systems of all systems work.
- *Avionics Tests:* once electrical power is available throughout the aircraft, the avionics systems (computers, displays, radio, navigation and

instrumentation equipment.) are switched on and checked to ensure they work and show no failure flags, and in-flight radio aid conditions (VOR/ILS/DME, ATC...) are simulated using test equipment.

- *Hydraulic Tests:* using external hydraulic mules and internal pumps, all hydraulic systems and the aircraft systems they feed (ramp, landing gear, flaps...) are checked
- *Fuel Tests:* the electrical fuel systems (refueling, engine feed, transfer...) are tested, in some cases with simulated fuel (Shellsol).
- *Flight Control Tests:* the flight control and autopilot systems are tested.
- *Pressurization Tests:* the aircraft is completely closed and pressurized air is introduced, simulating flight conditions and checking that there are no leaks or cracks.

The tests are normally carried out at a specific station using computerized systems that indicate the actions the operator must perform on the aircraft and even read from the aircraft through data access points.

In the event of a system failure, the test technicians, with the help of support engineers, locate and repair it (broken wires, faulty equipment, …)

As no digital assistance system is available, execution and troubleshooting times fluctuate greatly depending on the experience of the operators and support engineers.

This sometimes causes delays when the fault is difficult to resolve. In addition, support engineers are required on both work shifts (morning and afternoon), which makes the activity much more expensive.

After leaving the FAL, once on the FL, some of the tests are repeated under real conditions (real fuel...) and others are done for the first time (engine start) because it is impossible to do them inside the FAL.

Finally, in production flights, the pilots perform a complete typical flight profile in which all systems are tested under real conditions.



Figure 3-1 Computer-Assisted Functional Tests (Airbus) (Source: Merkur)

Station 5: Interiors and Final Close-up

Once the operation of all systems has been checked, the aircraft can be closed up and given its final customization at the last station.

Activities include the installation of thermal and acoustic insulation (blankets), interior panels, seats, lavatory and galley.

The final quality check (physical and documentary) is also carried out and the aircraft is made ready for delivery to the Flight Line.

One of the risks at this station is that handling during close-up may affect functional systems already tested, causing defects to appear on the FL that did not exist in the FAL.



Figure 3-13 A321 Final Close-up Mobile (Alabama) (Source: Airbus)

3.2 Aerospace Industrial Model. Industry 4.0 and Industry 5.0

3.2.1 Introduction

The aircraft manufacturing model is a complex process involving design, material selection, component manufacturing, assembly and rigorous testing. This process requires the collaboration of engineers, designers and technicians, as well as the use of advanced technologies that ensure efficiency and safety. It also has the following characteristics:

The manufacturing of components, systems and subsystems is usually carried out independently, on demand, by numerous suppliers around the world before being integrated into the final assembly line (FAL).

Production and assembly planning, as well as external and internal production supply logistics, are therefore a very complex but integral part of aircraft production, which nevertheless depends on a complex global supply chain.

The wide variety of systems, such as the aircraft cabin, where each customer has different requirements, and the low order volume further increase the complexity of planning and production tasks.

Manual work predominates in aircraft production, and final assembly takes place in a fixed position or in pulses rather than on a continuous moving line like car assembly lines.

All design, production and testing processes must be qualified, documented and supervised by the competent authorities, original equipment manufacturers and customers to guarantee aircraft safety.

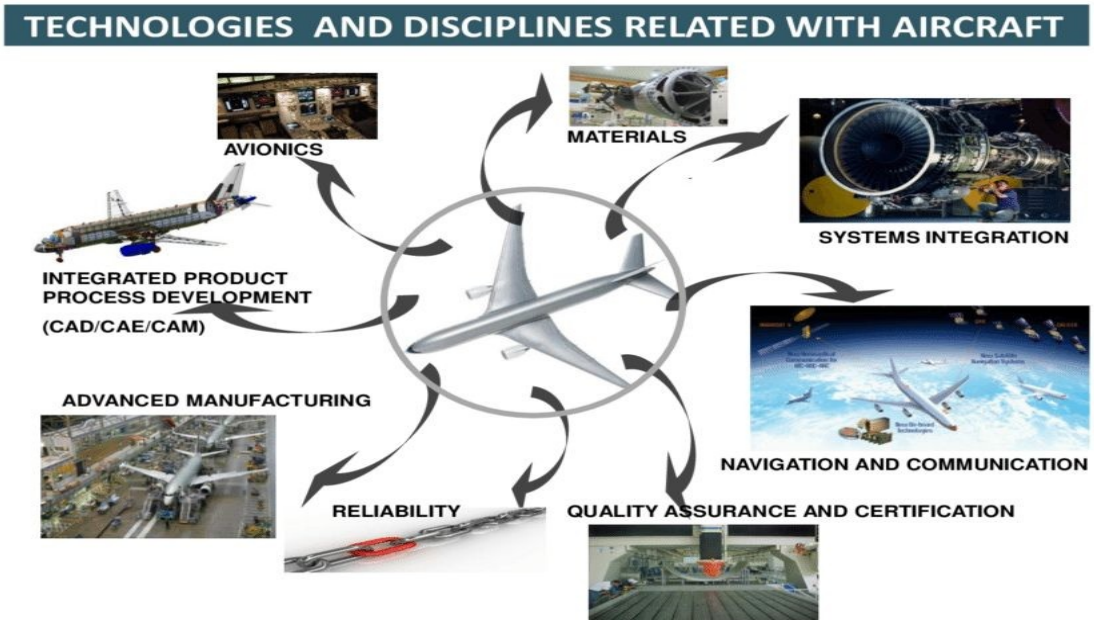


Figure 3-14 Aerospace Industrial Model (Source: Kemal 2013 - A survey of Aircraft Materials)

3.2.2 Industry 4.0

In the last decade, the industrial sector has undergone a complete transformation linked to technological progress derived from digitalization and interconnectivity.

This process, which formally emerged in 2011 in Germany, falls under the so-called Industry 4.0, and its fundamental principle is that machines, systems and people are interconnected, optimizing production and decision-making.

To this end, the industrial model is transformed through Cyber-Physical Systems (CPS), which are an evolution of traditional industrial systems combining physical elements with advanced digital capabilities, allowing more direct and flexible interaction between machines and human operators

The technologies required for this transformation are [47- 48]:

- *Internet of Things (IoT):* enables devices and machines to connect over the Internet, allowing real-time data collection and exchange.
- *Cloud Computing:* this technology supports IoT, providing access to data storage resources from different sources over the Internet.
- *Data Analytics (Big Data):* enables the analysis of large datasets to obtain valuable insights and make better-informed decisions.
- *Augmented and Virtual Reality (AR/VR):* overlays digital information on the physical environment, improving operator perception and providing support tools that increase efficiency and accuracy in manufacturing tasks.
- *Classic Artificial Intelligence (Classic AI):* plays a fundamental role in processing and cleaning large volumes of data generated by industrial sensors, facilitating predictive analysis and decision support in complex, highly dynamic production processes.
- *Robotics and Automation:* the integration of robots and automated systems increases efficiency and precision.

However, this model, while more efficient than past industrial organizations, has serious problems:

- *High Costs and Slowness:* industrial automation requires substantial investment in hardware, software and specialized engineering. Long PLC programming cycles delay adoption, deployment and scaling. In addition, traditional industrial controllers are obsolete and cannot harness the transformative power of industrial AI.
- *Rigid Systems:* traditional solutions struggle to adapt to changing production demands.
- *Labor shortage:* skilled labor for advanced automation is scarce (software engineers, robotics programmers...).

- *Loss of the Operator's Central Role:* machines still set the pace of the process, and the operator lacks control and feels overwhelmed by new technological changes that lead to a loss of identity in the process.
- *Sustainability Concerns:* inefficient systems and high energy consumption conflict with modern sustainability goals.

In summary, Industry 4.0, compared with the previous industrial model (Industry 3.0), has increased process efficiency and has also improved analysis and decision-making, the effectiveness and flexibility of industrial processes and the human-machine relationship, but always from a perspective in which the machine is the center.

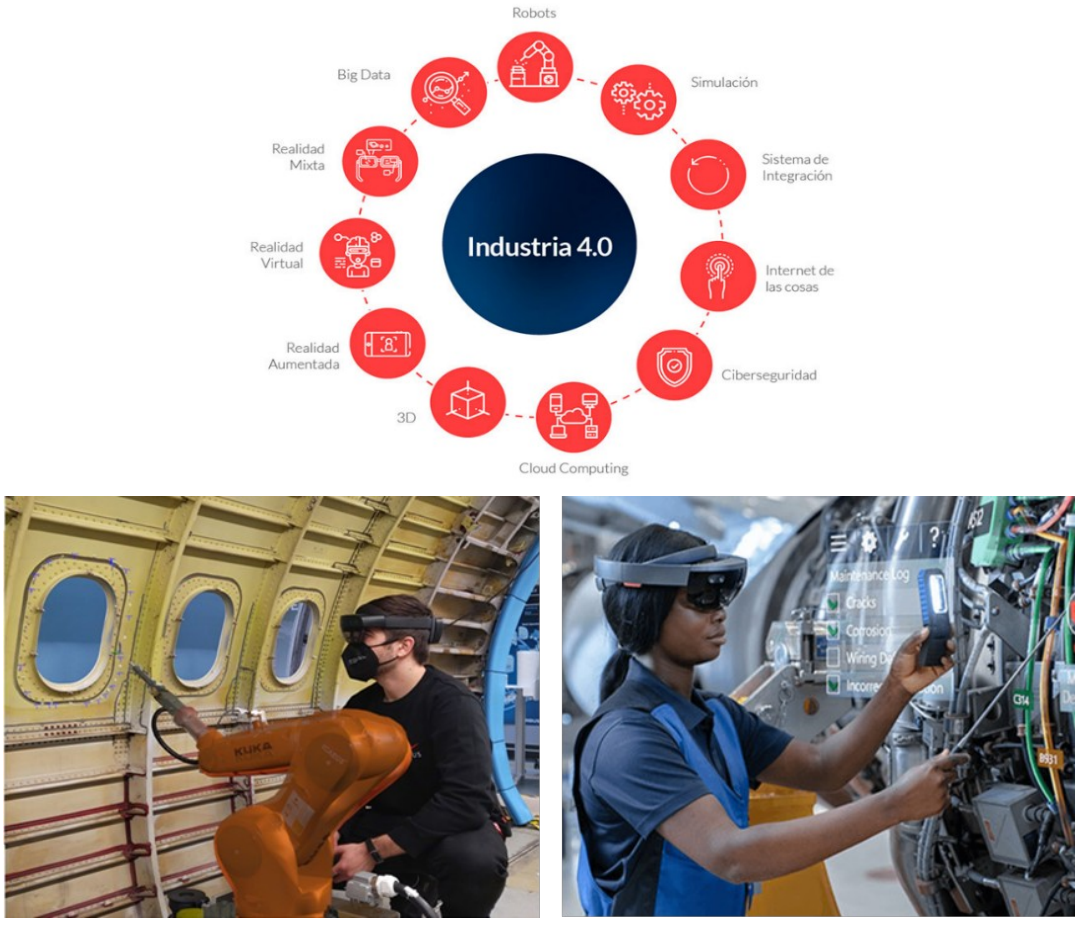


Figure 3-15 Industry 4.0 Technologies (Source: Keno Moenck 2024 - Industry 5.0)

3.2.3 Industry 5.0

With the evolution of society, which increasingly demands sustainability and production ethics, together with recent technological advances such as Generative AI, Europe (Industry 5.0) and Japan (Society 5.0) have raised the need to develop new, more human industrial models where the operator once again becomes the center of production [47-48].

In addition, events such as the Covid-19 pandemic, as well as the emergence of new geostrategic environments, wars, etc. make it necessary to have an industry that can withstand these situations and therefore has a more resilient design.

In short, the main points that would define the Industry 5.0 model would be:

- *Human-centric approach:* places the human at the center of the production process and emphasizes collaboration between humans and machines, leveraging human skills and creativity together with the precision and efficiency of robots and AI. It requires not only a change in the work environment and philosophy but also in the role of the operator in this production model. This role has evolved in line with the industrial model:
    - **Operator 1.0 (Industry 1.0-1770):** the operator who manually used tools based on experience and skill, but aided by steam-powered machines.
    - **Operator 2.0 (Industry 2.0-1870):** the operator whose work was determined by their place on the line (monotonous and non-creative), aided by electric machines.
    - **Operator 3.0 (Industry 3.0-1970):** the operator whose work was assisted by information systems (computers) and numerically controlled machines.
    - **Operator 4.0 (Industry 4.0-2010):** the operator who works collaboratively with classic robots, automated machines and computers within an interconnected cyber-physical system.
    - **Operator 5.0 (Industry 5.0-2025):** the operator who, building on 4.0 cyber-physical systems, is oriented towards the symbiosis of the human system with the cyber-physical one, aided by artificial intelligence (AI), natural language processing (NLP), intelligent robotics…

- *Industrial resilience:* the term resilience refers not only to a system's ability to recover from disruptions (supply chain failures, geopolitical crises, natural disasters.), but also to its ability to adapt to new production or competitive environments and continue operating effectively. Implementing resilience in manufacturing systems begins with designing robust systems that consider strategic redundancy and flexibility, together

with strategic management to identify possible scenarios and establish contingency plans.

- *Sustainability:* reducing the environmental impact of industrial activity through efficient use of resources, energy optimization and the incorporation of responsible practices throughout the product life cycle.



Figure 3-16 Industry 5.0 Values (Source: Keno Moenck 2024 - Industry 5.0)

These objectives are achieved through the following technologies:

- *Internet of Everything (IoE)*: defined as a network of connections between people, organizations, systems and physical objects. It focuses on increasing customer satisfaction and loyalty, as well as creating personalized experiences from the data generated.
- *Digital twins:* connection of product and process simulations through the transfer of information from physical to digital devices via IoT. Evolution towards an approach more oriented to personalization, support for human decision-making and improving the user experience. They also reduce maintenance costs and improve system performance.
- *Big Data Analytics (BDA):* to better understand consumer behavior, which will make it possible to optimize product prices, improve production efficiency and reduce operating costs.
- *Artificial Intelligence (AI):* this technology supports decision-making and the automation of repetitive, low value-added tasks, freeing up the operator's time to focus on more relevant activities, thus reinforcing a human-centric approach in the industrial environment. In addition, natural language communication with machines.
- *Collaborative Robots and AI Humanoid Robots*: a cobot is a robot designed to work safely alongside a human in a shared space. Its goals are to be safe (incorporating sensors to detect human presence), flexible

(lightweight and adaptable to different applications), easy to use and to enable process automation. A humanoid robot is a machine designed to imitate human form and functionality. It seeks to interact with humans naturally through artificial intelligence, computer vision and locomotion systems to walk and recognize faces.



Figure 3-17 Humanoid robots building an aircraft (Source: Gemini)

3.3 Artificial Intelligence and Industry 5.0

The recent emergence of generative AI, in language models as well as visual models and physical AI, has led to a resurgence of this type of technology which, although it has been with us for a long time, has had very limited use in industry.

3.3.1 Artificial Intelligence: Classic and Generative

Artificial intelligence (AI) evolved from the theoretical concepts of the early 20th century and its formal beginnings in the 1950s to modern, complex systems, marked by periods of great optimism and "AI winters" in which progress stalled.

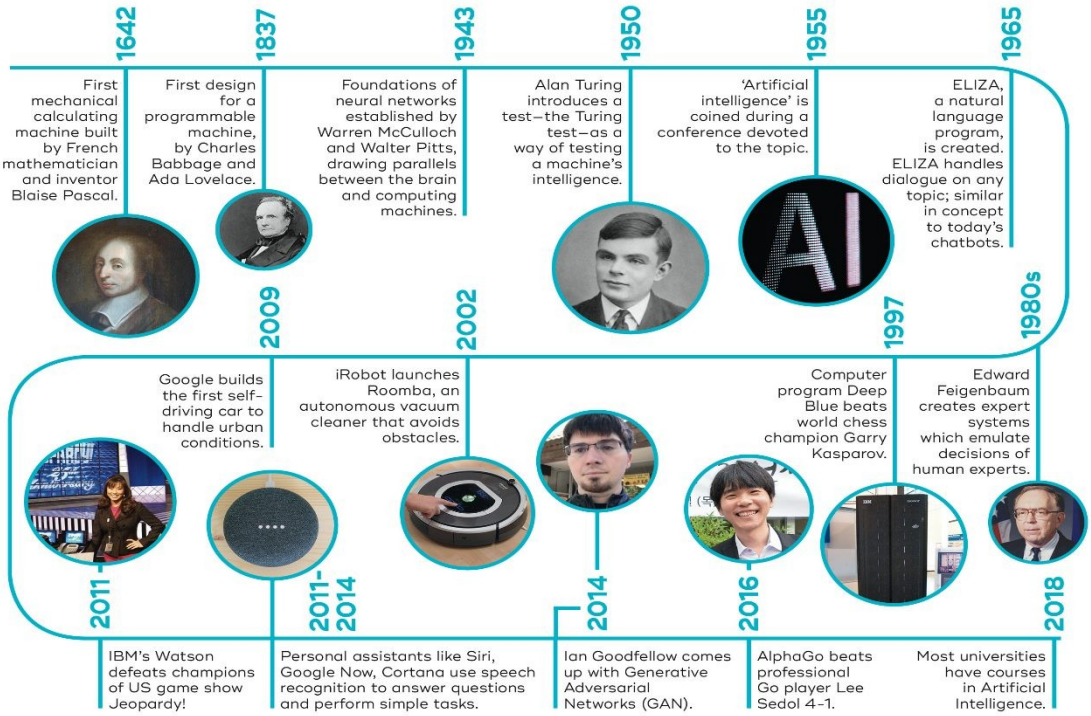


Figure 3-18 Evolution of Artificial Intelligence (Source: Medium)

Key milestones include Alan Turing's Turing Test, the 1956 Dartmouth Conference where the term AI was coined, the rise of expert systems in the 1980s and the advances, in recent years, in machine learning and deep learning driven by big data and greater processing power.

Coming back to the present, two types of AI can be distinguished:

- *Classic AI (Classic):* focuses on predictive analysis, classification and the automation of logical rules. It learns from structured data through supervised learning to optimize specific processes.
- *Generative AI (GenAI):* created to produce original content from natural language (text, images, video, code…).

In this project we will focus exclusively on the use of generative AI to understand, translate and generate natural language text.

To that end it uses a large language model (LLM), trained by reading billions of internet pages to learn to predict the most logical word to follow another. There are many types of LLM (ChatGPT, Gemini, Claude, Mistral, …) both for local use and cloud-connected.

It should be noted that an LLM does not "think" like a human; rather, it calculates statistical probabilities to respond coherently. If it does not know an answer, it will confidently make one up ("hallucination").

To improve its behavior, different techniques have been developed, such as Retrieval-Augmented Generation (RAG), which connects the LLM with external

databases or private documents so that the model answers based on specific data and documents and is therefore more accurate.

3.3.2 Impact of AI on Current I4.0 Industry

Generative artificial intelligence is transforming Industry 4.0 by moving from rigid, rule-based automation to smart factories capable of autonomously creating, coding, designing and diagnosing. While traditional AI predicted failures, generative AI designs the solutions and generates the code to execute them.

Some examples of the impact on the current I4.0 industrial model would be:

Cognitive Control

- *Let the machines program:* automates code generation for robots, CNC and hybrid systems, with no need for manual coding.
- *Single Control for all machines:* generates control intelligence on commercial hardware, from robots to CNC and everything in between.
- *Built for change:* adapts instantly to production shifts and product variation, minimizing downtime and manual intervention.
- *Always smarter:* continuously learns and optimizes for maximum machine performance.

Rapid Deployment

- *Faster than ever:* reduces deployment time with end-to-end, fully automated programming.
- *No more manual coding for robots/PLCs:* automates programming generation for industrial robotics.
- *Integrates with any line, old or new:* Integrates with legacy equipment and cutting-edge systems.
- *Accelerates commissioning and simplifies changes:* reconfigure and scale production in hours, not weeks

Smarter Manufacturing Control

- *Designed to scale across all your operations:* from a single cell to global factories.
- *Optimized for any production style:* supports both high-volume, low-variation runs and high-mix, high-variation environments.
- *Fewer errors, less rework:* consistent and precise logic execution on every machine.

- *Reusable logic, transferable intelligence:* move automation logic between machines and lines with minimal effort.
- *Connected:* from the shop floor to the supply chain.

3.3.3 Role of Artificial Intelligence in Industry 5.0

Artificial intelligence (AI) is fundamental to Industry 5.0, which evolves from Industry 4.0's focus on automation to a human-centric approach that prioritizes collaboration between humans and advanced machines (such as AI-powered robots) to achieve more sustainable, resilient and customized production systems.

AI enhances human capabilities by handling repetitive tasks, while humans contribute creativity and critical thinking to solving complex problems and to innovation. This integration results in intelligent, adaptive production lines, predictive maintenance, greater efficiency and sustainable practices such as waste reduction and better energy management.

Key Aspects of AI in Industry 5.0:

- *Human-machine collaboration:* AI and robots work alongside humans, while machines, increasingly humanized, perform routine tasks and free workers to concentrate on more creative, strategic and innovative activities.
- *Customization:* AI enables hyper-personalization, allowing products and solutions tailored to customers' individual needs and preferences.
- *Intelligent and adaptive production:* AI makes production systems smarter and more adaptable, going beyond simple automation towards real-time decision-making and self-regulation.
- *Predictive maintenance:* AI analyzes data to predict equipment failures, enabling proactive maintenance that minimizes downtime and improves overall efficiency.
- *Resilience:* AI-driven adaptive production lines can respond in real time to market changes, resource availability and unexpected disruptions, improving supply chain resilience.
- *Sustainability:* AI optimizes resource consumption, reduces waste and facilitates the integration of renewable energy sources and circular economy principles into production systems.

4 STANDARDS AND REFERENCES

4.1 Legal provisions and applied standards.

Installing a virtual assistant in a factory for technical support is subject to a legal ecosystem that combines Artificial Intelligence (AI) regulations, data protection and occupational safety.

4.1.1 Artificial Intelligence Regulation (EU) 2024/1689

This is the most recent and critical regulation [34]. Depending on how the assistant works, it will be classified by its risk level:

- *Transparency:* if the assistant is a chatbot or voice system that interacts with operators, the EU AI Act requires the worker to be clearly informed that they are interacting with an AI.
- *Training:* under Article 4, the company must ensure that operators using the system have adequate knowledge to understand how it works and its risks.
- *Human Oversight:* the design must allow a human to supervise or correct the assistant's decisions to avoid safety incidents.

4.1.2 Data Protection (GDPR and LOPDGDD)

When processing employee data, the strict obligations of the General Data Protection Regulation must be met [35]:

- *Data Protection Impact Assessment (DPIA):* mandatory before installation if the system involves systematic monitoring or large-scale data processing.
- *Privacy by Design:* the system must be configured to collect only the minimum data needed for technical support.
- *Digital Rights:* under the LOPDGDD (Spain), workers have the right to digital disconnection and to these systems not being used for abusive workplace monitoring without prior notice.

4.1.3 Occupational Risk Prevention (ORP)

The assistant becomes a work tool and must comply with physical safety and mental health regulations:

- *Royal Decree 1215/1997:* the work equipment (tablet, speaker or device hosting the assistant) must be safe and workers must receive training on its use.
- *Psychosocial Risks:* it must be assessed whether the assistant causes stress, mental fatigue or technostress in the operator due to excessive cognitive load or pressure from automated support.
- *Digital Ergonomics:* if the assistant requires the use of screens, it must comply with distance and viewing angle standards to avoid eye strain.

4.1.4 Digital Accessibility (Law 11/2023)

The Digital Accessibility law requires digital tools in the workplace to be accessible to employees with disabilities, following standards such as UNE-EN 301549.

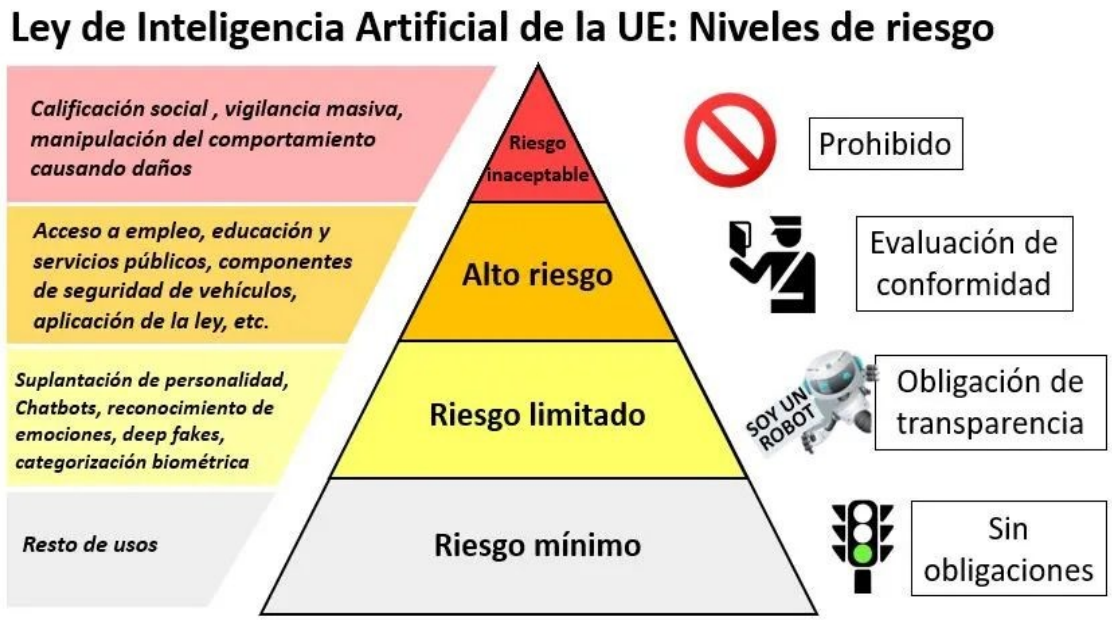


Figure 4-1 AI Risk Levels (Source: Juspax)

4.2 Base Software

Different software programs have been used in carrying out this project. They are detailed below.

4.2.1 N8N

The n8n software [36-41] is a "low-code" workflow automation platform that allows applications and data to be connected with draggable nodes, without the need to program.

It has currently positioned itself as a leading tool for integrating Large Language Models (LLMs) and Artificial Intelligence into business processes.

Unlike traditional automations, which are purely deterministic, n8n makes it possible to create "intelligent" systems that can reason, decide and execute actions based on natural language.

Main capabilities with AI and LLMs

- *Autonomous AI Agents [38-40]:* allows the creation of agents that not only answer questions but can also use "tools" (APIs, databases or code) to complete tasks autonomously.
- *Integrated LangChain Chains:* It internally uses LangChain concepts to structure complex flows, allowing models, memory (chat history) and document loaders to be connected visually.
- *"Brain & Arm" Architecture:* n8n acts as the "arm" that executes actions and retrieves data, while the LLM acts as the "brain" that processes and understands the information.
- *Specialized AI Nodes:* Includes nodes for specific tasks such as information extraction, summarization, classification, sentiment analysis, etc.

Supported Integrations and Models

- *Proprietary Models:* OpenAI (ChatGPT), Anthropic (Claude), Google (Gemini), etc.
- *Local/Open Source Models:* Integration with Ollama and Hugging Face to run AI on in-house servers, ensuring greater data privacy.
- *Vector Databases:* Connection with Pinecone, Weaviate or Supabase to implement RAG (Retrieval-Augmented Generation), allowing the AI to answer based on the organization's own documents.

Common Use Cases

- Customer Service: Chatbots [41] that consult internal manuals and answer questions or escalate to humans.
- Content Generation: Flows that research a topic on the web, write a draft and publish it on social media.
- Lead Qualification: Automatic analysis of a potential customer's profile to decide whether it should be sent to the sales team.
- Market Research: Agents that perform real-time web searches and generate structured reports.



Figure 4-2 Automation with n8n (Source: Fiverr)

5 DEFINITIONS AND ABBREVIATIONS

5.1 Definitions

5.1.1 Fundamental AI and Data Concepts

*Artificial Intelligence (AI): f*ield of computer science focused on creating systems capable of performing tasks that usually require human intelligence, such as learning, reasoning and decision-making.

Machine Learning (ML): subfield of AI that allows systems to learn and improve from historical data without having been explicitly programmed for a specific task.

Industrial Data: set of variables, sensor readings (IoT), maintenance history and operating parameters generated in a manufacturing plant. They are the main data feeding any industrial algorithm.

5.1.2 Generative Artificial Intelligence

Generative AI (GenAI): subset of AI based on machine learning and deep learning models capable of analyzing existing data and creating new, original content such as text, code, images or simulations.

Large Language Model (LLM): advanced Generative AI algorithm trained on huge amounts of text, enabling it to understand, summarize, translate and generate natural language.

Hallucination: phenomenon whereby a Generative AI model generates incorrect, made-up or logically unfounded information, but presents it in a coherent and convincing way.

5.1.3 Manufacturing and Implementation Environment

Production / Manufacturing Line: sequence of processes, operations and workstations organized to assemble or manufacture a final product.

Industrial IoT (IIoT): network of interconnected industrial devices, sensors, actuators and machines that collect and share data in real time within the plant.

Industrial AI Agent: AI-based software system capable of perceiving its environment (line parameters), reasoning and executing actions autonomously or semi-autonomously to optimize a process.

Prompt (Instruction): structured text input or instruction that an operator or system sends to a Generative AI tool to obtain a specific response or result (e.g. "Summarize the error on machine 4 and generate an action plan").

Digital Twin: real-time virtual replica of a real physical product, process or system. In this context, it is fed by AI to simulate production scenarios and predict bottlenecks.

5.2 Abbreviations

Avionics systems use a large number of abbreviations [42] defined by ICAO (International Civil Aviation Organization), mostly in English, to define navigation, communication, flight control systems, etc.

5.2.1 Reference and Flight Systems

- **AHRS** (Attitude and Heading Reference System).
- **ADI** (Attitude Director Indicator).
- **EFIS** (Electronic Flight Instrument System).
- **HSI** (Horizontal Situation Indicator).
- **INS** (Inertial Navigation System).
- **IRS** (Inertial Reference System).
- **PFD** (Primary Flight Display).

5.2.2 Air Navigation

- **ACAS** (Airborne Collision Avoidance System).
- **ADF** (Automatic Direction Finder).
- **DME** (Distance Measuring Equipment).
- **FMS** (Flight Management System).
- **GBAS** (Ground-Based Augmentation System).
- **GNSS** (Global Navigation Satellite System).
- **GPS** (Global Positioning System).
- **ILS** (Instrument Landing System).
- **NDB** (Non-Directional Beacon).
- **VOR** (VHF Omnidirectional Range).
- **XPDR** (Transponder).

5.2.3 Air Communications

- **ACARS** (Aircraft Communication Addressing and Reporting System).
- **ATC** (Air Traffic Control).
- **ATIS** (Automatic Terminal Information Service).

- **HF** (High Frequency).
- **VHF** (Very High Frequency).

5.2.4 Engine Control and Avionics Systems

- **EICAS** (Engine Indicating and Crew Alerting System).
- **ECAM** (Electronic Centralized Aircraft Monitor).
- **FADEC** (Full Authority Digital Engine Control).



Figure 5-1 ICAO (Source: Wikipedia)

6 FIELD OF APPLICATION

6.1 Case study: King Air 360 Aircraft

The King Air 360 is a high-performance turboprop aircraft (578 km/h) that offers the range of some jets (3,345 km) with the operational flexibility of a smaller propeller aircraft. It is certified for single-pilot operation and can carry up to 11 occupants in its standard version, while the King Air 360ER model allows up to 15 people [10].

It is a wide-body, fixed-wing aircraft that combines the power of turbine engines with the efficiency of propellers, placing it in a unique category between light piston aircraft and private jets.

It uses two Pratt & Whitney Canada PT6A-60A engines driving high-performance four-blade McCauley C780 propellers, as well as Collins Aerospace Pro Line Fusion avionics with three 14-inch touchscreens. It includes autothrottle and a digital pressurization system that reduces passenger fatigue by maintaining a lower cabin altitude.

It is used as a utility and business aircraft. Thanks to its extreme versatility, it can be used for corporate transport, cargo and special missions such as air ambulance.



Figure 6-1 King Air 360 Image (Source: Aircraftexchange)



Figure 6-2 PT6A-60A Powerplant and McCauley C780 Propellers (Source: Pratt & Whitney)



Figure 6-3 Cockpit and Cargo Cabin (Source: Aviapages)

6.2 King Air 360 Final Assembly Line (Wichita, USA)

The King Air 360 final assembly line [11] is a station-based flow production line located at Textron Aviation's facilities in Wichita, Kansas. The line has a capacity of about 30 aircraft per year.

The physical process on the final assembly line, from when the fuselage enters the line until the aircraft leaves the hangar, usually takes between 2 and 4 months. However, from purchase order to final delivery, the lead time is much longer (12 to 24 months) due to the supply chain and the waiting list.

On this line, the aircraft is moved manually or by tow through several positions:

- *Primary Structure Assembly:* the main fuselage sections, wings and tail are joined; they are often manufactured in adjacent areas of the same complex.
- *Basic Assembly and Systems Integration:* at dedicated stations the Pratt & Whitney PT6A-60 engines are installed, including the new IS&S ThrustSense autothrottle system and the Collins Aerospace Pro Line Fusion avionics.
- *Functional Tests:* the previously installed systems are rigged and checked.
- *Interiors and Final Close-up:* since it is a business aircraft, a large part of the final process takes place inside the aircraft, where the seating configuration and finishes chosen by the customer are installed.



Figure 6-4 King Air 360 Assembly Line (Wichita – Kansas) (Source: Arkansasonline)

6.3 Integrated Digital Avionics

To operate, aircraft have a set of electronic systems [14-18], that make up what is called avionics (from "aviation" and "electronics"), including indication, communications, air navigation, flight management systems, etc.…

Traditionally, all these systems were installed independently, and the pilot operated them individually from cockpit controls. Traditional avionics involved an overwhelming workload, with non-integrated information and increasing workloads, leaving little margin for error during critical phases of flight.

Today's pilots face a high-pressure environment where they have to make split-second decisions, with complex airspace and changing weather conditions that can significantly impact flight safety and efficiency.

Modern, integrated digital avionics, such as that fitted to the King Air 360 (Collins Pro Line Fusion), address these challenges by offering a flight deck designed to empower pilots with clarity, efficiency and unmatched situational awareness [19-20].

Some of the advantages of this type of avionics would be:

- *Operational Simplification:* the touch interface replaces the complexity of traditional controls with intuitive, smartphone-like screens that reduce pilot workload and make access to critical information faster and easier.
- *Enhanced Situational Awareness:* synthetic vision systems (SVS) see through weather and terrain with a real-time 3D view, helping pilots maintain situational awareness even in complex environments.
- *Smarter Flight Planning and Navigation:* performance-based navigation (PBN) helps pilots navigate more efficiently with advanced capabilities such as Required Navigation Performance (RNP) and area navigation (RNAV), giving them access to preferred airspace and providing more fuel savings. Integrated flight planning capabilities let pilots spend less time on tedious pre-flight tasks with simplified tools that guide them through the processes, step by step.
- *Future Connectivity:* The updated design keeps pilots ahead of changes in regulations and technology trends with a modular system built for easy upgrades and improvements.

The Collins Pro Line Fusion system in the King Air 360 [19-20], features three large 14-inch LCD touchscreens integrating flight, navigation, weather and engine displays.

It uses a high-speed open architecture to connect sensors, autopilot systems and radios, enabling intuitive, aware and automated flight management.

The main components are:

- *Touchscreens (MFD/PFD):* Three interconnected displays that act as the main contextual interface, showing graphics, maps and engine data.
- *Flight Guidance Computers (FGC):* The "brain" that manages flight guidance and autopilot modes.
- *Attitude and Heading Reference System (AHRS):* Provides accurate data on the aircraft's attitude.
- *Air Data Computer (ADC):* Measures pressure and temperature to determine altitude and speed.
- *Flight Management System (FMS):* Manages flight plans and navigation.
- *Sensors and Radios (CNS):* Communication, Navigation and Surveillance equipment.



Figure 6-5 Cockpit Avionics "Collins Proline Fusion" (Source: Kingairnation)

Communications between these components take place through:

- *High-Speed Network:* The components communicate over a robust, high-speed data bus network (typically ARINC 429 and Ethernet), enabling real-time information integration.
- *Touch Interface:* allows the pilot to interact directly with the displays to control the autopilot and radios, eliminating scattered physical controllers.

- *Unified Display:* the system fuses data from multiple sources (GPS, weather radar, traffic) and presents it on the main displays to increase situational awareness.

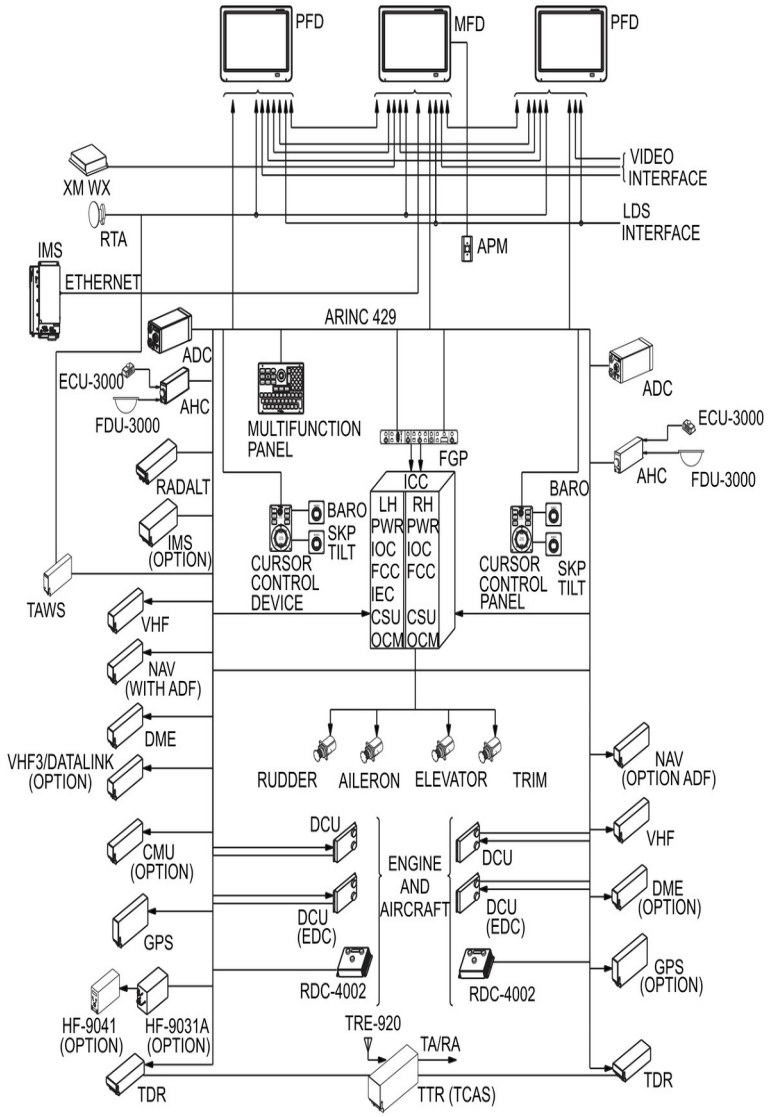


Figure 6-6 Avionics Block Diagram (Source: Proline Fusion for King Air AMM)

6.4 Air Navigation Systems

Due to the extent and number of systems that make up this avionics suite, we will focus only on those related to air navigation [21-22].

These systems allow aircraft to determine their position, plan their trajectory and move safely through airspace.

They are essential for coordinating flights on national and international routes, avoiding collisions, optimizing fuel consumption and reducing flight times.

They range from traditional visual methods to complex systems based on radio frequency and satellites, as well as the control and support services provided by aviation authorities.

In this project we will study the main navigation systems [23-27] and the tests carried out on them on the King Air 360 Final Assembly Line:

ATA	TITULO
34-10	PRUEBA FUNCIONAL DEL SISTEMA ADS (Air Data System)
34-20	PRUEBA FUNCIONAL DEL SISTEMA AHRS (Attitude Heading Reference System)
34-40	PRUEBA FUNCIONAL DEL SISTEMA WRS (Weather Radar System)
34-42	PRUEBA FUNCIONAL DE SISTEMA RALT (Radio Altimeter System)
34-43	PRUEBA FUNCIONAL DEL SISTEMA TCAS (Traffic Alert and Collision Avoidance System)
34-52	PRUEBA FUNCIONAL DEL SISTEMA GNSS (Global Navigation Satellite System)
34-53	PRUEBA FUNCIONAL DEL SISTEMA VHF NAV (VOR/ILS/MKR/ADF Navigation System)
34-54	PRUEBA FUNCIONAL DEL SISTEMA DME (Distance Measuring Equipment)
34-55	PRUEBA FUNCIONAL DEL SISTEMA XPDR (Diversity Transponder)

Figure 6-7 King Air 360 Air Navigation Systems

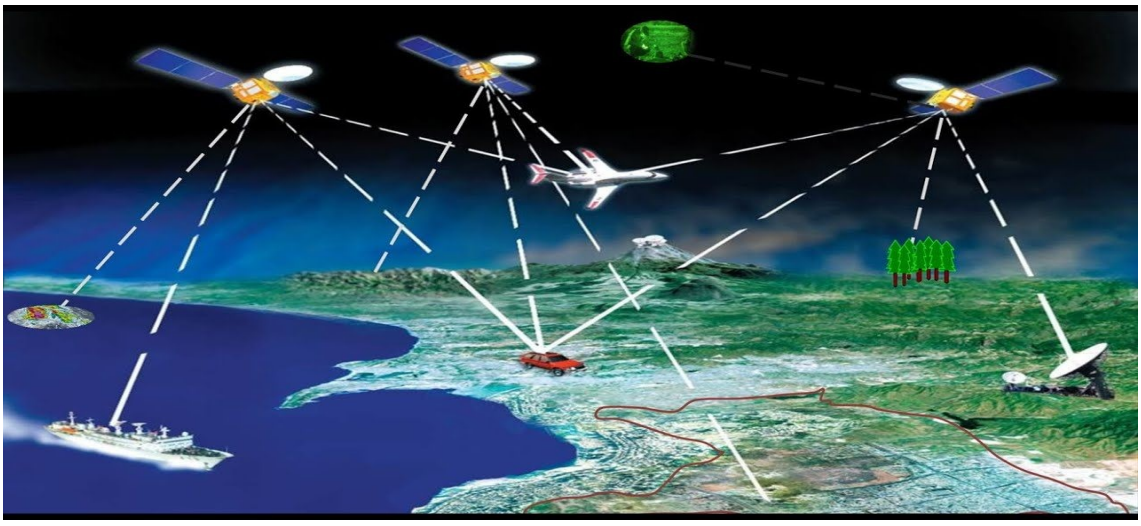


Figure 6-8 Air Navigation Aid Systems (Source: Rewiresecurity)

6.4.1 Air Data System (ADS)

The air data system consists of aerodynamic and thermodynamic sensors that measure the characteristics of the air surrounding the aircraft and allow the Air Data Computers (ADC) to calculate flight parameters such as airspeed (true and indicated), vertical speed, altitude (total, static and barometric), outside air temperature (total and static) and angle of attack [28-32].

The King Air aircraft has two ADC computers.

Each of them connects to the Pitot/static air inlets and to a temperature sensor, processes the air data inputs and provides the air data parameters to each

Primary Flight Display (PFD) and to the avionics core data input/output concentrators (IAPS).

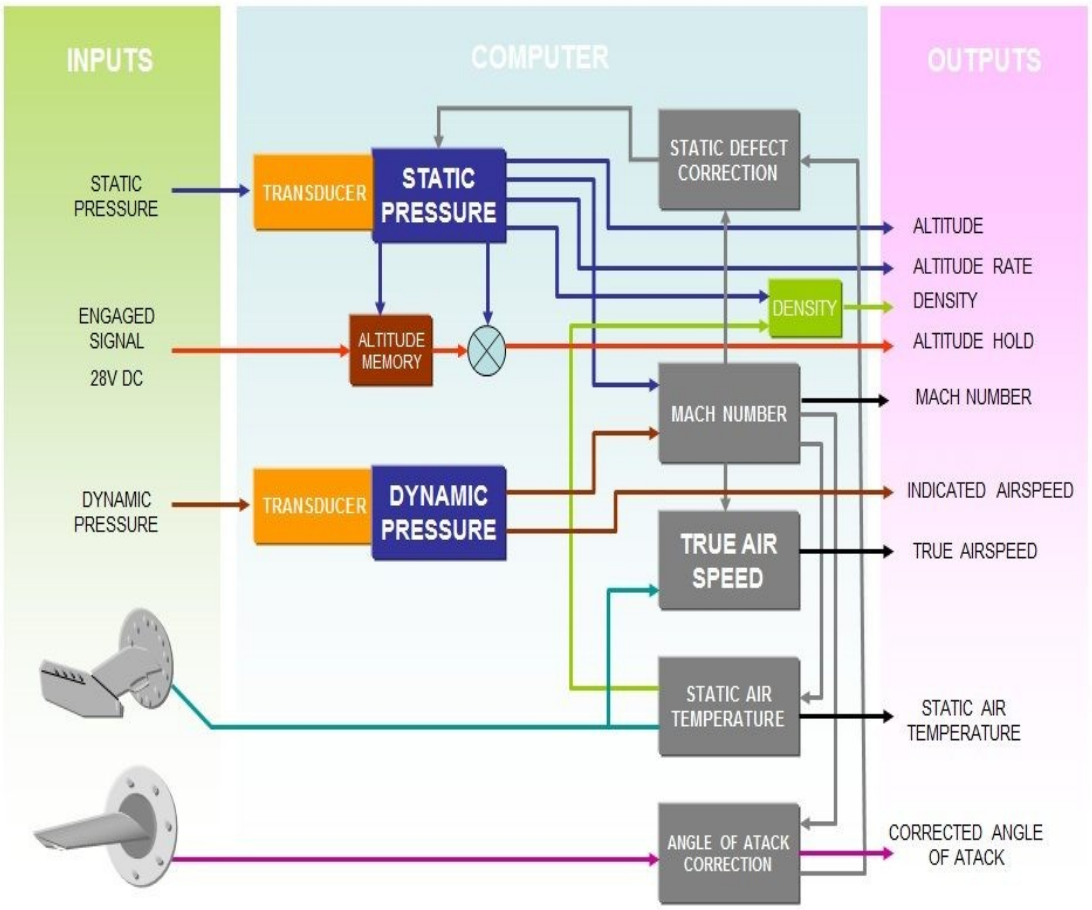


Figure 6-9 ADS Operation (Source: Instagram)



Figure 6-10 Dual Pitot Tube System & Temperature Sensor (Source: Spool-up)

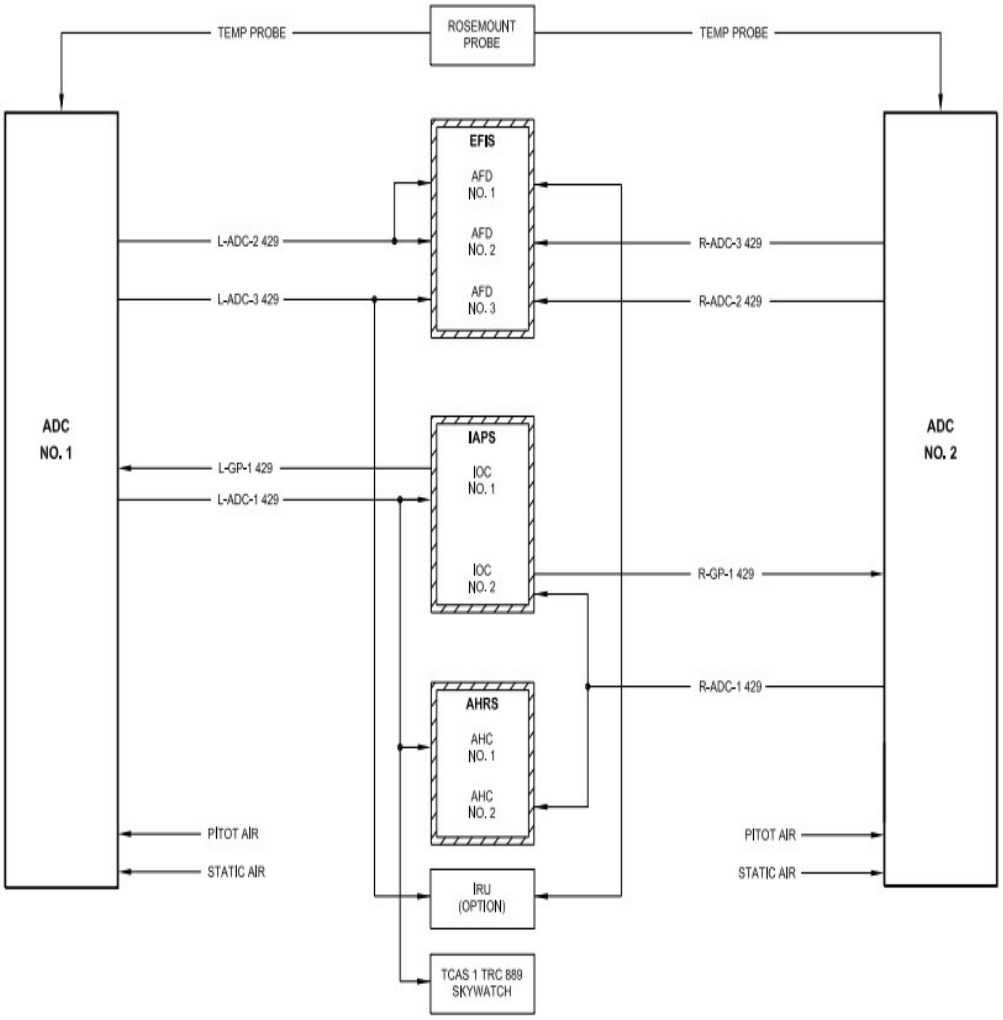


Figure 6-11 ADS Block Diagram (Source: Proline Fusion for King Air AMM)

Test Systems – ADS

To check the air data system in the FAL, flight speeds and altitudes must be simulated on the aircraft. An Air Data Test Set is used for this.

This equipment basically consists of a digital controller and a vacuum pump which, connected through hoses and caps to the Pitot tubes, makes it possible to replicate air pressure and speed without moving the aircraft off the ground.

It also makes it possible to detect air leaks in the lines connecting the Pitot tubes to the air data computers.



Figure 6-12 Air Data Test Set (Source: AvionTeq)



Figure 6-13 Cap Kit and Aircraft Connection (Source AvionTeq)

6.4.2 Attitude and Heading Reference System (AHRS)

The attitude and heading reference system (AHRS) is an integrated system that provides three-dimensional orientation data, including pitch, roll and yaw angles, as well as magnetic heading information [28-32].

These data are vital for pilots to maintain control and situational awareness.

In the Pro Line Fusion avionics, both the pilot and copilot sides contain an Attitude Heading Computer (AHC), a Flux Detector Unit (FDU) and an External Compensation Unit (ECU).

The AHC provides pitch, roll and magnetic heading data to the Electronic Flight Instrument System (EFIS) displays and to the Flight Guidance System (FGS). It

also generates rate and acceleration outputs on all three axes for use by the FGS. A battery provides backup power to keep the AHC running during power drops.

To calculate magnetic heading, the AHC has a magnetic sensor (FDU) that detects the horizontal component of the Earth's magnetic field.

It also has an external compensation unit (ECU) that stores corrections for interference with the Earth's magnetic field caused by the aircraft structure and for FDU misalignment.

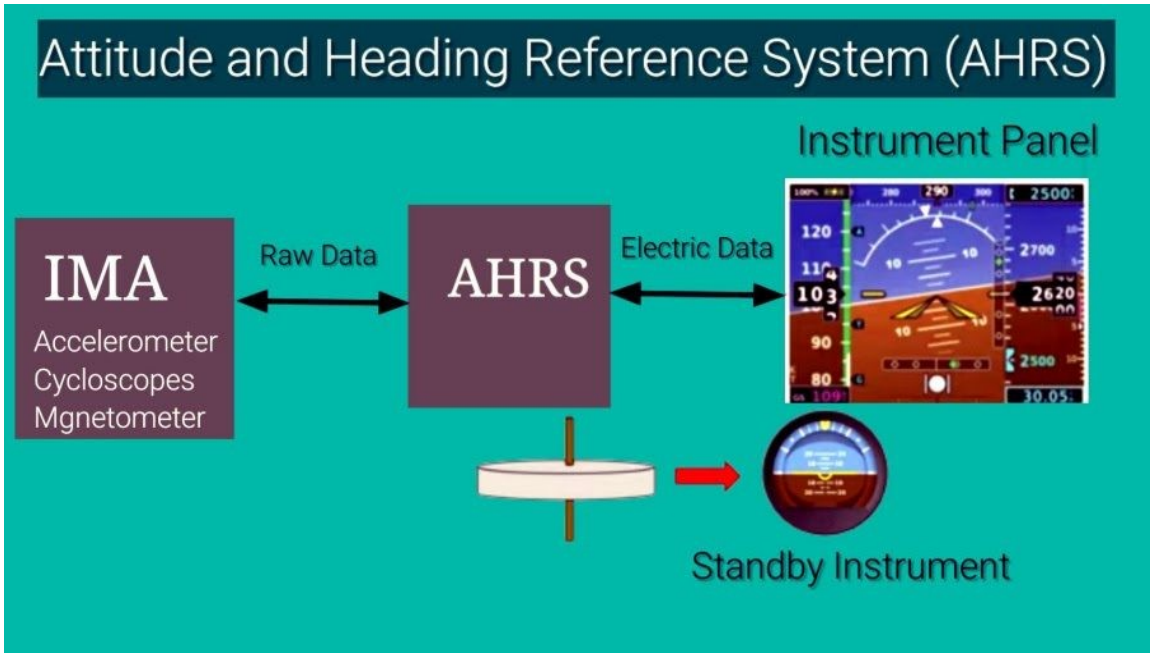


Figure 6-14 AHRS Operation (Source: Politicalfunda)



Figure 6-15 AHRS Controls (Source: Kingairnation)

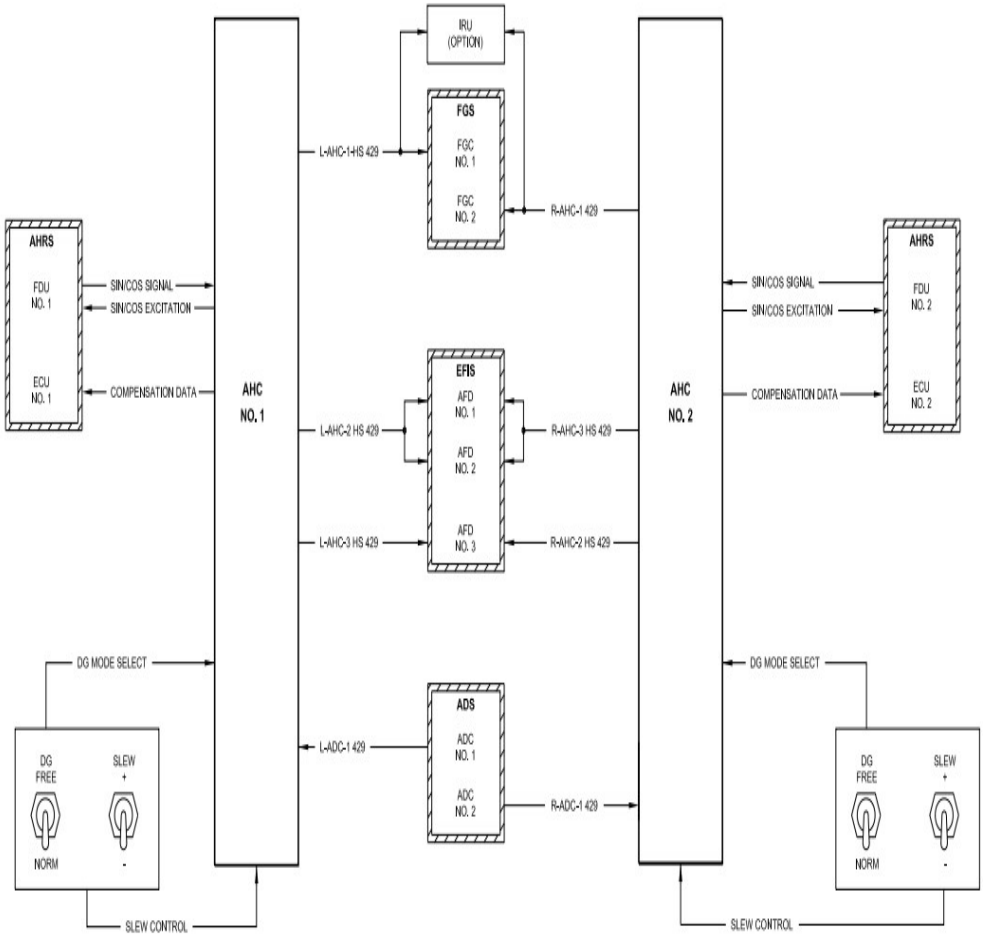


Figure 6-16 AHRS Block Diagram (Source: Proline Fusion for King Air AMM)

Test Systems – AHRS

The main objective is to check system operation and calibrate the attitude and heading alignment after the equipment is installed.

This requires various tools such as the leveling and compensation box, theodolite, inclinometer, laptop with software, connection cables and jacks.

The procedure includes leveling the aircraft, attitude alignment and compensation according to the functional test described in the Collins manual.

Finally, the system is restarted to ensure the new compensation values are permanently saved in the AHRS non-volatile memory.



Figure 6-17 AHRS System Test Equipment (Source DirectIndustry)



Figure 6-18 AHRS Test System Software (Source DirectIndustry)

6.4.3 Weather Radar System (WRS)

The onboard weather radar of an aircraft (Weather Radar System, WRS) is an electronic system located in the aircraft nose that emits microwaves and detects precipitation, storms, hail and turbulence along the flight path [28-32].

This system enables pilots to visualize hazardous weather conditions on a cockpit display using color codes (green to red) to avoid them and ensure safety.

The WRS consists of an X-band receiver/transmitter antenna (RTA) located in the radome that detects wet precipitation, moisture-based turbulence and ground returns ahead of the aircraft.

The information is shown as a stabilized three-color image (green, yellow and red) on the cockpit displays. The three colors indicate increasing rain intensities. The optional Doppler-based turbulence detection capability is shown in magenta. Each output reflects the state of the side controls for display range, mode, tilt and gain.

The single-knob tilt control panel allows the user to adjust the fixed RTA attitude by ±15 degrees.

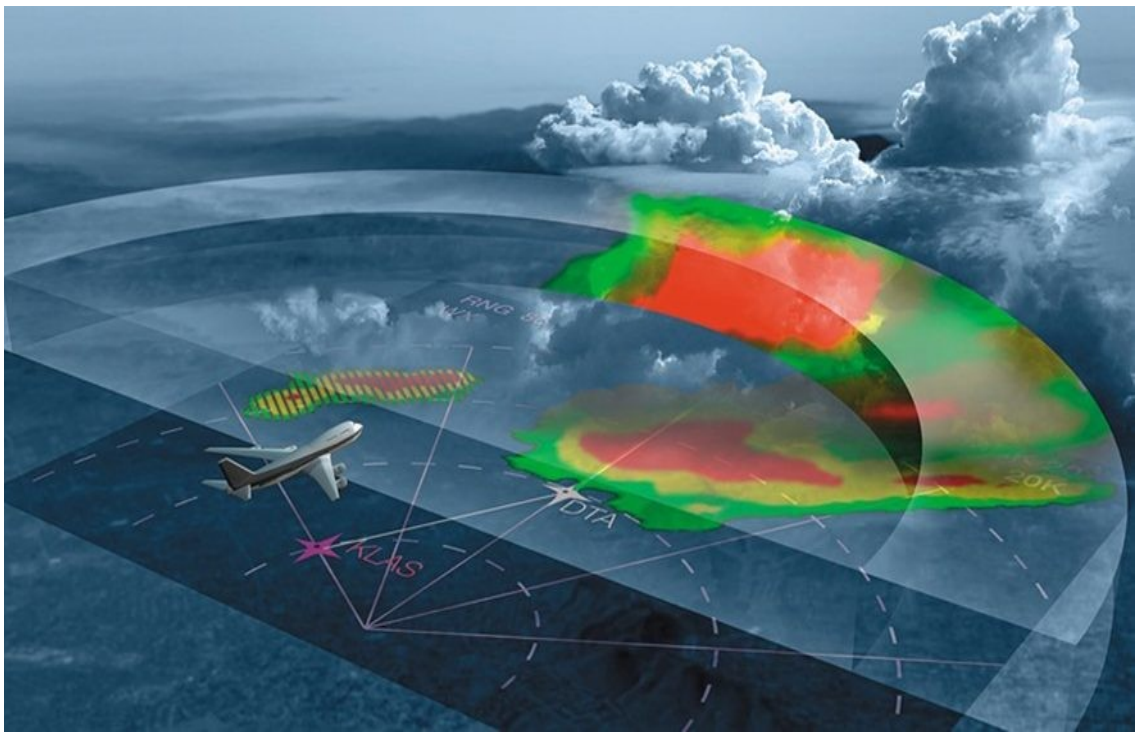


Figure 6-19 Weather Radar Operation (Source: Honeywellaerospace)

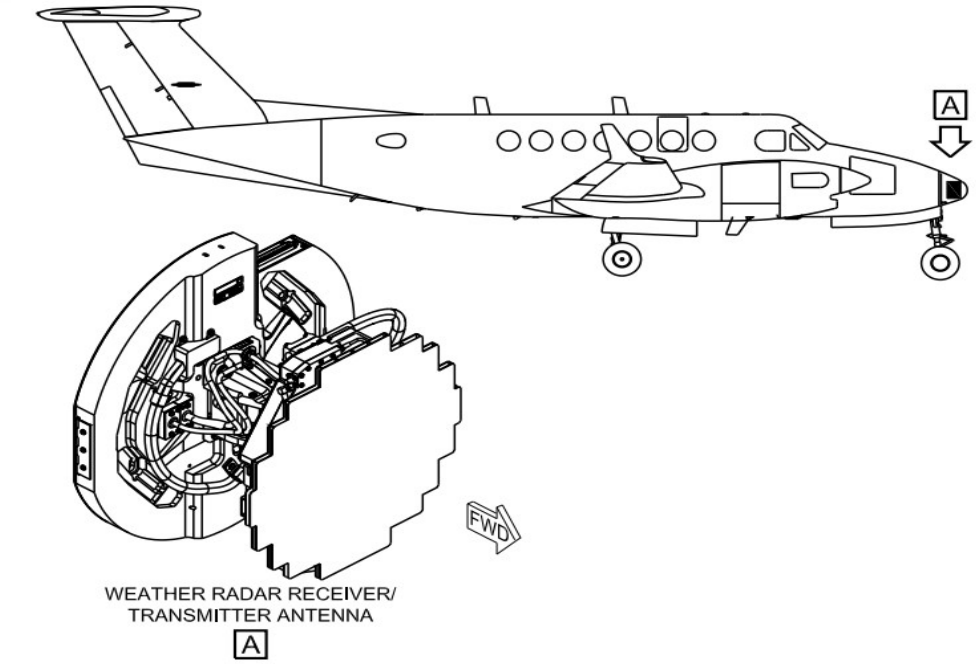


Figure 6-20 Weather Radar System (Source: Proline Fusion for King Air AMM)

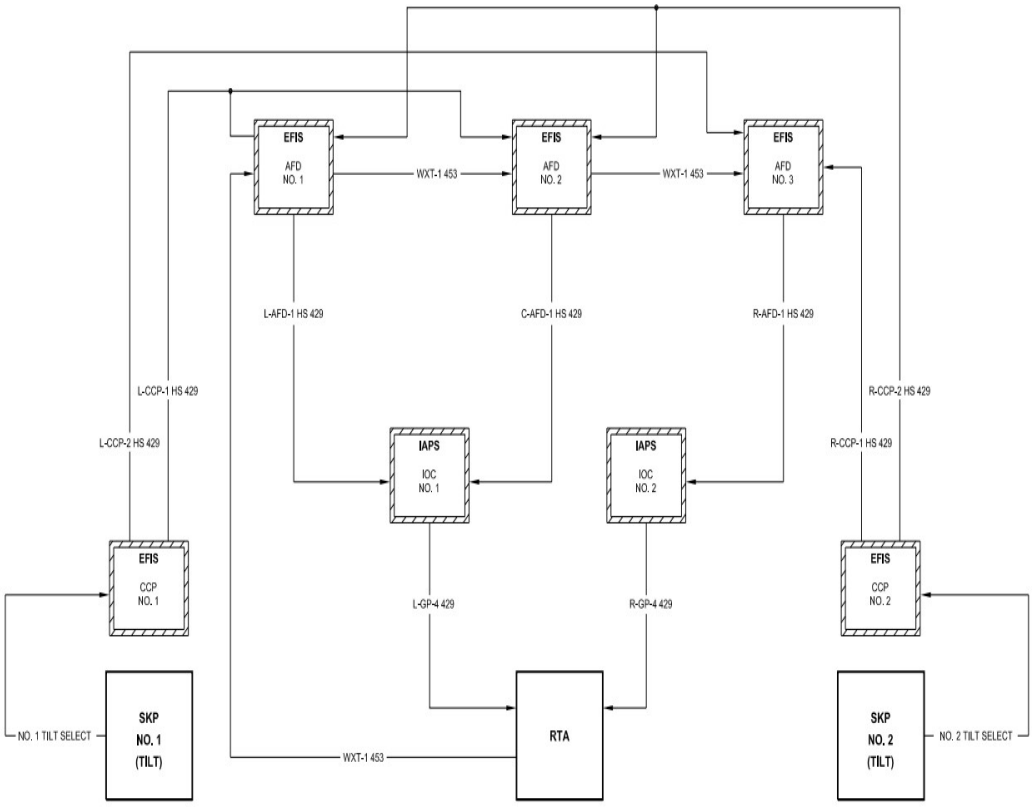


Figure 6-21 WRS Block Diagram (Source: Proline Fusion for King Air AMM)

Test Systems – WRS

The ground test of the weather radar is carried out outside the hangar according to the manual procedure using the cockpit controls, making sure the antenna points at clear areas to avoid radiation damage to nearby personnel or equipment (at least 30-50 meters clear).

It could also be carried out inside the hangar provided a radar simulator is available which, placed over the radome, prevents the radiation from affecting other people.

The test verifies the operation of the antenna, transmitter and receiver by displaying test patterns on the navigation displays, simulating precipitation echoes.



Figure 6-22 Digital Inclinometer (Source: Amazon)

6.4.4 Radio Altimeter System (RALT)

The radio altimeter (RALT) is an avionics instrument that measures the exact, absolute distance between the aircraft and the terrain directly below it [28-32].

Unlike the barometric altimeter, which measures atmospheric pressure relative to sea level, the radio altimeter provides the true height above the ground.

It works by emitting radio waves and measuring the time they take to bounce off the surface (max. 2,500 ft height), and is essential for automatic landings, low-altitude flight and safety systems.

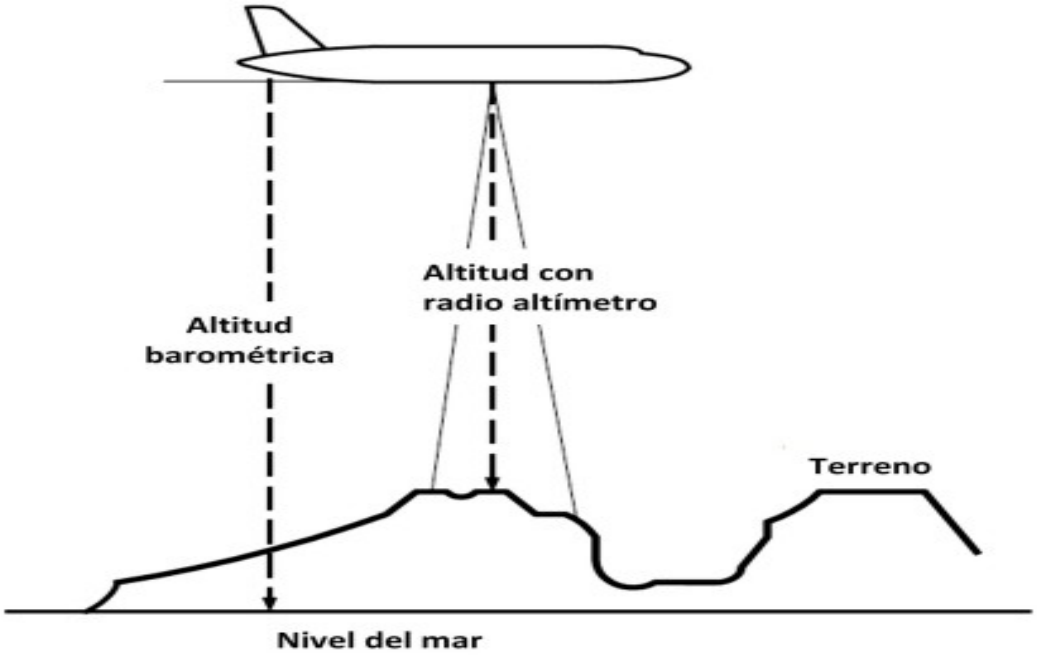


Figure 6-23 Radio Altimeter Operation (Source: UNAM)

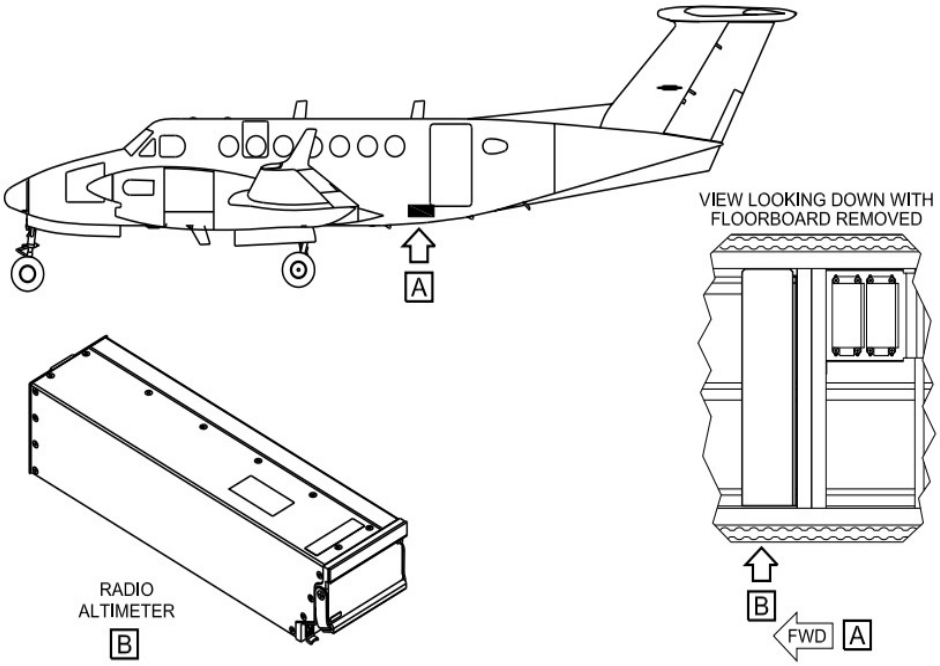


Figure 6-24 Radio Altimeter System (Source: Proline Fusion for King Air AMM)

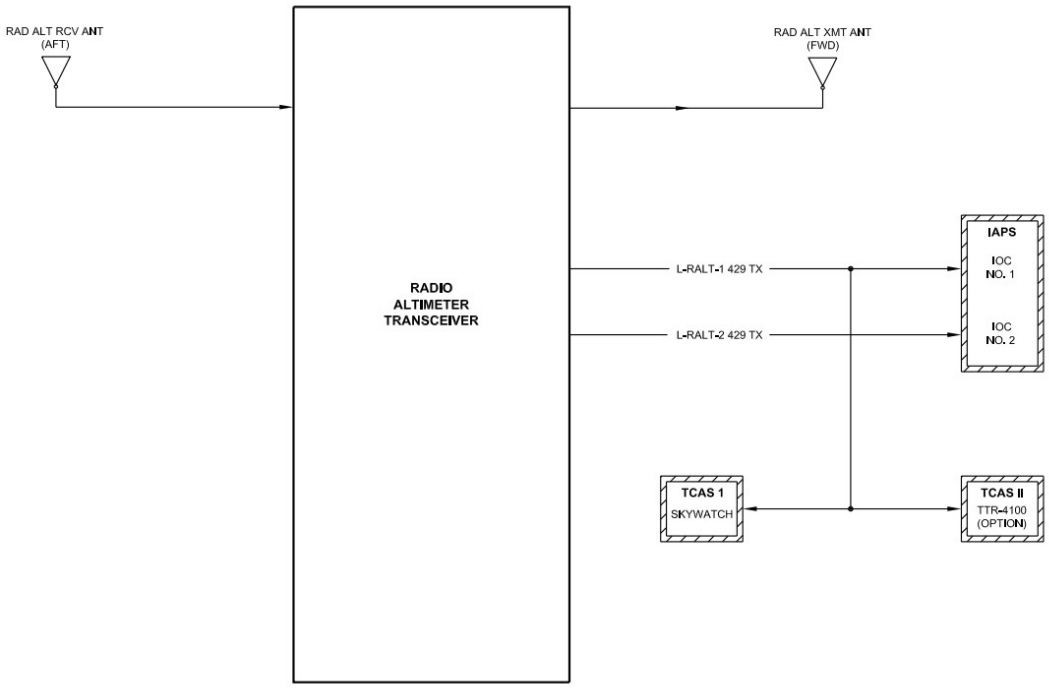


Figure 6-25 Radio Altimeter Block Diagram (Source: Proline Fusion for King Air AMM)

Test Systems – RALT

The system is tested simply using the cockpit controls and following the procedure established in the manuals or in the corresponding functional test operation.

It does not require specific equipment, although there are more accurate test methods that involve disconnecting the transmit antennas and fitting radio frequency delay lines such as the one shown in the figure below.

These delay lines simulate different heights in the range up to 2,500 ft. In this case the test is partial, since the antennas have been removed to connect the test equipment.

There is also equipment that allows altitude to be entered digitally through the buses (ARINC 429), although in this case neither the RALT equipment nor the antennas are checked.



Figure 6-26 Radio Altitude Simulator Delay Unit (Source: Neoncq)



Figure 6-27 ARINC Source (Source: Aviotools)

6.4.5 Traffic Alert and Collision Avoidance System (TCAS)

The TCAS (Traffic Alert and Collision Avoidance System) is an onboard aircraft safety system that autonomously prevents mid-air collisions, without relying on ground radars or controllers [28-32].

It detects other nearby aircraft through their transponders (Modes A, C, S) and issues alerts (TA) or maneuver instructions (RA) to the pilots if there is a risk of collision with another aircraft also equipped with TCAS.

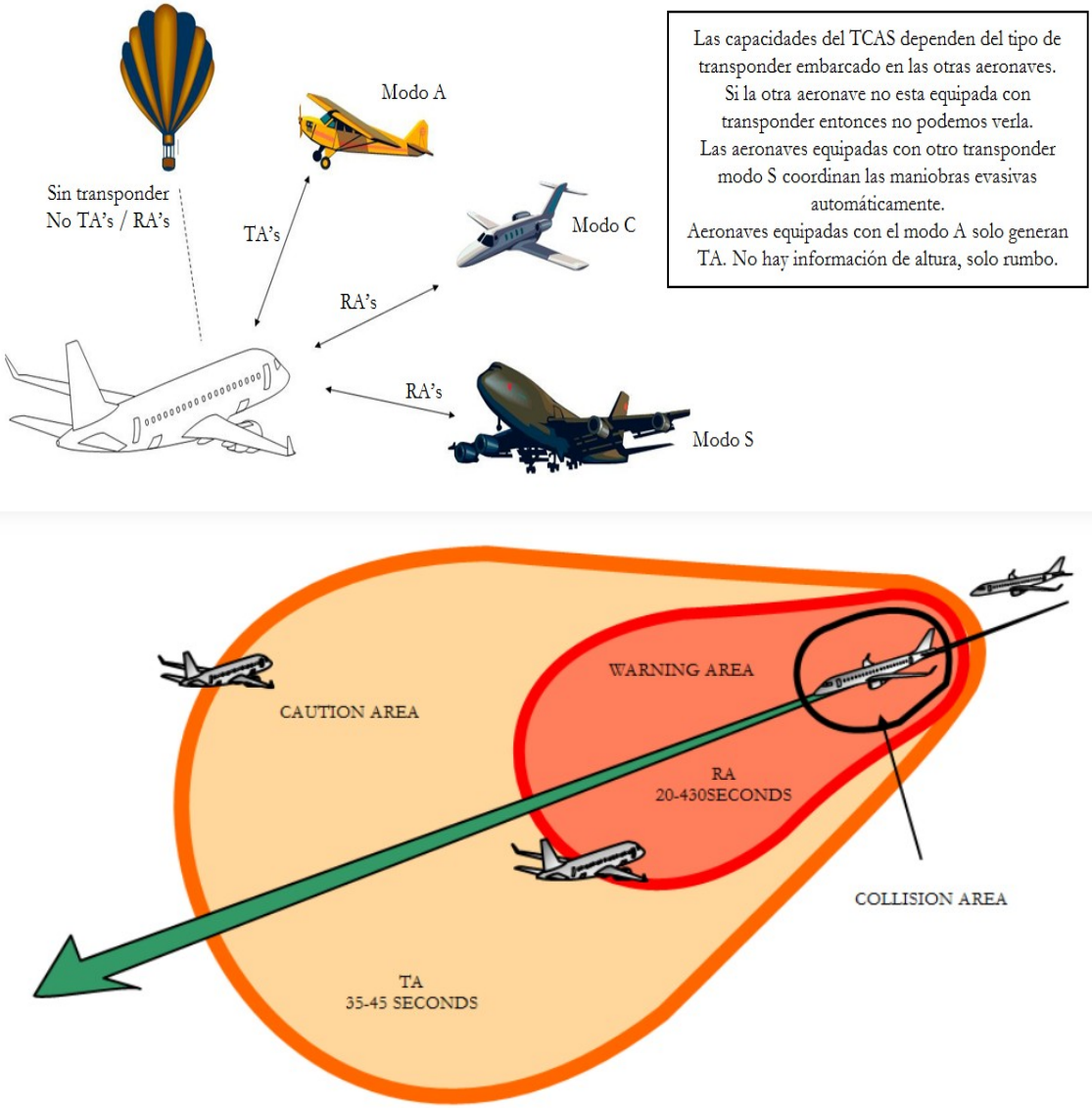


Figure 6-28 TCAS System Operation (Source: Instagram)

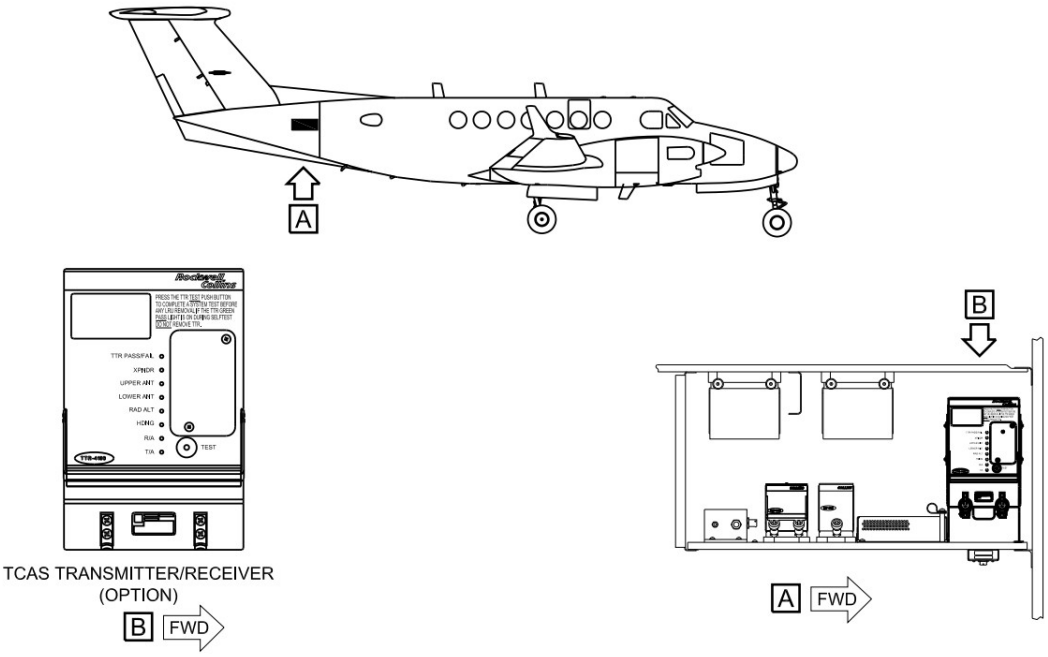


Figure 6-29 TCAS System (Source: Proline Fusion for King Air AMM)

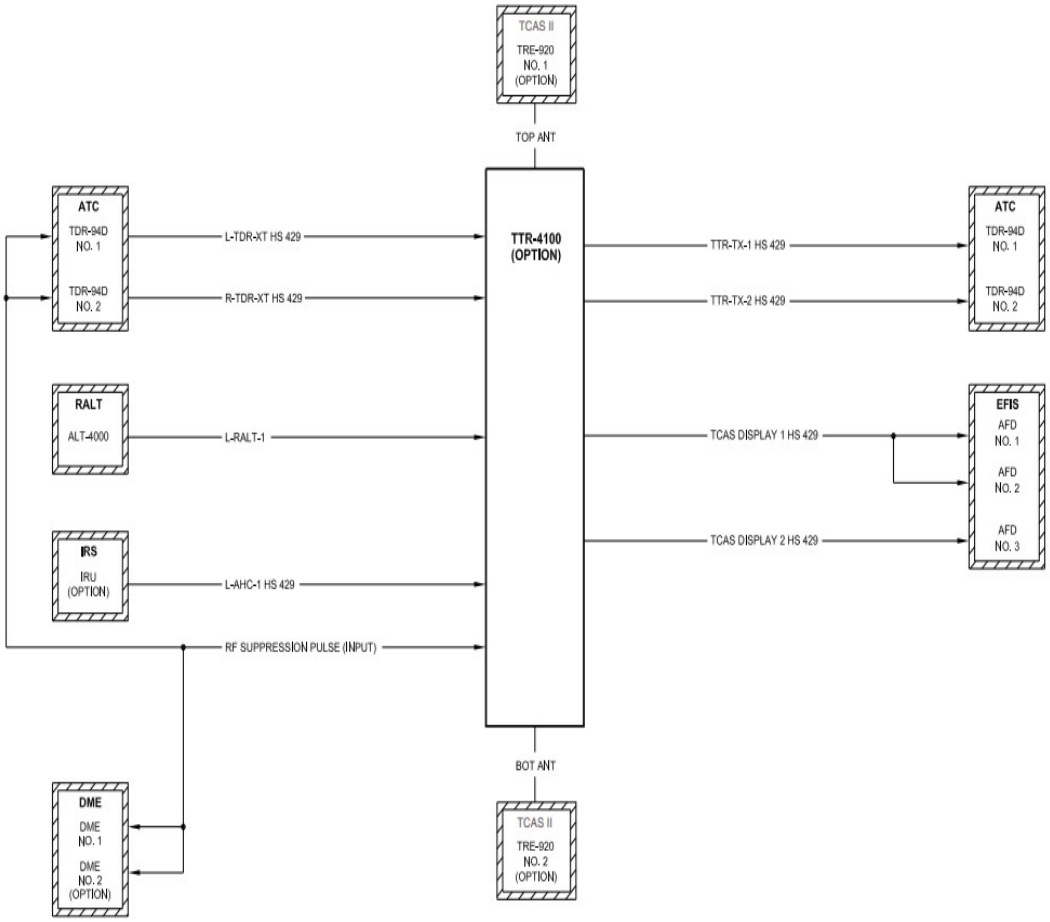


Figure 4-26. TCAS Block Diagram (Source: Proline Fusion for King Air AMM)

Test Systems – TCAS

The functional test of the TCAS (Traffic and Collision Avoidance System) is mainly performed on the ground through a "Self-Test" of the avionics system.

For complete functional tests, the figure below shows ramp test equipment such as the IFR-6000, which simulates intruder traffic to verify the 35-40 NM range and alert generation.

A weight-on-wheels simulator is also required to create ground/air conditions, although jacks could also be used to lift the aircraft.

It should be borne in mind that RA alerts are inhibited at low altitude (below 1,000 ft or 300 meters above the ground) to avoid false alarms during takeoffs and landings.



Figure 6-30 TCAS Test Equipment (Source: Aeroflex)

6.4.6 Global Navigation Satellite System (GNSS)

The aircraft satellite navigation system, known as GNSS (Global Navigation Satellite System, which includes GPS (American), Galileo (European) and GLONASS (Russian)), works through a network of satellite constellations that emit radio frequency signals [28-32].

A receiver on board the aircraft receives these signals from at least 4 satellites simultaneously to calculate the exact position, altitude and speed in real time.

GPS is complemented by a ground-based signal system called the Wide Area Augmentation System (WAAS), which improves and guarantees the integrity of GPS signals, improving signal accuracy, availability and vertical precision.

WAAS works as follows:

- A network of ground reference stations receives data from GPS satellites.

- These data are processed to determine integrity and differential corrections.

- The information is transmitted to International Maritime Satellite Organization (INMARSAT) satellites and rebroadcast on the GPS frequency.

The King Air has a GPS-4000S that can receive 12 channels although, as mentioned above, at least 4 satellites are required for a 3D position solution. The data obtained are supplied to the ATC, EFIS, IAPS and TAWS subsystems.

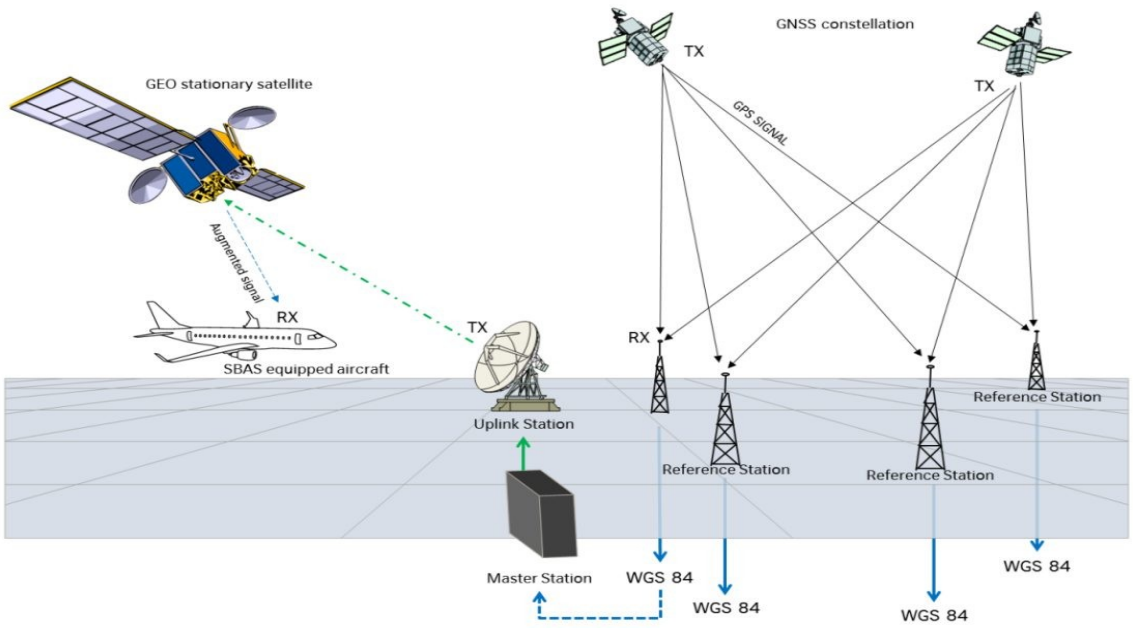


Figure 6-31 GNSS System Operation (Source: Adenilsongiovanini)

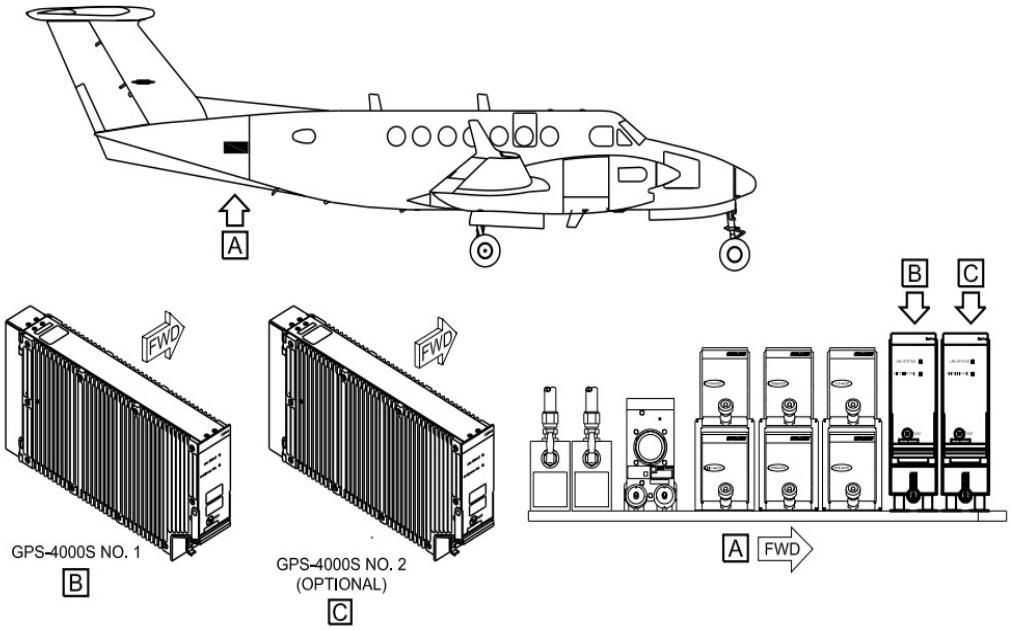


Figure 6-32 GNSS System (Source: Proline Fusion for King Air AMM)

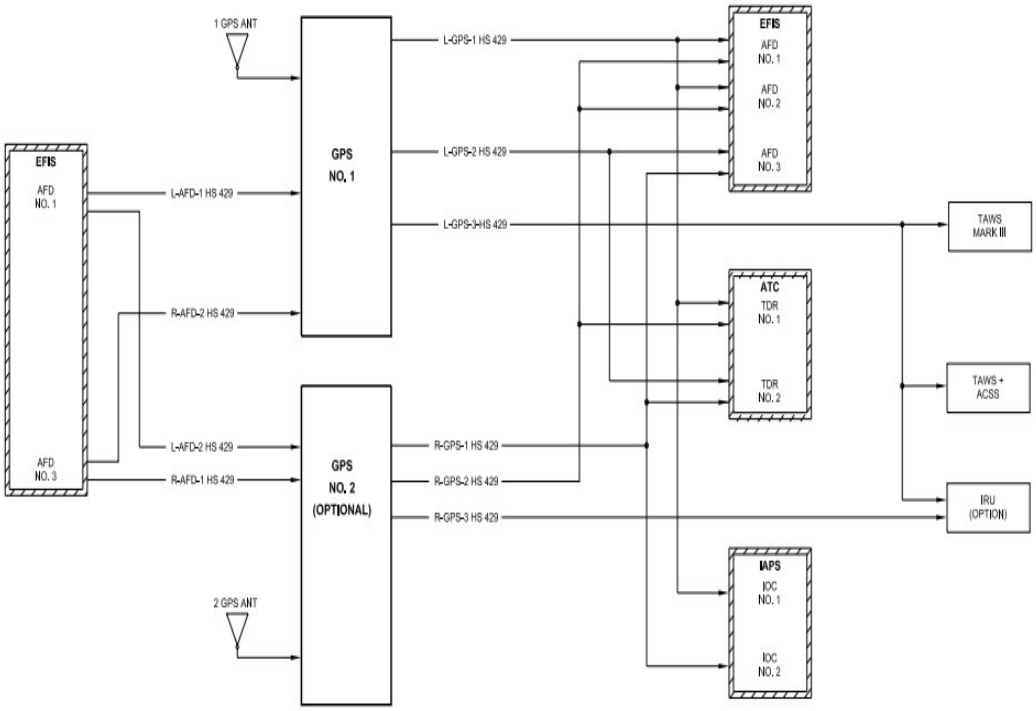


Figure 4-30. GNSS Block Diagram (Source: Proline Fusion for King Air AMM)

Test Systems – GPS

To test an aircraft's GPS inside a hangar (where the structure blocks natural satellite signals), two main technical methods are used that avoid having to tow the aircraft outside:

- *GPS Repeaters:* this is the most common method. It consists of installing an external antenna on the hangar roof that picks up the real satellite signals. This signal is sent by coaxial cable to an indoor unit that radiates it inside the hangar, allowing the aircraft's navigation systems to obtain a real position as if they were outside.
- *GNSS Simulators or Signal Generators:* instead of using real signals, these devices generate artificial radio frequency (RF) signals identical to those of the satellites. They allow technicians to program specific coordinates, altitudes and motion scenarios to verify the accuracy of the Flight Management System (FMS) under controlled conditions.

Shielded "hoods" are placed directly over the aircraft's GPS antennas to inject a test signal in isolation, avoiding interference with other systems or the emission of unauthorized GPS signals inside the hangar.



Figure 6-33 GNSS Hangar Test Equipment (Source: Pendulum)

6.4.7 VHF Radio Navigation System (RNAV VOR/ILS/MKR/ADF)

The radio navigation system (also called radio aids) is the set of radio signals emitted by ground stations or satellites that

allow an aircraft to determine its position, heading and distance without the need for external visual references [28-32].

The systems are divided according to the information they provide:

- VOR (Omnidirectional Range): the most widespread radio aid. It emits signals in 360 directions (radials), allowing the pilot to know which "path" or radial they are on relative to the station

  on the ground.

- *ILS (Instrument Landing System):* Provides vertical (glide slope) and horizontal (runway centerline) guidance for safe landings in low-visibility conditions.

- *DME (Distance Measuring Equipment):* Measures the slant distance between the aircraft and the ground station using the time a signal takes to travel there and back. It is used together with the VOR.

- ADF (Non-Directional Beacon): The oldest and simplest system. A ground antenna emits a signal that the aircraft's ADF equipment uses to point a needle directly at the station.

The aircraft has electronic systems that use the transmissions from the ground systems mentioned above to determine relative bearings, approaches...



Figure 6-34 Radio Navigation System Operation (Source: Wikimedia)

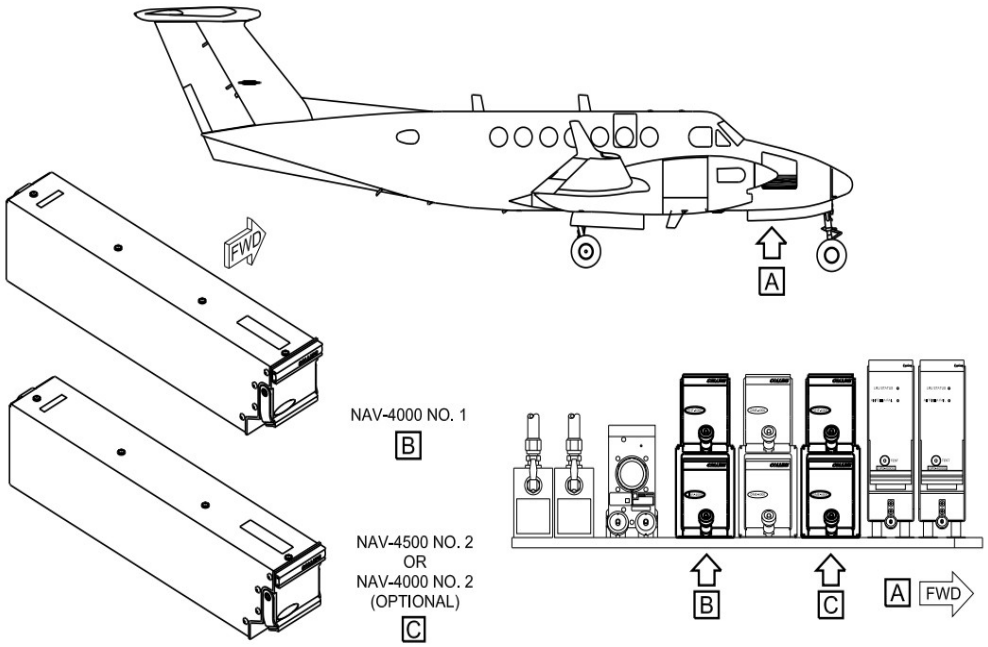


Figure 6-35 Radio Navigation System (Source: Proline Fusion for King Air AMM)

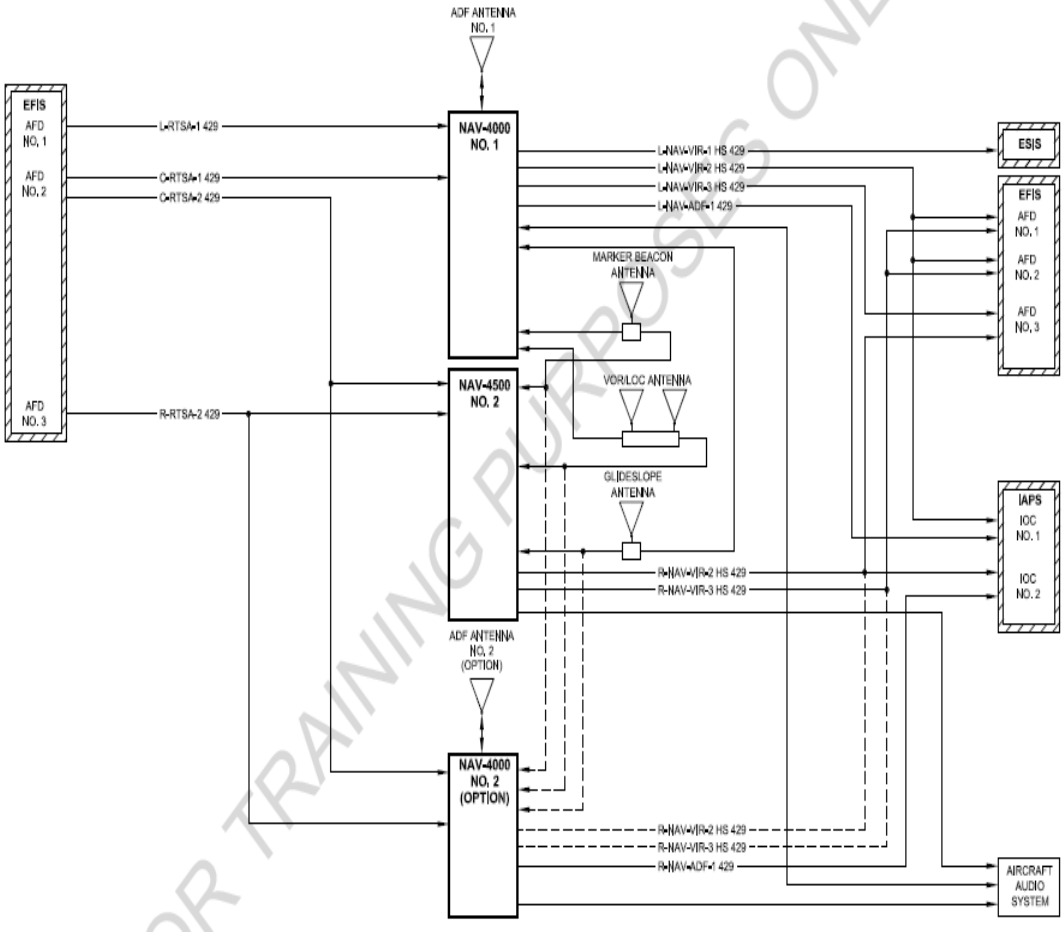


Figure 6-36 VHF Radio Navigation System Block Diagram (Source: Proline King Air AMM)

Test Systems – Radio Navigation

To test the radio navigation system, we will simulate inside the hangar the different signals coming from the ground stations. For this we will use equipment such as the IFR-4000 or the TIC VOR/ILS T30-B.



Figure 6-2 Radio Navigation Test Equipment (Source: Aircraft & Ground Ltd)

6.4.8 DME System (DME)

The DME (Distance Measuring Equipment) is a radio-based air navigation system that measures the time a signal takes to travel from the aircraft to a ground station and back, thus calculating the slant range in real time between the aircraft and the station [28-32].

It operates in the UHF band (960-1215 MHz) and is crucial for navigation, instrument approaches (ILS) and safety, informing the pilot of the exact distance to a point.

It is frequently paired with VOR radio aids or ILS landing systems, being selected automatically when the radio is tuned.

The system works using the "question-answer" (interrogation) technique, similar to radar.

The aircraft sends pulses, the ground station receives them and replies after a fixed delay, allowing the onboard equipment to calculate the distance from the round-trip time.

It should be noted that the system measures the straight line between the aircraft and the antenna. Therefore, if the aircraft is directly above the station at high altitude, the DME will not indicate zero but the height.

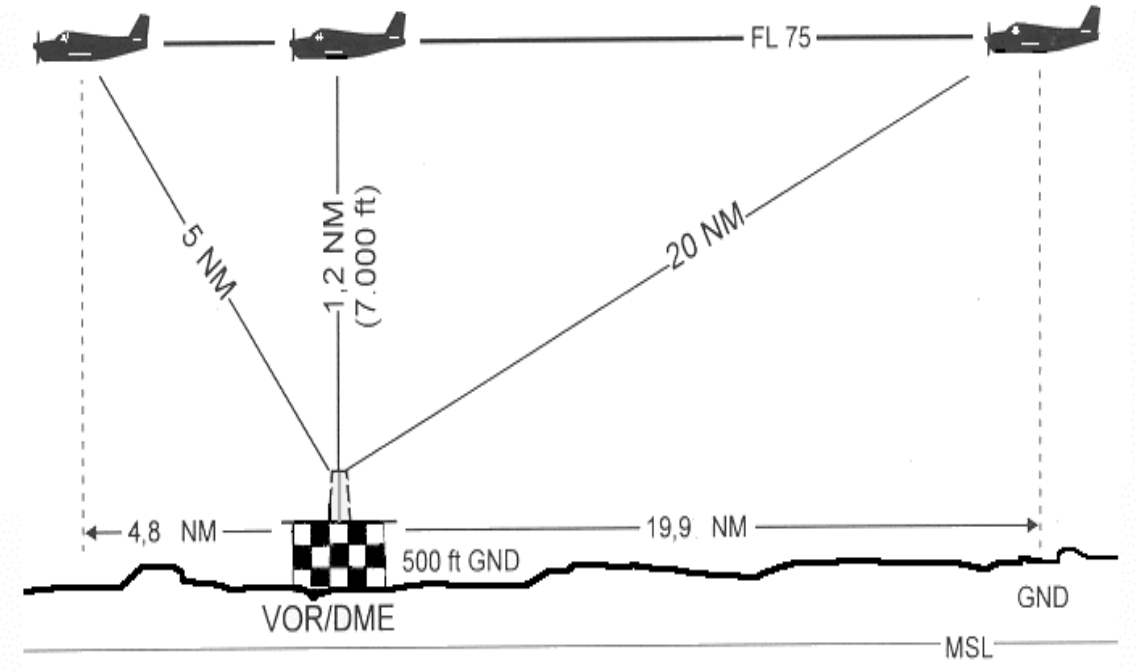


Figure 6-3 DME Operation (Source: Wikimedia)

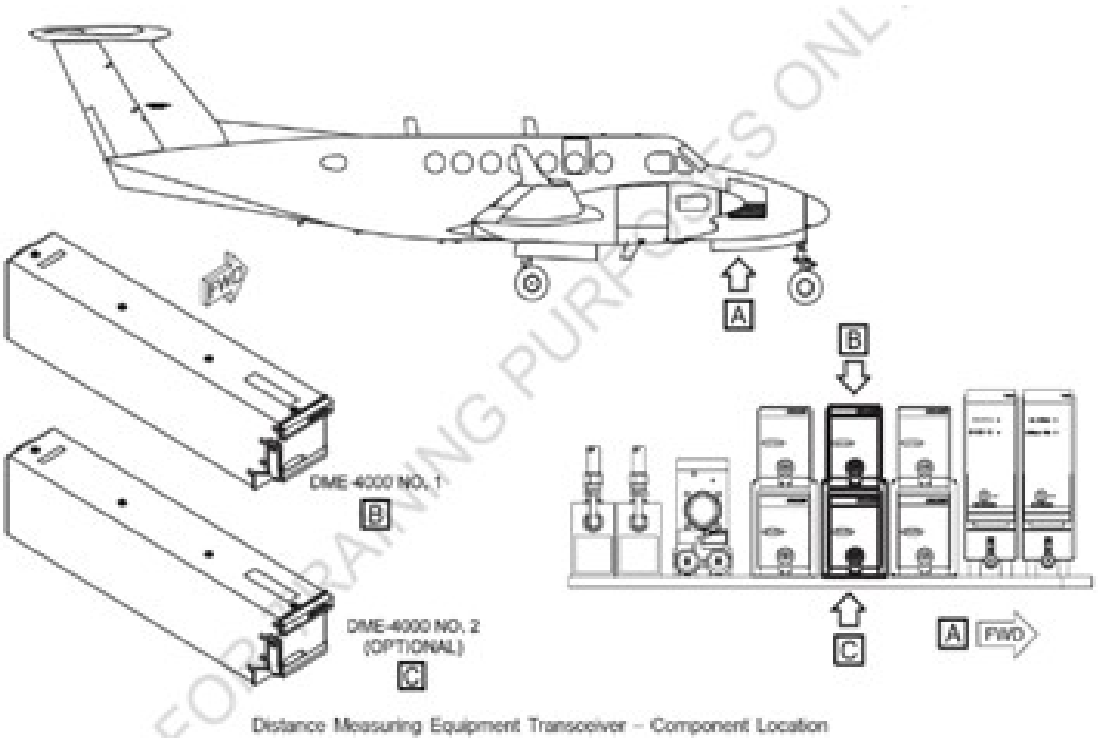


Figure 6-39 DME System (Source: Proline Fusion for King Air AMM)

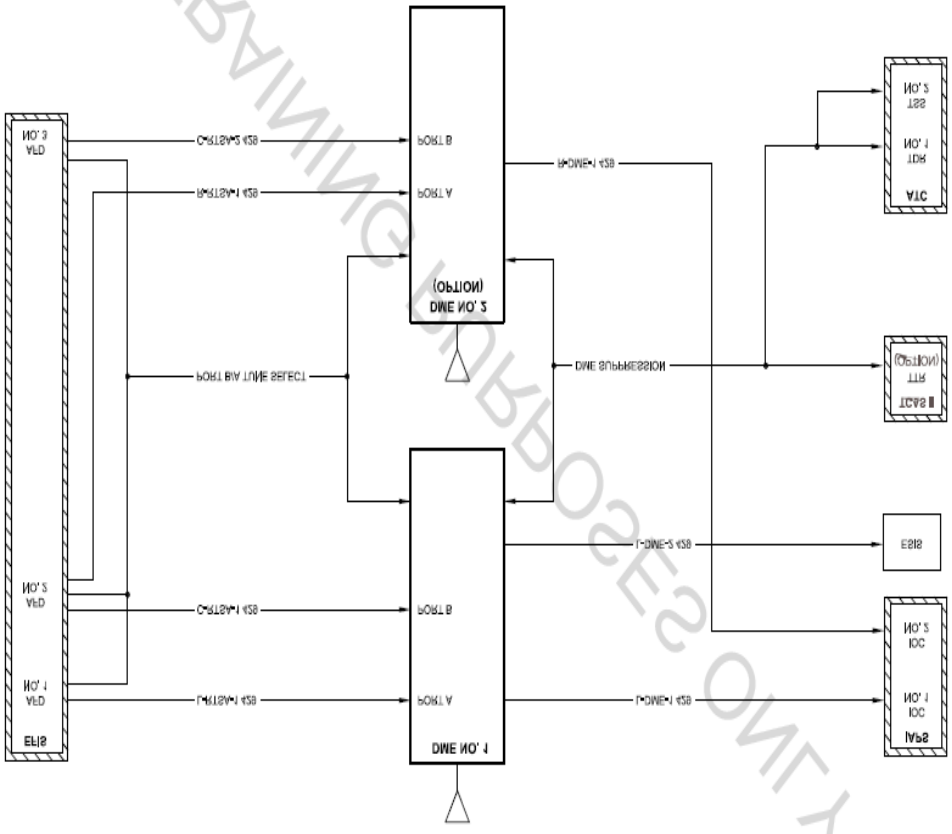


Figure 6-40 DME Block Diagram (Source: Proline Fusion for King Air AMM)

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Test Systems – DME

The aircraft DME (Distance Measuring Equipment) is tested mainly by two methods: the use of specialized ground test equipment (ramp tests) and operational checks in flight or at specific checkpoints.

To verify the accuracy and operation of the system in the hangar, signal simulators are used that "trick" the aircraft into believing it is near a ground station. A very common piece of equipment is the IFR-6000, although there are different models from other companies.







Figure 6-41 DME Test Equipment (Source: Laversab Inc.)

6.4.9 Transponder System (XPDR)

The XPDR (short for Transponder) is the electronic device on board an aircraft responsible for automatically "replying" when interrogated by air traffic control secondary surveillance radar (SSR) or by other aircraft [28-32].

As shown in the figure below, its main function is to turn the aircraft into a visible, digital entity for the controller, allowing its identification and precise tracking.

The King Air aircraft has a TDR-94D transponder whose main functions are:

- *Identification (Squawk):* the pilot enters a 4-digit code (from 0 to 7) assigned by the controller.
- *Data Transmission:* depending on the mode, it sends the aircraft's pressure altitude, speed and identity.
- *Collision Avoidance:* it is the basis for systems such as TCAS (Traffic Collision Avoidance System), allowing aircraft to "see" each other and avoid collisions.

The operating modes are:

- *Mode A:* sends only the 4-digit identification code.
- *Mode C:* adds information on the aircraft's pressure altitude.
- *Mode S (Selective):* the modern standard; allows more complex data exchange and a unique address for each aircraft, avoiding interference in congested areas.

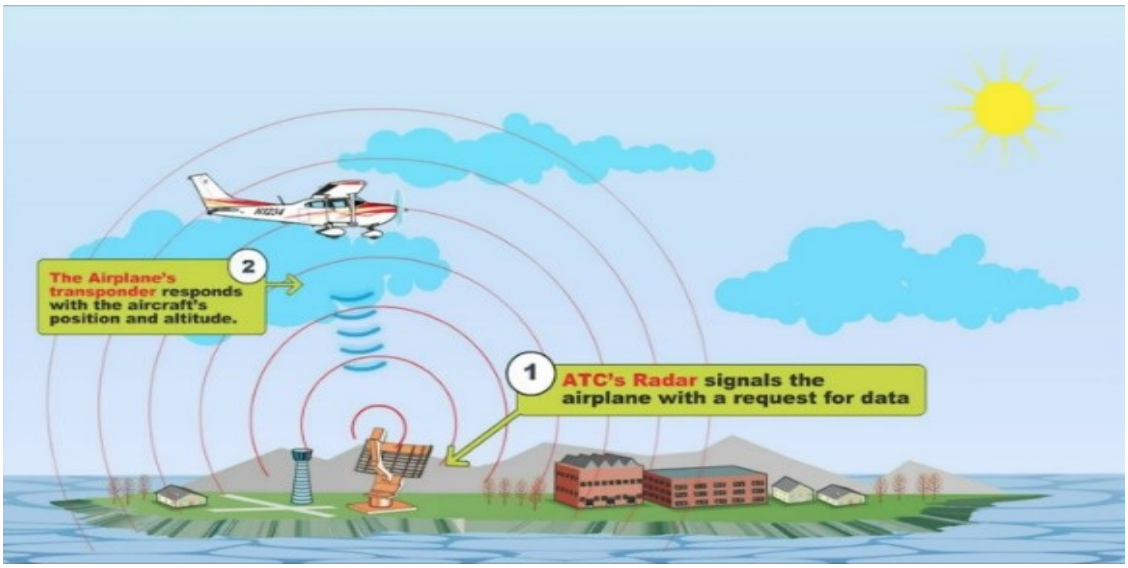


Figure 6-42 XPDR System Operation (Source: Youtube)

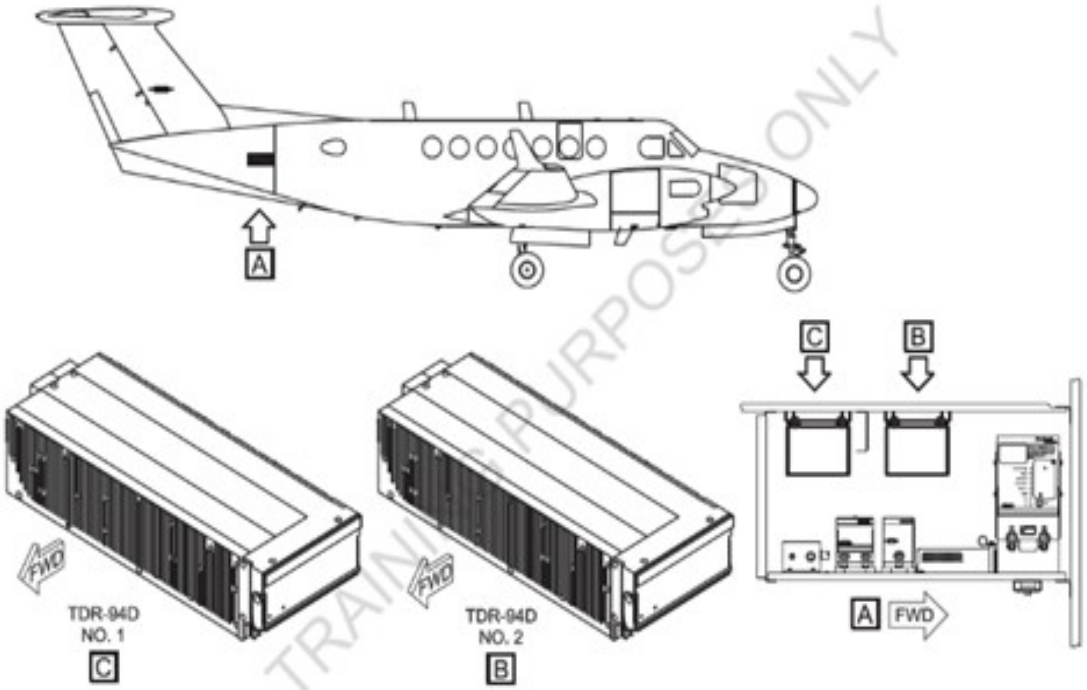


Figure 6-43 XPDR System

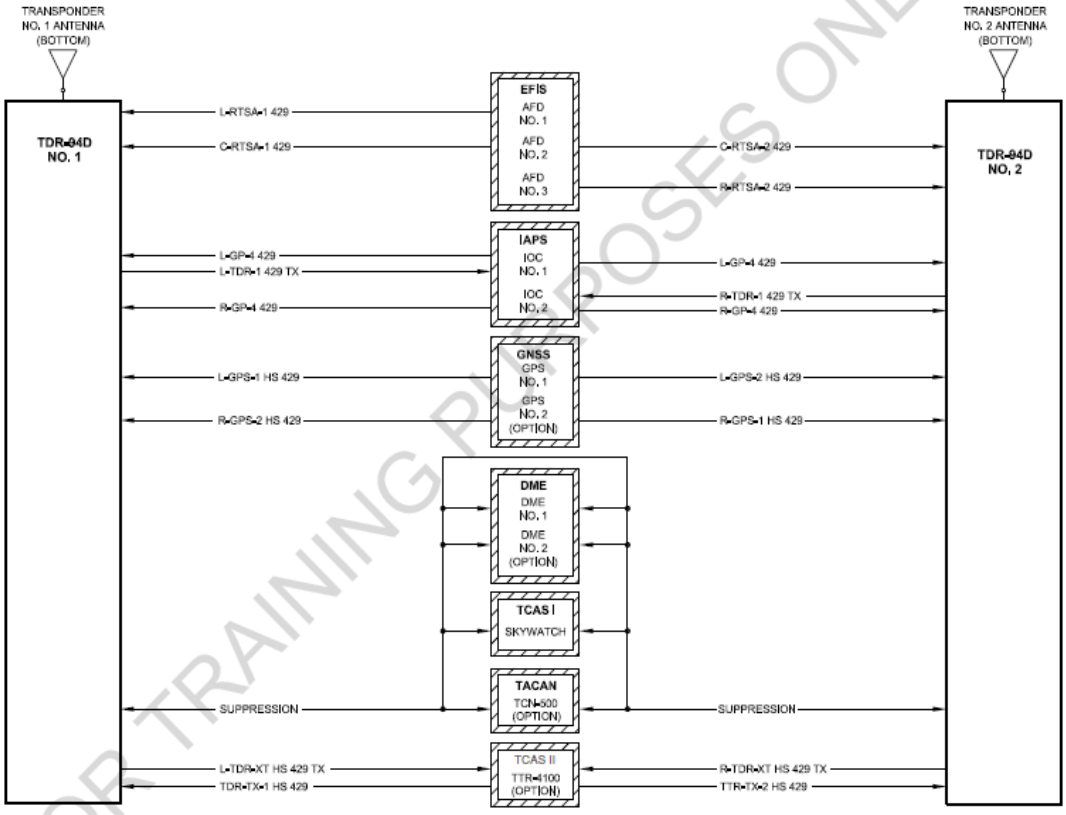


Figure 6-44 XPDR Block Diagram (Source: Proline Fusion for King Air AMM)

Test Systems – XPDR

The aircraft XPDR (Transponder) test is an essential technical procedure to verify that the aircraft correctly communicates its identity, altitude and position to air traffic controllers (ATC) and to other aircraft equipped with collision avoidance systems (TCAS).

The system test is performed by simulating interrogations using test equipment such as the IFR-6000, an air data test set and a weight-on-wheels simulator.



Figure 6-45 XPDR Test Equipment (Source: AvionTq)

6.5 Air Navigation Test Execution

6.5.1 Overview

The King Air 360 functional tests, also called “Ground Tests”, follow a logical sequence based on how the aircraft systems are integrated [33].

The test sequence would be:

- *Electrical Power-Up (“Electrical power on”):* electrical power (batteries or external power) is supplied to the aircraft to verify that current is distributed throughout the electrical distribution system and is available to all electrical equipment (avionics, hydraulics, engines...)
- *Fluid Tests:* the hydraulic (landing gear, brakes…), fuel and pneumatic (air conditioning and pressurization) circuits are checked to detect leaks and ensure correct operation.
- *Avionics Tests (“Avionics power on”):* the Rockwell Collins Pro Line Fusion avionics is switched on and the navigation, communication and flight control software is loaded and tested.
- *Engine Tests*: carried out outside the hangar to verify thrust and temperature and vibration parameters at different power settings.



Figure 6-46 Technician Performing Avionics Tests (King Air 360) (Source: Aviation Int. News)

6.5.2 Work Team

The required team is multidisciplinary and highly specialized.

In the case of avionics tests, the following roles are established:

- *Systems Engineers:* responsible for establishing the test sequence so that all the functionalities the system was designed for are verified. They analyze the data obtained during the tests and help with troubleshooting.

There is one engineer for the avionics tests.

- *Test Technicians:* operators or aircraft maintenance technicians trained in two specialties, electromechanics and avionics. They carry out the different steps set out in the work order (step-by-step test sequence that also indicates the safety measures and the test equipment to be used).

For avionics tests, there is usually one pair per shift (morning and afternoon), made up of an expert who leads the test and an apprentice who prepares the aircraft, the test equipment, etc. and follows the senior operator's instructions.

- *Quality Specialists:* they supervise that each step strictly complies with certification regulations.

Normally, in functional tests this quality process is delegated to the technicians who perform them, since their high level of training guarantees that operations are carried out in accordance with regulations.



Figure 6-47 Avionics Specialists Testing Systems (Source: Asiaeducationreview)

6.5.3 Planning

Planning an avionics test system is a rigorous technical process that guarantees flight safety by verifying all subsystems and associated components.

For the Air Navigation Tests, King Air sets a period of 5 days with 2 operators on each 8-hour shift (2+2), with a final review on the last shift of the last day.

It takes approximately 160 hours in total, considering that a small number of incidents (3 or 4) may arise from system failures (wiring, aircraft equipment or test equipment) and waiting times due to interference with other test teams (hydraulics, landing gear…) or other incidents (tooling not available, replacement materials, waiting for equipment to be brought...).

Occasionally, faults that are difficult to locate may occur, so the total time can extend by one or two more days, and the aircraft must be removed from the line to a specific area (“buffer”) so that its problem does not affect the rest of the aircraft.

The total cost depends heavily on the hourly rate set by the factory, but it can be between $100/h and $150/h depending on whether the operator is junior or senior.

The work is normally done in pairs including an apprentice operator and an experienced one.

In total, the testing process can reach $20,000 per aircraft.

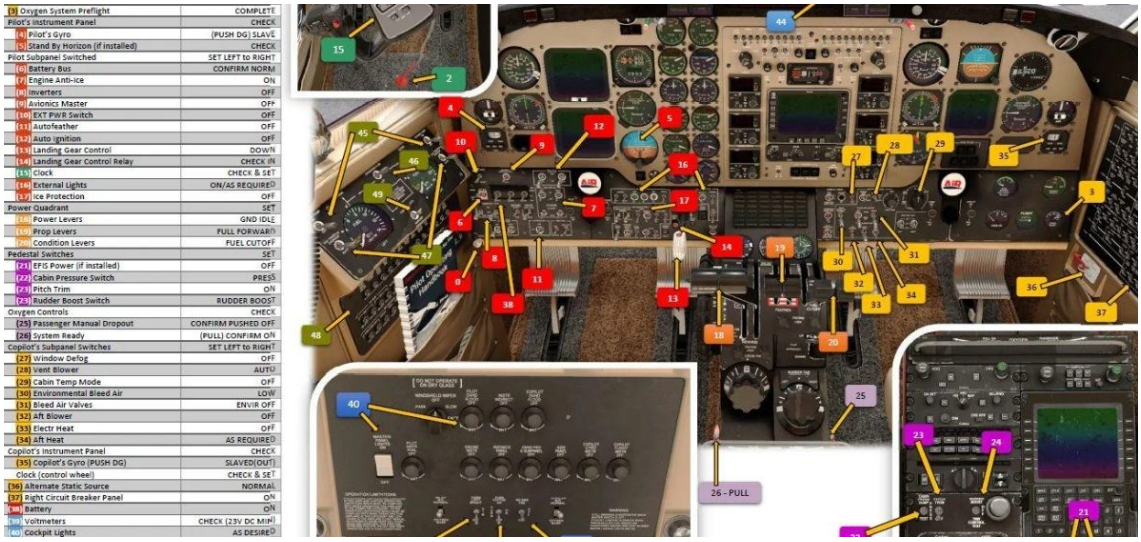


Figure 6-48 King Air Systems Checklists (Source: X-Plane)

Process

Going into detail, the process requires the aircraft to be powered (the “power on” has been carried out) and normally starts with the tests of the air data (ADS) and attitude (AHRS) systems, since they are needed to simulate altitudes and speeds (essential for the rest of the avionics tests). The system/day/operators/shift, hours and cost plan is as follows:

System	Day 1	Day 2	Day 3	Day 4	Day 5	Hours (h)	Cost ($)

ADS	2					8+8	800+1200
AHRS	2					8+8	800+1200
WRS		2				8+8	800+1200
RALT		2				8+8	800+1200
TCAS			2			8+8	800+1200
GNSS			2			8+8	800+1200
VHF N				2		8+8	800+1200
DME				2		8+8	800+1200
XPDR					2	8+8	800+1200
Final					2	8+8	800+1200
Total	**2+2**	**2+2**	**2+2**	**2+2**	**2+2**	**80+80**	**20.000**

The cost takes into account that in each pair the salary is different for the apprentice ($100/hour) and for the experienced technician ($150/hour).

Apart from any problems in the aircraft systems, the plan may be altered by variations in the specialization of the workforce. For example, if on one or more shifts the senior technician is not available and is replaced by a less qualified one (who does not know the aircraft systems or how the test equipment works), the duration increases and even more intensive help from the systems engineer may be needed during test execution.



Figure 6-49 Technicians Troubleshooting a Fault on a King Air 360 (Source Gamaaviation

7 DESIGN REQUIREMENTS

7.1 Introduction

Design requirements are the technical, functional and aesthetic specifications essential for a product, system or service to meet user needs and project objectives.

They range from functional to ergonomic aspects, costs, safety, etc.

In our case, the design is based on Generative Artificial Intelligence (GenAI), a powerful tool that can transform functional testing on the aircraft final assembly line (FAL) by acting as an intelligent "copilot" for both technicians and engineers.

Unlike traditional AI, which only detects patterns and requires extensive knowledge of data analytics, GenAI can help execute tests, facilitate troubleshooting, assist in the use of complex test equipment or ensure the safety of the people working at the station.

As the range of possible project objectives is very broad, we will begin by studying the most relevant problems of the testing process.

From there, possible solutions will be determined, and these in turn will define the design requirements.

7.2 Avionics Functional Testing Issues

Functional tests on an aircraft final assembly line are a series of rigorous technical ground checks performed on each system of the assembled aircraft to ensure that they work according to their design and safety standards before its first flight and subsequent delivery to the customer.

They are essential to ensure airworthiness before the first flight.

Problems detected during these tests are usually due to the complexity of system integration, assembly/rigging defects, faulty equipment and human error.

In avionics tests in particular, problems usually stem from defects caused by improperly seated connectors/pins, incorrectly connected wiring, badly installed antennas or electromagnetic interference (EMI).

Likewise, incorrectly installed or configured equipment (software versions.) causes errors during execution.

These problems often generate discrepancies that delay aircraft delivery and increase production costs, making additional tests (“re-tests”) necessary to ensure system operation.

7.2.1 Defects in Electrical Installations

Aircraft wiring, also called EWIS (Electrical Wiring Interconnection System), is a system that connects all the aircraft's electronic equipment, sensors and antennas.

It includes the electrical conductors (wires grouped into harnesses), connectors, supports and clamps, and all the elements that ensure power and signals reach where they should, under the right conditions and throughout the aircraft's operating life.

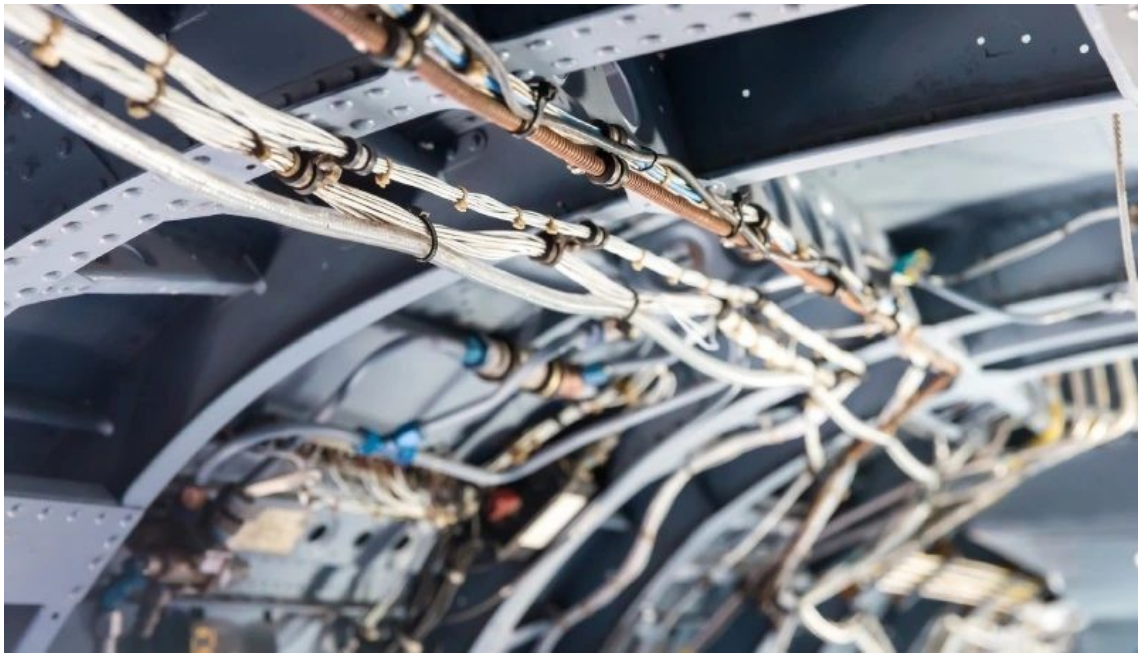


Figure 7-1 Installation of Electrical Harnesses in an Aircraft (Source: Siemens)

From a technical point of view, EWIS operates subject to vibration, thermal cycles, humidity, contaminants, mechanical stress and, in addition, ever-increasing electrical demands. Degradation is rarely immediate. Usually, latent faults appear that are difficult to detect and remain hidden for years until several conditions coincide and the system stops behaving as it should.

Depending on the behavior of the fault, they can be:

- *Intermittent Faults:* the most difficult to diagnose, where the system works and fails randomly, often related to vibration or temperature changes that cause an incorrectly manufactured or installed connection to connect/disconnect.
- *Electrical Arcing:* sparks produced by contact between high-voltage wires (115 VAC) or between a wire and the structure, which can damage equipment or start fires.

- *Electrical Overload:* wires carrying more current than designed, causing overheating and melting of the insulation.

Classifying faults according to where they occur, we have:

- *Electrical Harness Defects:* aircraft electrical harnesses present the following typical problems:
  - *Mechanical/thermal:* damage caused by contact/chafing of harness against harness or harness against structure or pipes. Also damage due to mechanical impact, drilling punctures, overheating. Missing or incorrectly installed sleeves and/or ties. Deformation or incorrect installation of the wiring protective sleeve.
  - *Contamination:* may be due to accumulation of dust and lint, metal swarf (after mechanical work in the area) or spilled liquids.
  - *Inappropriate Repairs:* these faults derive from the deterioration of previous repairs and production splices.

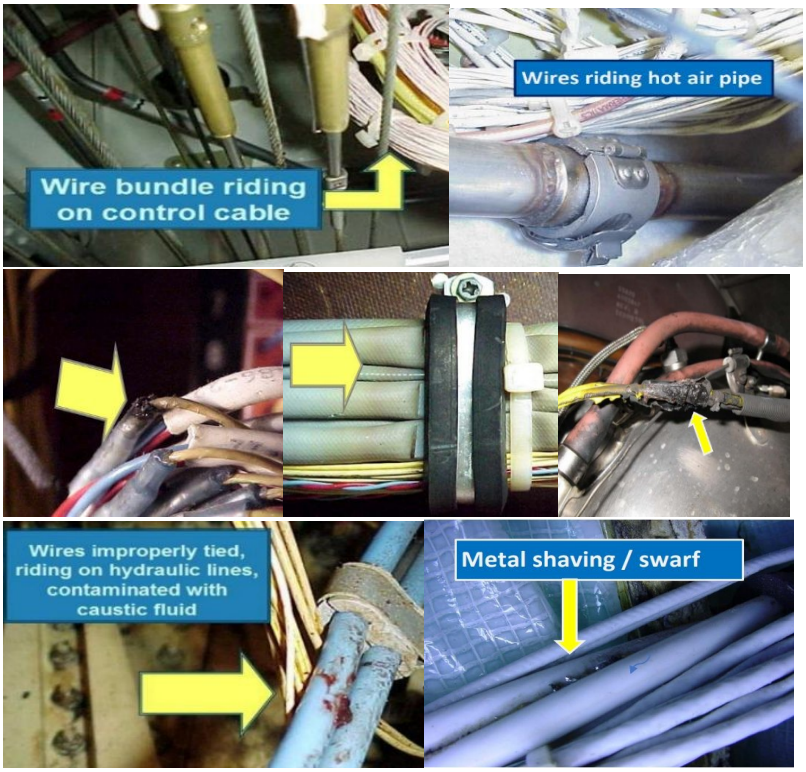


Figure 7-2 Electrical Harness Defects (Source: FAA)

- *Defects in Electrical Connectors:* the typical problems they present are:
  - *Damaged Connectors:* damaged mechanical elements can be seen visually: rear backshell missing, broken or without rubber grommet.

- *Loose Connectors:* detected when the connector feels loose or moves when handled, indicating it is not properly secured. The cause may be incorrect installation (not fully locked) or removal in earlier phases for production reasons (harness rework, cleaning…). They can cause intermittent electrical problems, such as signal loss, power fluctuations or even system failures.
- *Recessed or Damaged Pins*: a common problem that can occur due to poor pin crimping or poor insertion. It can happen during the harness manufacturing phase or through handling during harness routing, signal measurements, electrical continuity checks, ... Improperly seated, bent, damaged or broken pins can cause electrical problems such as intermittent signal loss, power fluctuations or even short circuits.
- *Signal Interference*: damaged or incorrect connector shielding can cause electromagnetic interference (EMI) or radio frequency interference (RFI). Signal interference can cause electrical problems such as signal loss, noise or distortion.
- *Overheating:* due to excessive current, poor ventilation or other factors. Overheating can cause connectors to melt, which can lead to electrical problems such as short circuits or even fires. To solve overheating, start by checking the connectors' current rating.



Figure 7-3 Amphenol Connector, Inserter/Extractor and Crimper (Source: Amazon)

7.2.2 Equipment Defects

Errors in the installation of an avionics unit, also called a Line Replaceable Unit (LRU), directly impact functional tests, causing failures ranging from simple error codes to erratic system behavior. The most common failures would be:

- *Physical and Connection Errors: they cause f*aulty connections such as bent pins, poorly seated connectors or corrosion on contacts, which

generate intermittent faults that are difficult to detect on the ground (No Fault Found) and may later appear in flight with vibration, pressurization and aircraft movement. There is also damage caused by lack of space or inadequate tools when inserting the unit into its rack or mount.

- *Configuration and Software Errors:* installing an LRU with a software revision level that does not match the operator's approved configuration or current airworthiness directives. Failing to perform the "configuration verification" after software loading to ensure it matches the aircraft type design.
- *Data Corruption:* errors when loading navigation databases or aircraft-specific configuration parameters (registration…)

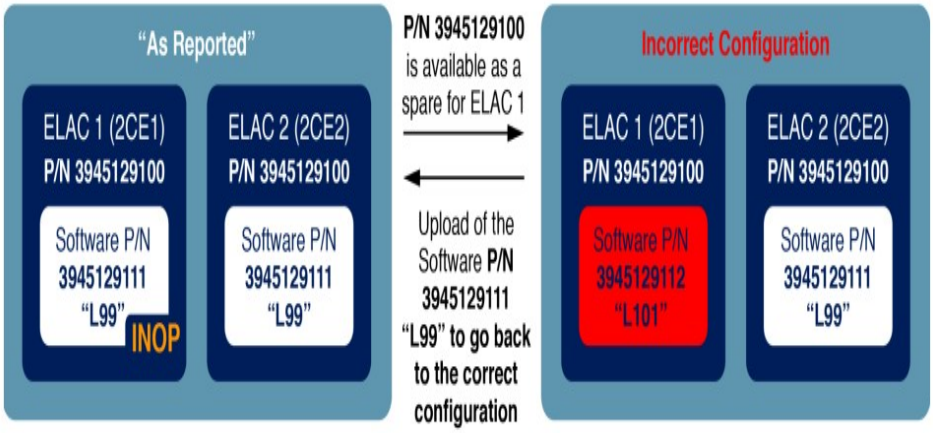


Figure 7-4 Incorrect Software Configuration (Source: ELAC Computers)

7.2.3 Human Factors

Human factors in the functional testing area refer to the physical, psychological and environmental influences that affect test execution, performance and the safety of the staff involved on the platform. In short, those elements that, depending on the operator, play a crucial role in process efficiency and in the functional quality of the aircraft.

Importance of Addressing Human Factors:

- *Improves Safety:* reduces accidents both during tests and in flight.

- *Improves Efficiency:* ensures there are no stoppages and the process runs smoothly.
- *Supports Well-being:* creates a healthier work environment for FAL staff.

By type, they can be classified as:

- *Fatigue and Workload:* long shifts and irregular hours can lead to fatigue, reducing alertness and increasing the risk of errors. In general, adequate rest periods and shift rotations help mitigate fatigue-related risks.
- *Stress and Pressure:* high-pressure environments, especially during peak hours, can affect judgment and performance. Stress management techniques and support systems help staff handle demanding situations.
- *Communication and Coordination:* poor communication between the different test teams and any fitters doing outstanding work during the test phase can cause occupational accidents (electrocution, entrapment...).
- *Situational Awareness:* staff operating the systems must stay aware of the actions they perform on the aircraft and their visible or invisible impact on the environment and on the people sharing the platform. For example, if an HF transmission system is being tested, people near the aircraft's transmitting antenna may be exposed to electric shocks. These people see no apparent danger in the antenna cable, but the test technician operating the radio must be aware of it.
- *Human Error and Decision-Making:* errors in operating certain systems can cause unexpected accidents. One example is the use of automatic or semi-automatic power or motion systems (hydraulic actuators, ramps, landing gear, flaps...) where, once actuated, the system completes the whole process and can cause injuries such as entrapment or amputation. Implementing checklists and automation reduces the likelihood of errors.
- *Environmental and Physical Conditions:* although these problems are limited in the hangar, noise and hazardous materials (fumes, confined spaces...) can affect concentration and efficiency. Protective equipment (masks, gloves...) and ergonomic workplace designs improve safety and comfort.

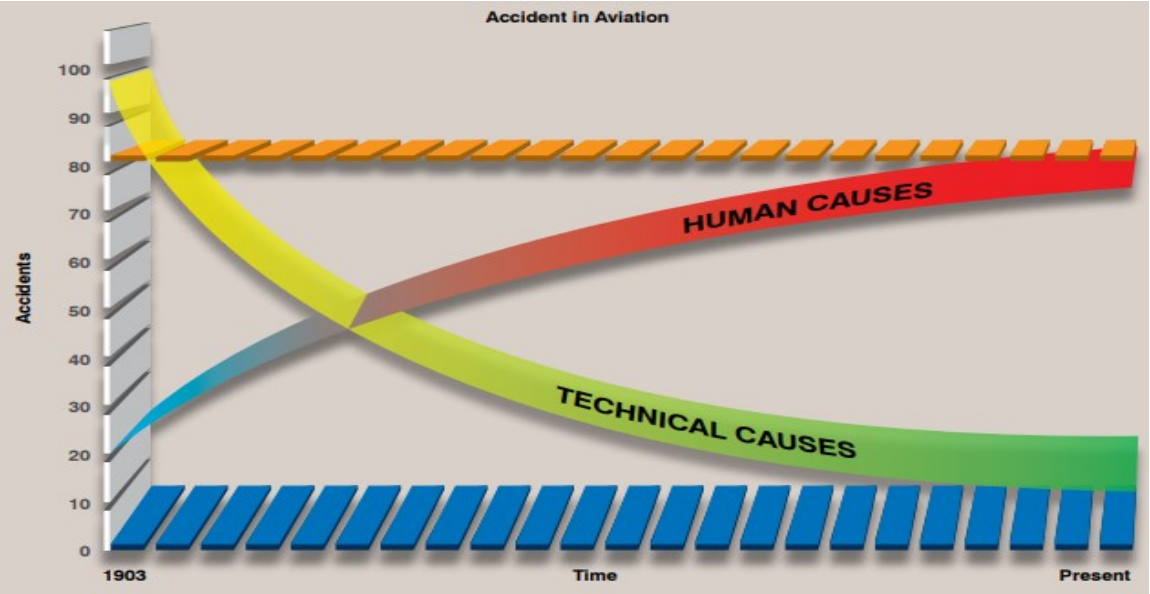


Figure 7-5 Evolution of Aviation Accident Causes (Source: Hispaviacion)



Figure 7-6 Work on Aircraft with PPE (Source: FORMAL)

7.2.4 Training and Knowledge

Knowledge on the assembly line is essential to guarantee quality, safety and efficiency in the aircraft assembly process. The operators' technical and practical know-how makes it possible to minimize errors, reduce waste and optimize cycle times.

The importance of knowledge in this environment shows in:

- *Error Reduction and Quality Improvement:* experienced workers identify faults and signs of failure that would go unnoticed by newly hired staff,

avoiding non-quality issues and reducing rework. Technical knowledge ensures precise and consistent assembly.

- *Operational Efficiency (Lean Manufacturing):* applying the Lean methodology makes it possible to identify sources of waste and optimize workflow (value added). Qualified staff understand how to apply work standards to improve performance.
- *Cycle Time Optimization:* process knowledge enables correct "line balancing", assigning tasks so that the workload is similar at each station and the time limit per product is not exceeded.
- *Safety and Maintenance:* trained operators not only perform the tests but also operate and help maintain the test equipment, reducing occupational risks and unplanned stoppages.
- *Adapting to Industry 4.0:* In modern environments, knowledge is not only manual but also digital. Operators must understand how new technologies and automation systems work to maximize production.



Figure 7-4 Assembly Line Training (Source: FID Aeronautica)

7.2.5 Documentation

Aircraft ground test documentation requires a rigorous, structured approach focused on test objectives, safety protocols, configuration control and precise data management to ensure airworthiness before takeoff.

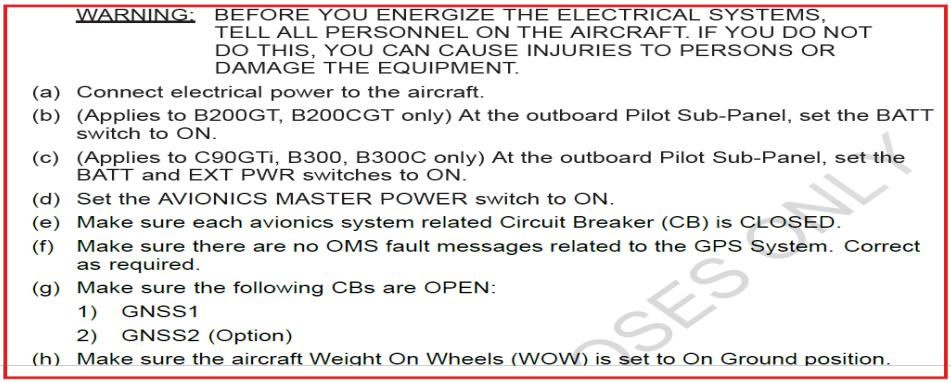
Essential factors include detailed test plans/cards, hazard analysis (risk mitigation), instrumentation calibration and complete, auditable test results.

The key documentation factors are categorized as:

- *Test Planning and Safety (Pre-test):* there must be sequential planning that avoids interference with other tests or incompatible aircraft system operations. For example, it is not possible to simulate that the aircraft is in

flight while testing the landing gear. The steps to be performed must also be available physically or digitally, clearly indicating the actions, configuration and details on the aircraft and the ground equipment to be used or its equivalents. They must identify risks and mitigation procedures (hazards, consequences, probability and emergency measures).

- *Test Execution (Conduct):* Execution must be step by step to make it easier to follow the sequence, and there must be a real-time record of test actions, anomalies, conditions and equipment used. Execution through computer-assisted systems facilitates all this work and allows multiple records of the different executions, failures and troubleshooting actions.
- *Final Report and Data Analysis (Post-test):* Test results (steps performed and data obtained) must be documented in accordance with the product's design and quality requirements. They must also be accessible in historical archives so that, if necessary, the recorded information and those responsible for execution can be retrieved.



WARNING: BEFORE YOU ENERGIZE THE ELECTRICAL SYSTEMS, TELL ALL PERSONNEL ON THE AIRCRAFT. IF YOU DO NOT DO THIS, YOU CAN CAUSE INJURIES TO PERSONS OR DAMAGE THE EQUIPMENT.

(a) Connect electrical power to the aircraft.
(b) (Applies to B200GT, B200CGT only) At the outboard Pilot Sub-Panel, set the BATT switch to ON.
(c) (Applies to C90GTi, B300, B300C only) At the outboard Pilot Sub-Panel, set the BATT and EXT PWR switches to ON.
(d) Set the AVIONICS MASTER POWER switch to ON.
(e) Make sure each avionics system related Circuit Breaker (CB) is CLOSED.
(f) Make sure there are no OMS fault messages related to the GPS System. Correct as required.
(g) Make sure the following CBs are OPEN:
1) GNSS1
2) GNSS2 (Option)
(h) Make sure the aircraft Weight On Wheels (WOW) is set to On Ground position.

Figure 7-8 Typical Functional Test with Warning (Source: King Air AMM)

7.2.6 Test Equipment

Test equipment, also called GSE (Ground Support Equipment), is essential for carrying out many of the functional tests.

For example, in avionics tests it is essential to have electronic measuring equipment (multimeters, power supplies...), aeronautical bus simulators (ARINC 429, MIL-1553,.), navigation signal simulators (VOR, ILS, ATC, TCAS...), etc.

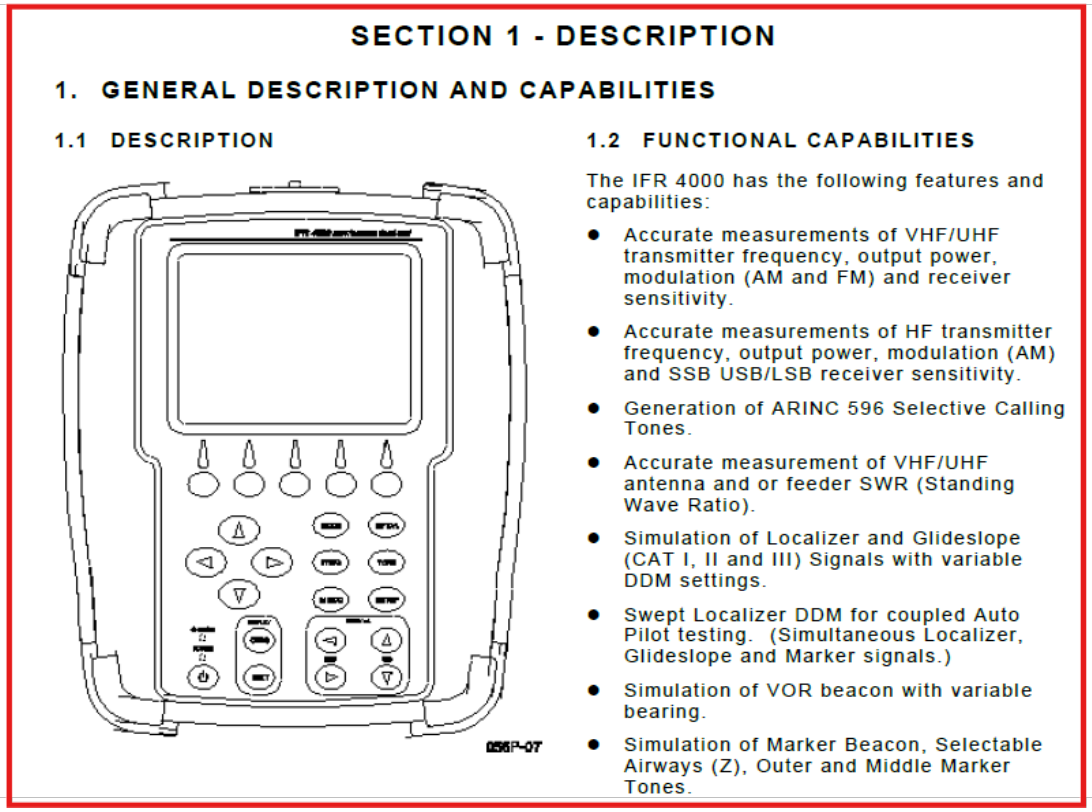
In the last three years, according to IATA, nearly 6,000 cases of aircraft damage caused by ground equipment have been reported. They are mainly problems associated with poorly maintained ground support equipment, lack of or non-adherence to ground procedures, and lack of adequate training.

At FAL level, the problems that may arise are:

- *Lack of Equipment:* in this case, either the tests must be stopped and restarted when the equipment is available, or other equipment must be used that may not be suitable for the required functionality. Equipment must be kept close to the point of use and in sufficient numbers in case it is

needed by two work teams simultaneously. For example, voltmeters, ammeters, …

- *Out-of-Calibration Equipment:* it is essential that any test equipment used in aircraft tests is calibrated. Using uncalibrated equipment can lead to incorrect rigging or checks that carry over to many aircraft and, when detected, require very costly actions on in-service aircraft. Not to mention, of course, the safety risks if the parameters are wrong.
- *Poorly Maintained Equipment:* in this case the test equipment itself or the aircraft may be damaged, and operator safety may even be put at risk. Examples include bursting of pressurized air or oil hoses (causing skin and eye injuries), electrocution due to poor “electrical grounds”, unexpected movement causing impacts…
- *Faulty Equipment:* this situation may prevent the test from being carried out or, at best, lengthen it, with the consequent increase in costs and interference with the rest of the tests. It is also possible that the equipment has an internal fault not detected by the operator, leading them to think the fault is in the aircraft, with the consequent investigation cost or even the replacement of equipment that, although working correctly, appears to be damaged. For example, the air data system requires simulating external air pressure. If the test equipment leaks or is not calibrated, the LRU will report an internal sensor fault when the problem is external.
- *Equipment Without Technical Documentation:* for equipment of some complexity, up-to-date operating documentation (latest revision) must be available, accessible and in language suited to the user. On certain occasions, the documentation is lost or not well understood (English, Chinese...), which can delay tests and, in the worst case, cause damage to the test system or the aircraft through incorrect use.



SECTION 1 - DESCRIPTION

1. GENERAL DESCRIPTION AND CAPABILITIES

1.1 DESCRIPTION

1.2 FUNCTIONAL CAPABILITIES

The IFR 4000 has the following features and capabilities:

- Accurate measurements of VHF/UHF transmitter frequency, output power, modulation (AM and FM) and receiver sensitivity.
- Accurate measurements of HF transmitter frequency, output power, modulation (AM) and SSB USB/LSB receiver sensitivity.
- Generation of ARINC 596 Selective Calling Tones.
- Accurate measurement of VHF/UHF antenna and or feeder SWR (Standing Wave Ratio).
- Simulation of Localizer and Glideslope (CAT I, II and III) Signals with variable DDM settings.
- Swept Localizer DDM for coupled Auto Pilot testing. (Simultaneous Localizer, Glideslope and Marker signals.)
- Simulation of VOR beacon with variable bearing.
- Simulation of Marker Beacon, Selectable Airways (Z), Outer and Middle Marker Tones.

Figure 7-9 Navigation Test Equipment Manual (Source: Aeroflex)

7.3 Functional Requirements

Considering the problems analyzed above and the classic criteria in the aerospace sector (quality, safety and efficiency), we can consider the following functional requirements for an AI-based support system:

7.3.1 Functional Test Assistance

- *System Information:* the AI must know the mission of the aircraft systems and how they are operated. This information must be provided to the technician as support for any doubts about operation.
- *Test Generation:* it must be able to indicate the steps to check a system if there is no documentation.

- *Equipment:* it can indicate the test equipment required according to the documentation (work order, maintenance manual...)
- *Performance:* it must be able to determine the number of technicians required and the estimated duration of the test if successful.

7.3.2 Diagnostic Assistance in Case of Failure

- *Instruction Generation:* based on the AI's knowledge of the system (equipment, schematics...) and the Interface Control Documents between subsystems, it must be able to provide the steps to follow to isolate the system failure.
- *Data Analysis:* with the data obtained and historical data (from the aircraft itself or similar ones), a probabilistic analysis of the origin of the failure must be generated.
- *Solution Proposals:* it must present one or several options taking into account the costs of opening areas, availability of replacement equipment, …

7.3.3 Repair Assistance

- *Repair Procedure:* description of how to carry out the work.
- *Required Tools:* determine the tools required, such as scissors, crimpers, soldering iron...
- *Available Materials:* determine which material should be used (wire, heat-shrink, connectors, pins…) and where it is located.

7.3.4 Assistance in the Use of Test Equipment

- *Selection of Equipment:* select the appropriate equipment for carrying out the test or troubleshooting. It must be within its calibration period and must not be faulty. Equipment that is commonly used is also preferable, as it is familiar to the technicians who will use it.
- *Location and Request:* know where the equipment is physically located or even automatically request the equipment from the industrial equipment laboratory.
- *Operating Information:* provide information on how the equipment is operated in the technicians' language.

7.3.5 Training Assistance

- *General Training:* aeronautics and air navigation systems.
- *Specific Training:* on test equipment, repair, safety…

- *Failure and Solution Simulation:* for technician training.

7.3.6 Human Factors

- *Risks:* determine and communicate the risks posed by the functional test both to the technicians performing it and to the rest of the staff at the station
- *Protective Measures:* monitor compliance with personal protective measures (cap, gloves…)
- *Fatigue/Stress:* it must be able to determine, from parameters such as tone of voice, language, etc.…, the state of the test crew, helping to ease the situation with appropriate language.

In short, the goal is to reduce the need for highly qualified staff, reduce downtime, optimize the use of test equipment, increase safety…

All of this leads to improved efficiency, without requiring additional effort from staff and reducing the technical stress arising from the use of sophisticated tools and the urgency of finding solutions to failures.

7.4 Ergonomic Requirements

As we are framing the project within what would be Industry 5.0, one of the design requirements would be ease of use and comfort.

This defines the characteristics our product must have:

- *Natural Language Processing (NLP):* allows the assistant to understand and interpret the questions it receives from the user and extract the relevant information to generate the answer.
- *Machine Learning:* based on feedback received from users, it can learn and adapt to improve its answers, making it more effective over time.
- *Continuous improvement:* its information sources receive updates or new sources are added, and it is able to integrate them into its knowledge.
- *Customization:* allows it to adapt to users' needs and preferences, enabling it to give them better answers.
- *Security:* has measures to guarantee access control and data security, so that data privacy can be guaranteed in accordance with legislation and its regulations.
- *Intuitive interface:* makes it easy for the user to understand its use and functions, so that they can make better use of the tool.

- *Availability:* allows queries to be made at any time and almost instantly, offering an advantage over a manual search.
- *Documentation:* the more there is, the easier the whole development and technical support process will be.

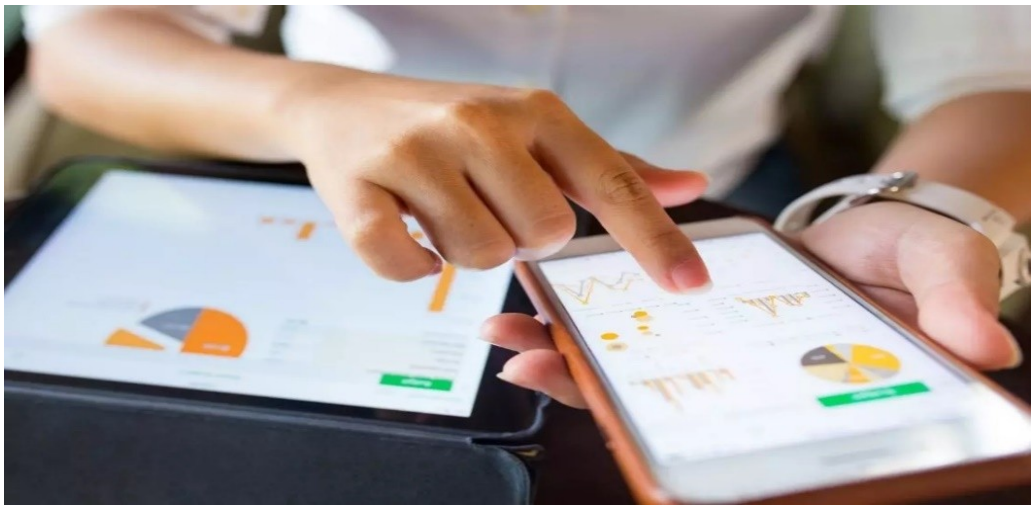


Figure 7-10 App Ergonomics (Source: Yahoo Finance)

7.5 Architecture Requirements

The architecture of an AI assistant for avionics functional tests requires robust specifications to minimize latency and must also take into account that it operates in an industrial environment. The requirements would be:

7.5.1 Hardware Requirements (Processing)

- *Processor:* minimum Intel Core i5/i7 or equivalent AMD Ryzen for smooth multitasking.
- *RAM:* minimum 16 GB as the current standard for optimal performance with multiple applications open.
- *Connectivity:* internet connection with speeds above 100 Mbps to avoid interruptions in the flow of critical data.

7.5.2 Hardware Requirements (Interface)

- *Microphone/Speaker:* for natural language communication. It must also allow audio recording.
- *Camera:* gives the AI access to the physical world, as well as documenting through photos and videos. Also visual recording.
- *Graphic Display:* apart from verbal communication, the nature of the tests (operations with multiple sequential steps) requires an easily viewable display.

- *Portability:* since operations, checks, etc. may be performed both in the aircraft cockpit (controls and indicators) and outside (sensors, antennas, etc.), the device must be portable. In addition, it must not be too heavy and must have a battery life consistent with the working day (8 h).

7.5.3 Software Requirements

- User Interface (Front end): website, mobile app, etc.) where the user interacts.
- AI/NLP Engine (Brain): Natural Language Processing to understand intents and extract entities. Includes LLMs (Gemini, ChatGPT, etc.) for response generation.
- Dialogue Management: component that maintains conversation context, controls the flow, manages state and decides the best response.
- Integration Layer (Middleware/Backend): connects the chatbot with external systems (databases) for real-time actions.
- Databases and Memory:
- Vector databases: essential for RAG, they enable semantic searches and access to up-to-date knowledge.
- Logs and Session Database: Store the history to maintain context.

7.5.4 Software Requirements (RAG Retrieval Augmented Generation)

- Data Ingestion: pipelines to process, clean and chunk documents.
- Embeddings: models to convert text into numerical vectors.
- Semantic Search: selection of the most relevant information to send as context to the LLM.

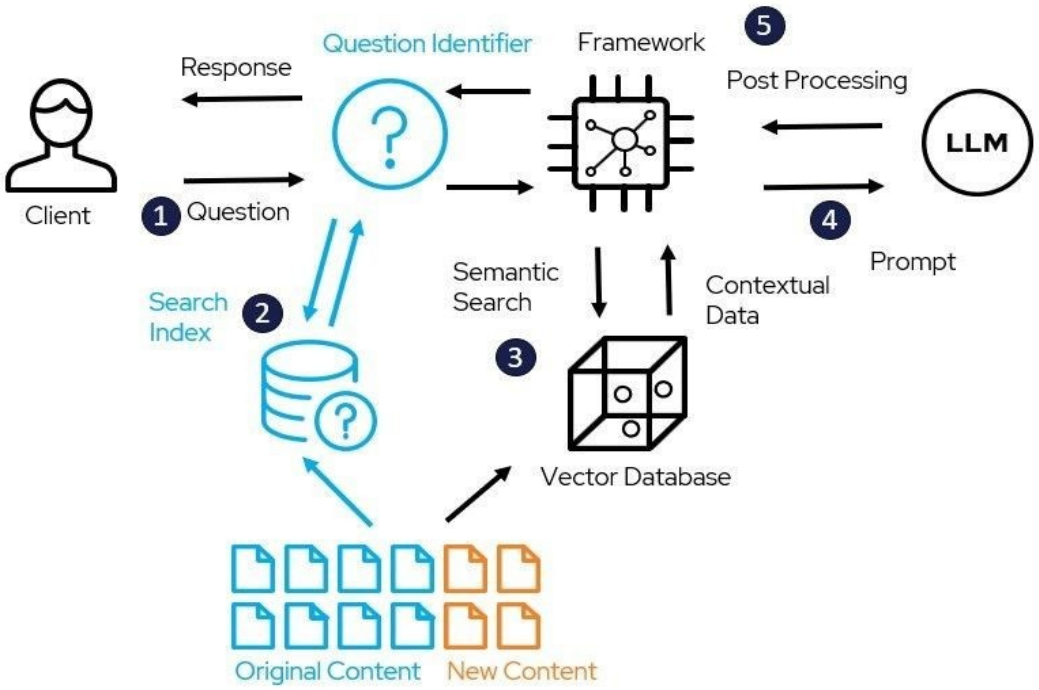


Figure 7-11 Typical Virtual Assistant Architecture (Source: Brahmasumm)

7.6 Requirements Verification

The test assistant must be validated as a Level 1 Artificial Intelligence application (human augmentation and support) under the regulatory framework of the European Union Aviation Safety Agency (EASA).

Validation requires strict traceability of information, continuous human oversight and "hallucination" tests.

It is vital to consider the EASA guidelines and the EU AI Act, which define the use of AI systems for manufacturing and maintenance processes as critical tools. It will be necessary to document:

- *Learning Assurance:* document how the model has been trained and tested.

- *Continuous Auditing:* keep a log of interactions to ensure the assistant does not drift into bias or misinterpretation.
- *Compliance Documentation:* generate the EASA certification documentation required for this type of assistance tool

The verification and validation methodology can be divided into three fundamental concepts:

- *Domain Knowledge Evaluation:* use metrics specialized in aircraft maintenance to measure the model's accuracy and reasoning before putting it into production.
- *Hallucination and Accuracy Verification:* test the model with a dataset of standard FAL queries (Maintenance Manuals, Operation…). It is evaluated that it does not make up information and that its document retrieval hit rate is above 90%.
- *Human Factors Testing:* validate the interface with real test technicians in the FAL. The time to resolve questions with and without the assistant is measured, and feedback is gathered on whether the answers are practical, safe and readable in the workshop environment.

In the “Final Results” chapter, the above parameters are checked and suitable metrics are applied to determine the validity of the assistant under the applicable standards.

8 ANALYSIS OF SOLUTIONS

8.1 Introduction

Before selecting the technology that forms the core of the intelligent assistant, an analysis has been made of the main existing alternatives for decision support and technical assistance systems.

The objective is to determine which of them best meets the functional and ergonomic requirements defined in the previous section.

The technologies considered are:

1. Expert Systems.
2. Document Search Engines.
3. Traditional Chatbots.
4. Classic Machine Learning Systems.
5. Large Language Models (LLMs).

8.2 Expert Systems

For decades, expert systems were the most widely used solution for industrial diagnosis. They work on a knowledge base made up of rules of the type: “IF A occurs and B occurs THEN do C”.

Advantages:

- Completely deterministic answers.
- Very reliable when the rules are well defined.
- Easy validation in critical environments.
- They do not generate made-up information.

Disadvantages:

- They require thousands of rules to be programmed manually.
- Very costly to maintain.
- Not very flexible with new equipment.
- They do not understand natural language.
- They cannot summarize technical documentation.
- Unable to reason about new documents.

They could be used to automate some test procedures, but not to answer open questions or interpret aeronautical manuals from different manufacturers.

8.3 Document Search Engines

A document search engine is an application or computer system designed to crawl, index and retrieve information contained in files (PDFs, Word…).

It allows users to search for and locate specific documents within internal databases or corporate repositories using keywords, filters and logical operators

Examples of commercial solutions include Elasticsearch, Solr or similar indexing engines.

The system only searches for documents where certain words appear.

Advantages:

- Very fast.
- Scalable.
- Easy to maintain.
- Very suitable for large amounts of documentation.

Disadvantages:

- They do not understand the meaning of questions.
- The technician must read the entire document.
- They do not generate answers.
- They do not perform reasoning.
- They do not relate information from several documents.

Although they make it possible to locate information quickly, they do not reduce the technician's cognitive load during functional tests.

8.4 Traditional Chatbots

Classic chatbots use decision trees or pre-programmed answers.

Advantages:

- Simple to develop.
- Low cost.
- Fast answers.

Disadvantages:

- Very limited conversations.
- Unable to understand complex technical documentation.
- They do not learn.
- They do not make inferences.

Suitable for answering frequently asked questions, but insufficient for advanced technical assistance.

8.5 Classic Machine Learning Systems

Classic Machine Learning (ML) systems are Artificial Intelligence algorithms that learn patterns directly from structured data to make predictions or decisions, without the use of deep neural networks.

Examples include Random Forest, SVM, conventional Neural Networks, XGBoost, etc.

Advantages:

- Very good for classification.
- Very good for failure prediction.
- Excellent performance when large amounts of data exist.

Disadvantages:

- They need structured data.
- They do not process natural language.
- They do not interpret manuals.
- They do not generate explanations.
- A different model must be developed for each task.

They are very interesting for probabilistic analysis of the origin of faults, but cannot act as a conversational interface for technicians.

8.6 Large Language Models

A large language model (LLM) is an artificial intelligence system based on deep learning and the Transformer architecture. It is designed to process, understand and generate human language through statistical analysis.

Models such as GPT, Llama, Mistral or Gemma are capable of understanding natural language, interpreting technical documentation and generating contextualized answers.

Advantages:

- They understand natural language.
- They generate technical answers.
- They summarize documentation.
- They integrate information from multiple documents.
- They support learning through new documentary sources (RAG).
- Very good user interaction.
- High scalability.
- They allow integration with vector databases (Qdrant).

Disadvantages:

- Higher computational consumption.
- Risk of hallucinations.
- They require validation mechanisms.
- They need more powerful infrastructure.

It is the only technology capable of simultaneously meeting the requirements for conversation, document search, reasoning, instruction generation and technical assistance.

8.7 Justification of the choice

Following the analysis, it can be concluded that none of the technologies studied on its own meets all the functional requirements defined for the assistant supporting aircraft functional test technicians.

- Expert systems are highly reliable, but require a very high maintenance effort and do not allow working with unstructured documentation.
- Search engines make it easier to locate information, although they force the technician to interpret the documents found manually.
- Traditional chatbots offer simple interaction, but lack reasoning capability.
- Finally, machine learning algorithms are suitable for specific classification or prediction tasks, although they cannot understand queries formulated in natural language or generate technical explanations.
- On the other hand, Large Language Models (LLMs) are the only alternative capable of integrating natural language understanding, contextual reasoning, answer generation, technical documentation analysis and continuous adaptation through Retrieval Augmented Generation (RAG) techniques.

  This capability makes it possible to build an assistant that answers technicians conversationally, consults up-to-date aeronautical documentation, proposes test and repair procedures and facilitates decision-making during maintenance operations.

Consequently, this work selects an LLM as the core of the intelligent assistant, complemented with a vector database for semantic retrieval of technical documentation and a RAG system, which allows answers to be grounded in the organization's specific information.

This combination provides the best balance between functionality, scalability, maintainability and capacity for future evolution, in line with the principles of Industry 5.0 and the operational needs of aircraft maintenance.

8.8 Intelligent Assistant

To create an aircraft test support assistant with Artificial Intelligence (AI), it is necessary to combine development tools, language models and a technical knowledge base.

Development tools are the set of frameworks and kits that allow programmers to build, integrate and manage software such as the AI chatbot system in our project.

The language model (LLM) gives the AI the ability to understand, process and generate text with a natural result. In a chat, it acts as an interlocutor, analyzing the question, understanding the context, searching its information base and writing a coherent, useful and personalized answer in real time.

As for the knowledge base, it must be as complete as possible, and for that it is necessary to gather records of previous tests, work documentation, aircraft and test equipment manuals, error and troubleshooting history, etc. All of this will be inserted through suitable vector databases.

Likewise, given the type of business we are dealing with (aerospace sector), it is essential that the platform allows the origin of the data to be controlled (RAG), guarantees its privacy and provides rigid logical structures.

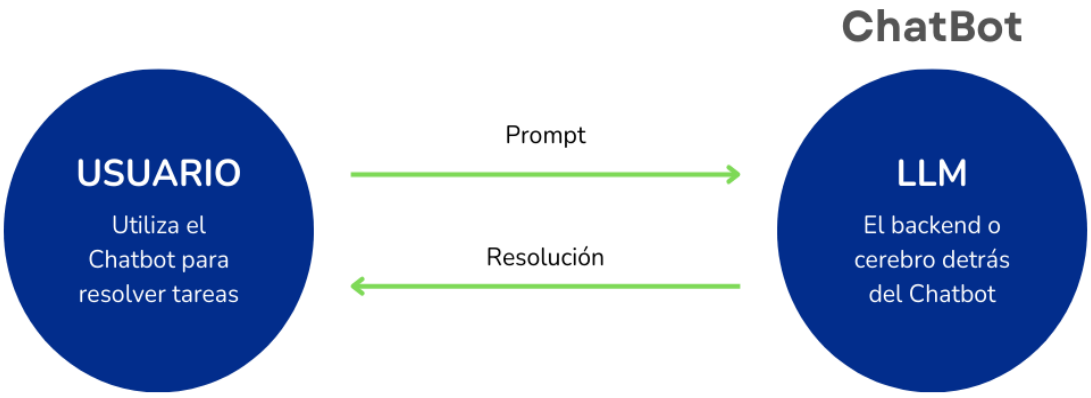


Figure 8-1 Chatbot with LLM (Source: ITCL Technology Center)

8.9 General Architecture

The architecture of a chatbot is the set of logical and technical components that structure how it works. It acts as the "blueprint" of the system, allowing it to interact with the user, understand their intentions, process information and generate a coherent answer.

It must have the following systems:

- *Orchestration platform:* connects the language model with the user interface and the data through techniques such as RAG (Retrieval-Augmented Generation). E.g. n8n, Langflow or LlamaIndex.
- *Language Model (LLM):* options such as OpenAI (GPT-4), Anthropic (Claude), or local open-source models (Llama 3, Mistral).
- *Vector database:* to store the chunked documentation (PostgreSQL, ChromaDB, Pinecone, Qdrant).

8.9.1 Orchestration Platforms

These platforms manage the conversation flow, connect the technical manuals and decide what to answer. Some of the available options:

- *n8n (Low-Code/Hybrid):* combines visual nodes with code (Python/JS) and has native support for AI agents (LangChain). It also allows strict rules to be structured.
- *Langflow (No-Code, 100% AI-focused):* visual tools designed exclusively for dragging and dropping Artificial Intelligence components (LLMs, memory, vector databases). They are excellent for rapid prototyping of RAG systems.
- *Rasa (Pure code / On-Premise):* does not depend on commercial models (such as OpenAI). It is extremely secure, runs 100% locally and processes language locally.
- *Dify.ai (All-in-One Platform):* combines visual chatbot creation, technical manual management (PDFs), data processing and bot evaluation in a single interface.



Figure 8-2 N8N Platform (Source: Medium)

8.9.2 Language Models

To create a technical chatbot, it is necessary to choose language models (LLMs) that excel in structured logic and understanding of technical documentation. Commercial and open-source models are available:

Commercial Models: the easiest to connect and offer the greatest technical accuracy, but they run in the cloud.

- *GPT-4o / GPT-4o-mini (OpenAI):* the industry standard option. It stands out for its speed, low cost (in the mini version) and excellent ability to write automation code.
- *Claude 3.5 Sonnet (Anthropic):* currently considered the best model in the world for development, code refactoring and complex logic.

- *Gemini 1.5 Pro / Flash (Google):* its main advantage is a gigantic context window (up to 2 million tokens). It is the best option if the chatbot needs to analyze extensive technical manuals.

Open-Source Models: if the data cannot be in the cloud:

- *Llama 3 (Meta):* the most powerful general-purpose open-source model. Its 8B (for modest servers) or 70B versions understand and process QA flows perfectly.
- *DeepSeek-Coder / Codestral (Mistral AI):* models specialized exclusively in programming. They are perfect if the chatbot will only focus on creating automated test scripts and analyzing syntax.
- *Phi-3 / Phi-4 (Microsoft):* small models, but extremely optimized for logical and technical reasoning. They consume very few server resources.

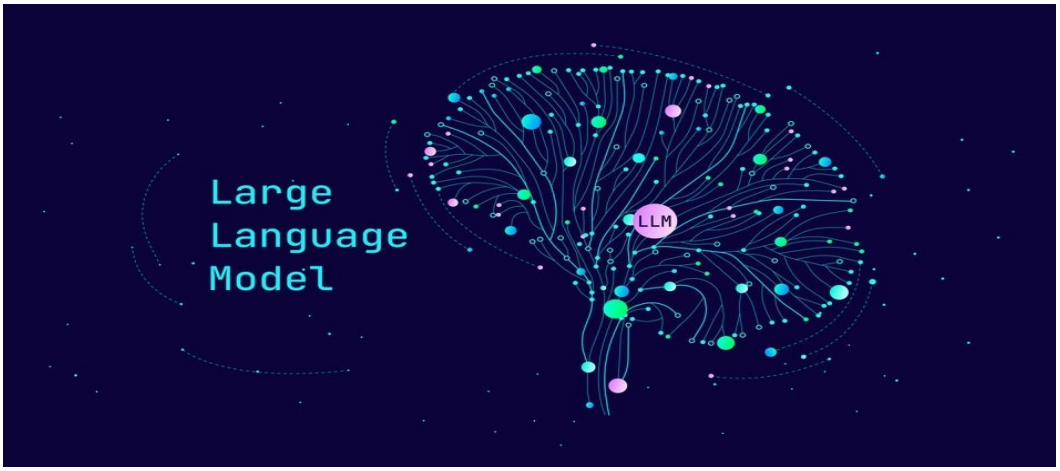


Figure 8-3 AI Language Models (Source: Shutterstock)

8.9.3 Vector Databases

A technical chatbot based on an LLM has general knowledge of aircraft but is not a specialist in any particular model, so it needs to search the appropriate documentation (manuals, records...).

To do this, the PDFs are converted into vectors and stored in databases.

Some of them are:

- *ChromaDB / Qdrant:* excellent open-source options that can be installed on local servers. Qdrant is extremely fast with millions of technical documents.
- *Pinecone:* the market leader in the cloud. It is ultra-fast and requires no maintenance, but the data leaves our infrastructure. To use Pinecone locally, an official emulator distributed as a Docker image must be used.

- *PostgreSQL:* allows business data and manual vectors to be stored on the same server.

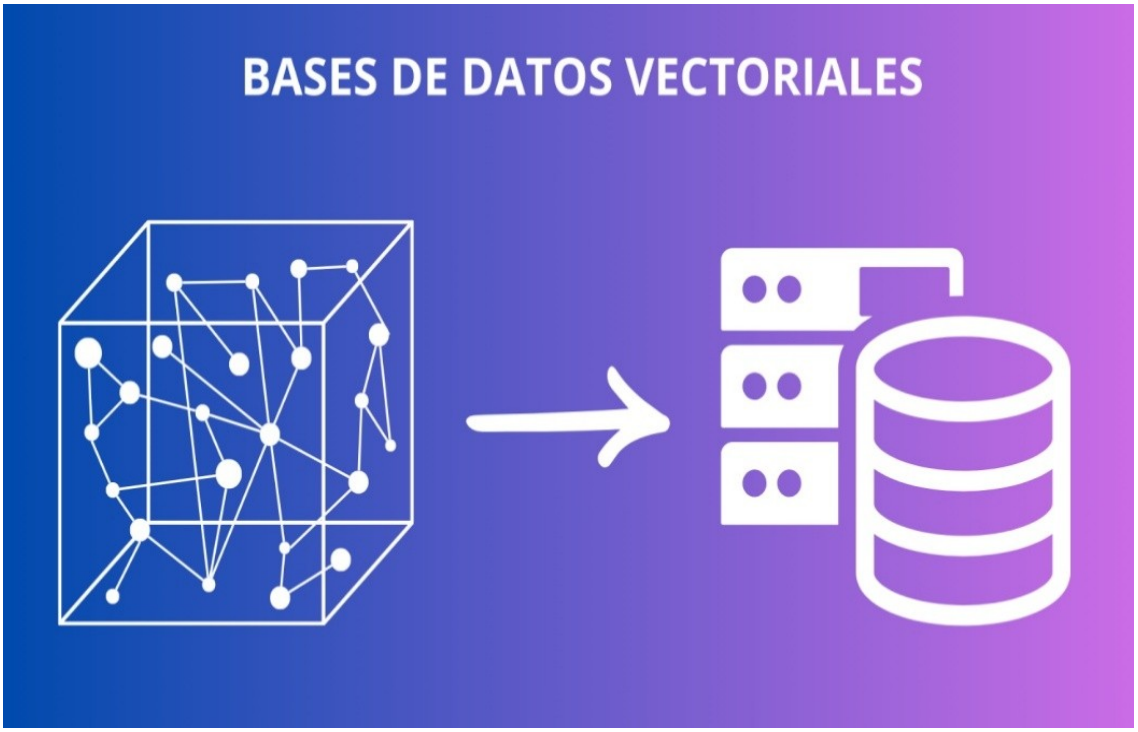


Figure 8-4 Vector Databases (Source: Casal 2025 – Vector Databases)

8.10 N8N Orchestration Platform

N8N [36-37] is a powerful node-based workflow automation platform that allows different applications, databases and APIs to be connected to perform automated tasks without writing code.

Its main differentiating factor is that it is distributed under an open-source model, which means it can be installed free of charge on our own server, guaranteeing absolute privacy and eliminating per-task volume costs.

8.10.1 Key Components

The platform uses an interactive visual interface based on a digital canvas:

- *Workflows:* the complete digital canvas where the automation is designed from start to finish.
- *Nodes:* individual visual blocks that execute a specific action. There are two main types: Triggers: responsible for starting the flow when an event

occurs (e.g. an email arrives, a webhook is received or a row changes in Google Sheets). Actions: the steps that do something with the information received (create a PDF, send a message, etc.).
- *Connections:* the lines that join the nodes together and determine the order in which data travels

8.10.2 Main Features

The main features are:

- *Native AI:* allows advanced language models (such as ChatGPT, Claude…) to be integrated natively. You can build intelligent agents capable of making autonomous decisions within the flow, classifying emails or summarizing documents logically.
- *Hosting Flexibility:* unlike cloud alternatives, you can download n8n and run it locally or on a virtual private server. This is ideal for companies that handle sensitive data and must strictly comply with privacy regulations.
- *Advanced Low-Code Approach:* although it is designed not to require programming, if a native automation falls short, it allows JavaScript or Python nodes to be inserted to transform or clean complex data manually.
- *Data Control and Complex Logic:* supports real logical architectures, meaning you can structure conditionals (IF), loops (LOOPS), time waits and human approval nodes (human in the loop).

8.10.3 Agent Architecture

When developing a technical assistant for aircraft systems, query handling for the different functional systems (hydraulic, electrical, avionics, fuel, etc.) changes radically depending on the chosen architecture.

Until recently, when we thought of "conversational AI", we imagined a single chatbot (single agent) with all the associated information answering the user's questions. Today, more complex architectures are emerging with different specialized agents that improve the answers.

Single-Agent Architecture

In the field of artificial intelligence, a single agent [40] is a centralized autonomous system that perceives its environment, makes decisions and executes actions on its own to achieve a goal, without collaborating with other AI systems.

In the context of an assistance chat, it means that a single model (a single "brain") processes all the information and answers the user.

This model contains the knowledge and business rules, receives the user's input, searches the database and generates the final answer directly.

The advantages of this approach are:

- *Simple implementation:* simple interface, shorter development and initial configuration time. It does not require a supervisor system to route the question. It also has a lower risk of internal communication failures.
- *Total consistency:* a single knowledge base avoids contradictory answers to the user.
- *Lower cost and latency:* consumes fewer memory and computing resources. Since there is no debate or data transfer between sub-agents, the answer is faster.
- *Easy maintenance:* data updates are applied in a single place.

The disadvantages are:

- *Context saturation (hallucinations):* aviation manuals are massive. A single agent can confuse error codes or procedures of different systems (e.g. confusing a pneumatic valve with a hydraulic one).
- *Complex critical updates:* if the manual of a single system changes (e.g. avionics update), retraining or updating the single agent's knowledge index can alter the performance of its answers on the other systems.

Multi-Agent Architecture

Multi-agent architectures [38] are fundamental in the field of artificial intelligence, allowing several agents to work together to achieve common goals in a coordinated way

Multi-agent systems divide responsibilities among several specialized agents that execute different tasks within a business process.

Although this system requires additional coordination and orchestration, and introduces complexity and increased latency, it has numerous advantages:

- *Deep specialization:* each agent masters a set of specific systems that are functionally related in the aircraft.
- *High Accuracy:* by limiting each agent's context, answers are more accurate, avoiding hallucinations or cross-contamination of data between different systems.
- *Modularity:* ideal for complex industrial processes with multiple subsystems.
- *Scalability:* allows new agents to be added to integrate systems not covered in the initial project (hydraulic, fuel, flight control systems, etc.)



Figure 8-5 Agent Architecture (Source: Economiable)

Agent Architecture Selection

In our case, the choice between a single-agent and a multi-agent system was not clear, so different tests were carried out.

After a first single-agent prototype and various queries, a large number of failures were detected.

Analyzing their cause showed that they derive from the enormous complexity, volume and criticality of the associated documentation.

Thus we see that:

- Aviation manuals (maintenance, operation) have thousands of pages, and when querying with a single agent, steps from systems with similar terminology were mixed up (for example, the test of one system with another that had controls with similar names, etc..).
- Loss of critical details: by processing all the aircraft information in a single "block", it may ignore safety warnings for a specific system buried under gigabytes of text.
- If the manufacturer publishes a revision anywhere in a document, the complete knowledge base for the whole aircraft must be reindexed, tested and validated.

- As more revisions and bulletins, fault histories, etc. are added, the single agent becomes slower, less accurate and more costly to maintain.

Following the failure of the previous tests, and taking into account that aeronautical documentation is structured and divided according to the ATA coding system, the use of multi-agent systems is proposed.

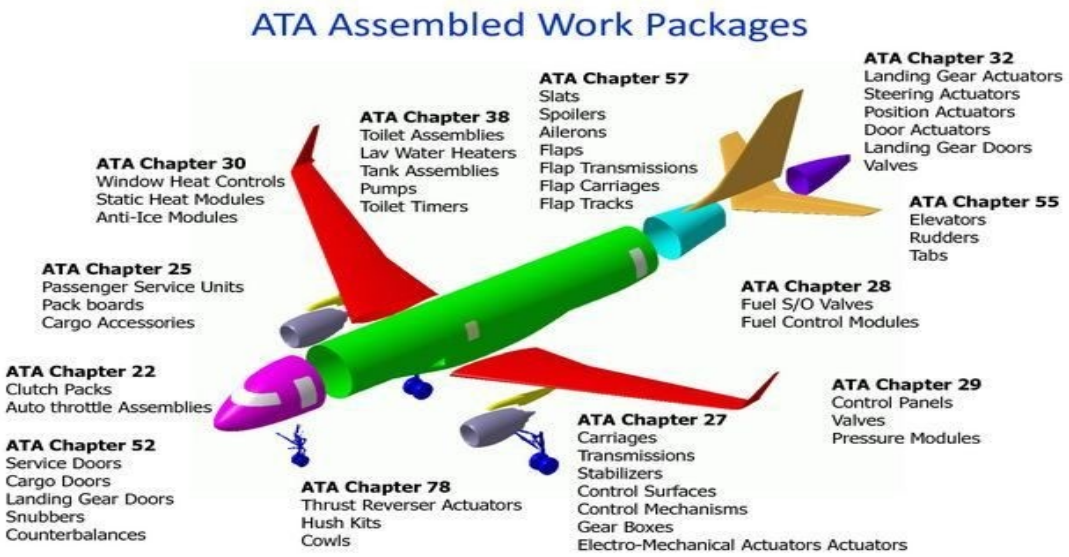


Figure 8-6 ATA Chapters (Source: Instagram)

Specialist Agents

In this model, each agent is associated with one or more ATAs, and an orchestrator agent is established that redirects questions to the specialist agent.

After studying the avionics tests and the affected systems and subsystems, it was decided to create the following agents and their knowledge scope:

ADS Agent:

- Air Data System (ADS)
- ADC (Air Data Computer)
- Pitot-static system
- Static ports
- Total Air Temperature (TAT)
- IAS, TAS, Mach
- Altitude, Barometric Pressure
- Pitot/Static Failures

AHRS Agent

- AHRS
- IRS / INS
- Gyroscopes
- Accelerometers
- Attitude (pitch, roll)
- Heading
- Magnetometer
- Inertial reference failures

WXR Agent

- WXR (Weather Radar)
- Weather radar
- Reflectivity (dBZ)
- Radar attenuation
- Tilt and Gain
- WX, WX+T, MAP Modes
- Storm detection
- Turbulence (TURB)
- Ground clutter
- Weather radar failures

DME Agent

- DME (Distance Measuring Equipment)
- DME Distance
- DME Navigation
- DME UHF Frequencies
- DME Interrogation/reply
- DME HOLD
- Pairing VOR/DME
- DME Failures

GNSS Agent

- GNSS (Global Navigation Satellite System)

- GPS
- Galileo
- GLONASS
- Satellite positioning
- RAIM
- FDE
- RNAV/RNP Navigation
- Satellite signal
- GNSS Failures

TCAS Agent

- TCAS (Traffic Collision Avoidance System)
- TA (Traffic Advisory)
- RA (Resolution Advisory)
- Air traffic
- Collision avoidance
- Mode S
- TCAS Coordination
- Intruder traffic
- TCAS Failures

RALT Agent

- RALT (Radio Altimeter)
- Radio Altitude
- Radio altimeter
- Height above terrain
- GPWS
- Callouts
- Low height indication
- Autoland
- RALT Failures

VHF Agent

- VHF (Very High Frequency Navigation)
- VHF Navigation
- VOR Systems (VHF Omnidirectional Range)
- Localizer (LOC)
- ILS Localizer
- VOR signal reception
- Aeronautical NAV frequencies
- Aeronautical radio navigation
- CDI (Course Deviation Indicator)
- OBS (Omni Bearing Selector)
- VOR signal deviation
- Morse identification of VOR/ILS aids
- Interference in NAV signals VHF Failures

XPDR Agent

- XPDR (Transponder System)
- Transponder
- Mode A/C/S
- Squawk
- IDENT
- ADS-B
- SSR
- Reported altitude
- XPDR Failures

Equipment Manuals Agent

- IFR6000
- IFR4000
- Druck_ADTS_405

8.10.4 Orchestrator Agent (version 1)

An orchestrator agent is the central Artificial Intelligence component that receives the user's query, interprets it and decides which specialist agents to invoke. It coordinates their answers and unifies the information to provide a coherent solution.

The operating dynamics are as follows:

- *Reception:* the test technician asks a question and the orchestrator deciphers the overall intent.
- *Subtask Delegation:* the orchestrator splits the query and assigns specific parts to independent specialist agents that know each subsystem in depth (XPDR Agent, RALT...).
- *Execution and Retrieval:* each specialist agent uses integrated tools to query databases in real time (maintenance manual, operation, etc.).
- *Synthesis and Answer:* the agents return their findings to the orchestrator, which filters out unnecessary data and writes a single, clear and precise answer for the user.

After various tests, a series of additional basic rules were determined and imposed that improve the answers:

- Always use the tool before answering: no exceptions.
- Answer only with the information found in the documents: avoid adding information from your general knowledge, which usually leads to non-applicable data.

- If you find no information, answer exactly: "I have no information in the manuals."
- Never make up information: it is better to give no information than inaccuracies that can have serious consequences.
- When using a vector tool, always formulate the “query” expanding the acronyms. Examples:
    - "ADC" → "Air Data Computer ADC"
    - "AHRS" → "Attitude Heading Reference System AHRS"
    - "WXR" → "Weather Radar WXR"
    - "DME" → "Distance Measuring Equipment DME"
    - "GNSS" → "Global Navigation Satellite System GNSS"
    - "TCAS" → "Traffic Collision Avoidance System TCAS"
    - "RALT" → "Radio Altimeter RALT"
    - "VHF" → "Very High Frequency VHF"
    - "XPDR" → "Transponder System XPDR"
- Include both the acronym and its full English name in the “query”.

You may slightly rephrase to improve the search, but keep the original meaning.

Likewise, the orchestrator agent has a simple memory of 10 questions. It can remember things for up to ten questions, and from the eleventh it forgets and starts the record again. This memory capacity covers most typical conversations.

The orchestrator's role is set through a “prompt” instruction, which works as a command that guides the AI to generate the exact result sought.

In our case it has been set as: “You are a technical assistant specialized in avionics”.

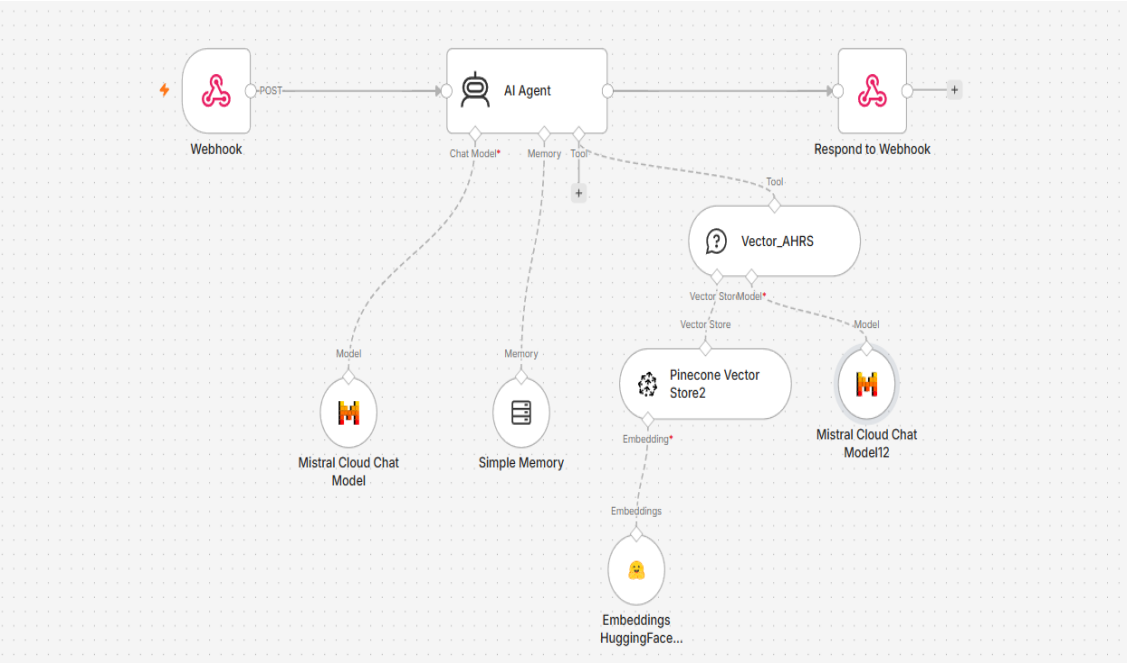


Figure 8-7 N8N Orchestrator Agent Version 1 (Source: Own Elaboration)

8.10.5 Orchestrator Agent (version 2)

The orchestrator agent presented in the previous section (version 1) was a first functional prototype of the system.

Its objective was to validate the proposed architecture and check the feasibility of making natural language queries on the technical documentation stored in the vector database.

Although this prototype generated correct answers, it had an important limitation in terms of information traceability, as it was unable to indicate the source PDF document or the specific pages or sections from which the information used to generate the answer had been extracted.

This limitation was because the Retriever only retrieved the textual content of the vectors stored in Pinecone, and the metadata associated with each chunk was lost during the retrieval process. As a result, the orchestrator agent only had each agent's answer, but not the context.

For this reason, it was necessary to develop a new orchestrator agent (version 2), which is the final version of the system.

This new implementation keeps the capabilities of the initial prototype but modifies the information indexing and retrieval process.

During indexing, each vector stores not only the textual content of the chunk but also its associated metadata, including the name of the PDF document, the page number and other relevant identifiers.

In this way, when the Retriever performs a semantic search on Pinecone, it retrieves both the text chunks most similar to the user's query and the metadata associated with each of them. Thus, the orchestrator agent simultaneously receives the content needed to generate the answer and the information that identifies its origin, and can indicate the PDF document and the corresponding page from which each chunk used was obtained.

Orchestrator Agent Flow

The processing flow of the new orchestrator agent is shown in the figure below and can be summarized in the following stages:

1. Webhook

Entry point of the workflow. It receives the external request via an HTTP POST request made from the user interface. Receiving the request automatically starts the flow execution.

2. AI Agent (Orchestrator)

Main node responsible for managing the entire query process. It interprets the question asked by the user and coordinates the use of the language model, the conversational memory and the tools connected to the agent.

Whereas the previous prototype's orchestrator received only each agent's answer and nothing else, this orchestrator obtains the metadata associated with each chunk, allowing it to identify the source PDF document and the page where the information used to generate the answer is located.

3. Mistral Cloud Chat Model

Language model used by the Orchestrator Agent to understand the user's query, interpret the context provided by the retrieved documents and produce a coherent answer in natural language.

4. Simple Memory

Conversational memory system responsible for storing the conversation context, allowing continuity between different questions asked during the same session.

5. Pinecone Vector Store (Vector_ADS)

Retrieval tool based on a vector database that stores the embeddings of all the previously indexed documentation.

When it receives a query, it uses the generated embedding to locate the chunks whose semantic meaning is most similar to the user's question. As a result it returns:

- The textual content of the retrieved chunks (pageContent).
- The metadata associated with each chunk (document name, page number, section, etc.).

Thanks to this information, the orchestrator agent can generate well-grounded answers and maintain traceability of the sources used.

6. Embeddings HuggingFace

Model responsible for transforming both the user's query and the document chunks into vector representations (embeddings).

These vectors are sequences of numerical values that capture the semantic meaning of the text, allowing documents to be compared by similarity rather than by exact word matching.

This vector is used internally by Vector_ADS to retrieve the most relevant documents. The user does not work directly with these numerical values; they only receive the retrieved chunks and the answer generated by the agent.

7. Respond to Webhook

Final node of the workflow, responsible for returning the answer generated by the Orchestrator Agent to the user interface via the corresponding HTTP response.

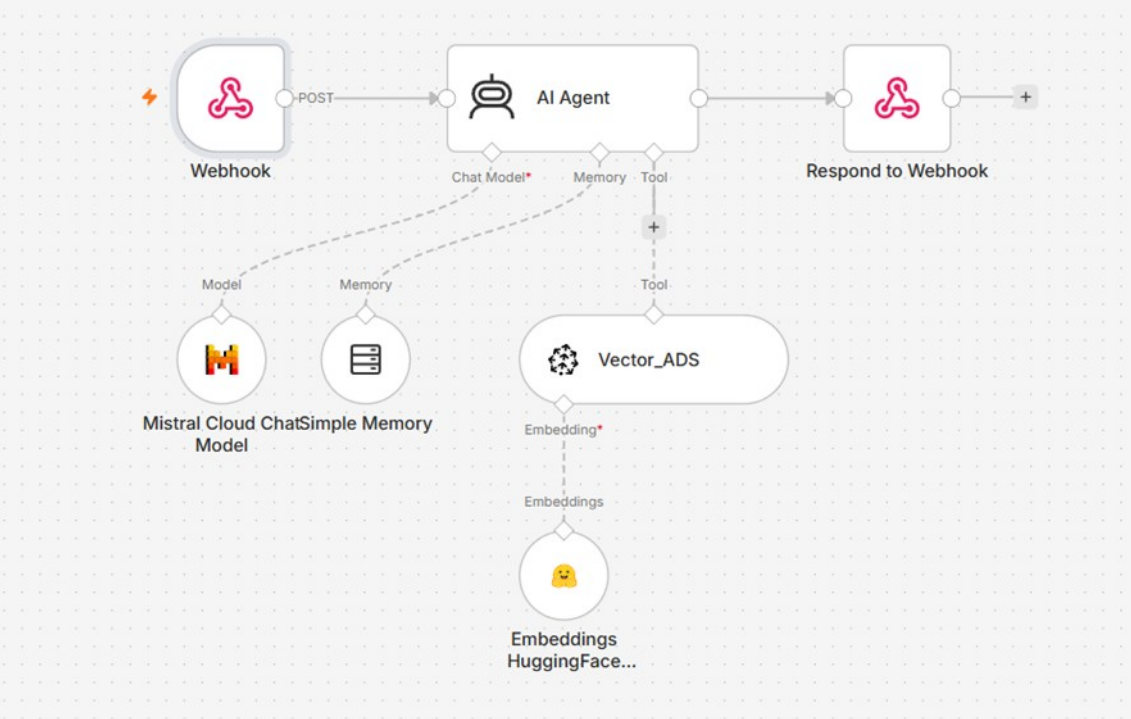


Figure 8-8 N8N Orchestrator Agent Version 2 (Source: Own Elaboration)

8.11 Mistral 14B Language Model

Mistral was selected over options such as Gemini or ChatGPT mainly because of data control, operating costs and customization capability.

Compared with the other LLMs, the Mistral model has the following advantages:

- *Absolute privacy:* Mistral offers an open model that can be installed on in-house servers, so confidential technical data never leaves my infrastructure.
- *No usage costs:* ChatGPT and Gemini charge for every question and answer (pay per token). With Mistral on an in-house server, the cost is fixed regardless of how many queries the user makes.

- *Real fine-tuning:* Mistral can be retrained internally with all the aircraft's technical documentation (manuals, records).

- *Hardware efficiency:* although it does not surpass the raw intelligence of ChatGPT 4.0, it offers far greater efficiency, speed and cost-effectiveness for specific tasks.

- *Excellent for programming:* Mistral's architecture has been shown to be especially good at understanding source code, logic and programming languages compared with other models of the same size.

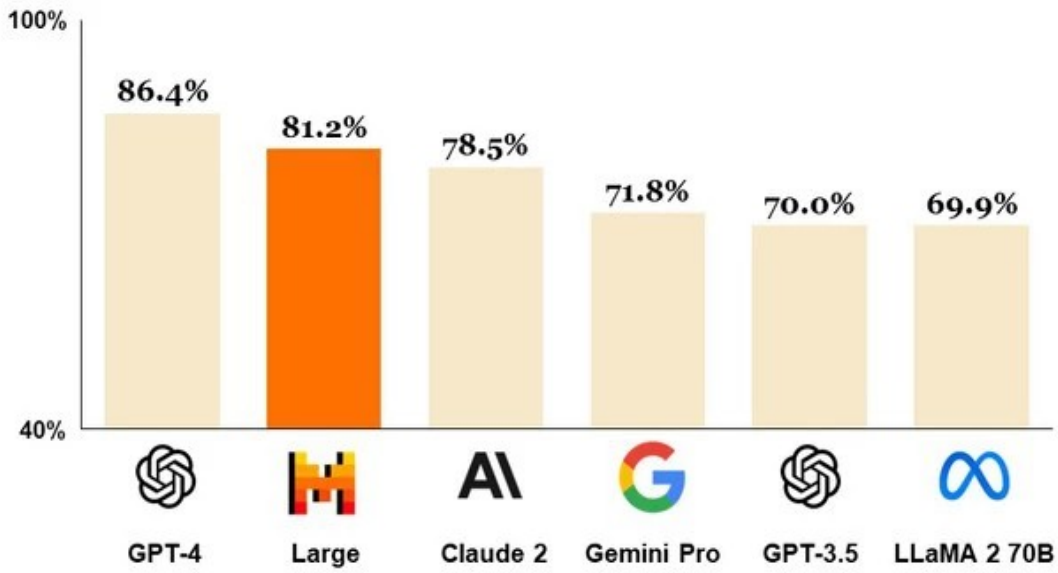


Figure 8-9 LLM Comparison (Source: PCMag)

8.12 Pinecone Database

Pinecone was chosen as the vector database for the aircraft functional test technical chatbot for one critical reason: aviation safety and technical complexity demand instant, accurate answers based on strict regulations.

Pinecone solves the specific technical challenges of this environment through the following advantages:

- *Management of Massive Technical Documentation:* Pinecone allows the chatbot to understand the meaning of a technical query (e.g. "procedure for bleeding the hydraulic system") and find the exact paragraph in milliseconds, overcoming the limitations of traditional keyword search.
- *Reduction of "Hallucinations":* in aircraft functional tests, the chatbot cannot make up data. Pinecone acts as the language model's "external memory". The chatbot will answer strictly based on the avionics manuals injected into the database, guaranteeing that tolerance limits, voltages or pressures are 100% real and extracted from the official documentation.
- *Scalability Without Server Maintenance:* Pinecone eliminates the need to manage database clusters. It automatically indexes millions of vectors (technical manual chunks) and maintains millisecond response times.
- *Native, Fast Integration with n8n:* when building the chatbot in n8n, there is a native Pinecone node. No complex coding is required; just drag the node, connect the API credentials and the chatbot will be ready to query the manual vectors in minutes.

Comparison with other databases

During testing, PostgreSQL was tried, but when compared with Pinecone the latter proved better for the following reasons:

Speed

- *Pinecone:* built from day one solely to search Artificial Intelligence vectors. It uses cloud indexing algorithms that find the exact text chunk in milliseconds, regardless of whether you have uploaded 1,000 or 10 million pages of technical manuals.
- *PostgreSQL:* a traditional relational database (tables, rows and columns) adapted for AI. When the number of documents grows, response speed drops noticeably.

Zero Configuration

- *Pinecone:* a fully managed cloud service. There is no need to worry about configuring RAM, disk space, servers or updates. You create an account, connect the API and it works.

- *PostgreSQL:* requires installing a server, configuring system memory (`shared_buffers`), creating vector indexes manually and managing backups.

Simplicity with n8n

- *Pinecone:* has a native n8n node designed specifically to connect to the "AI Agent" in a single click. No code needs to be written.
- *PostgreSQL:* Although n8n connects to Postgres, combining vector logic, writing custom SQL queries and structuring the chatbot's memory requires designing much more complex n8n flows that are harder to maintain.

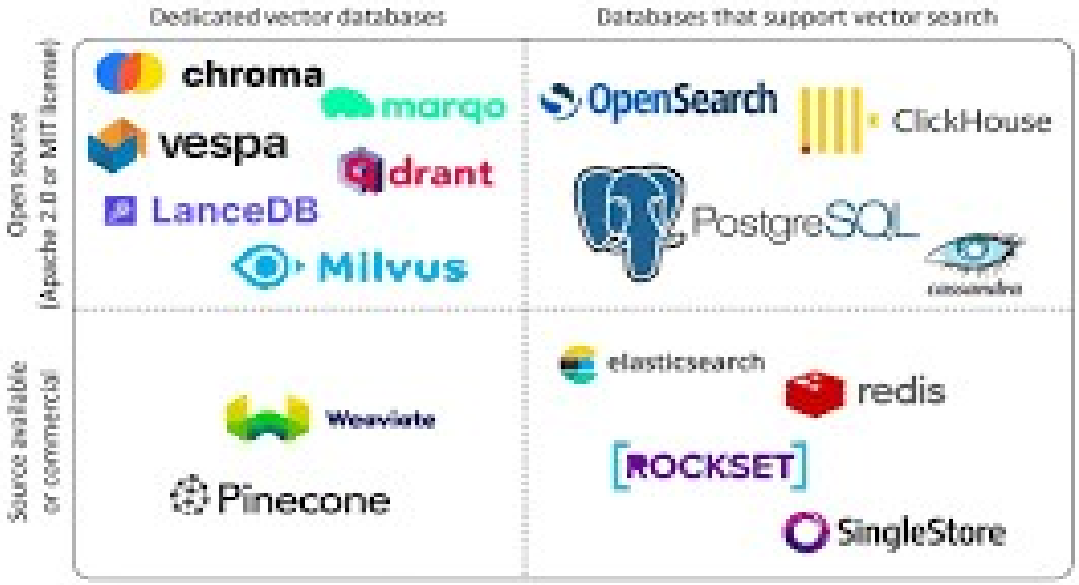


Figure 8-10 Database Comparison (Source: V7labs)

Data flow from Drive to Pinecone

1. When clicking “Execute Workflow”

Manual start of the workflow that triggers the whole automated processing sequence. This is only done the first time to load all the file texts into the database; once stored, it is not needed every time a question is asked, unless new files are added or changed and the database needs updating.

2. Google Drive – Search File/Folder (AHRS, ADS…)

Automatic search for PDF files within specific Google Drive folders, where each folder represents a documentary agent (ADS, AHRS, etc.).

3. Download File

Automatic download of the PDF documents found in the selected folders, keeping the original file name (example: ADS_Prueba.pdf).

5. Default Data Loader

Processing and extraction of the textual content of the PDF document, converting the file into text readable by the system. Associating each text with its file name as well as its corresponding category (ADS, AHRS…)

6. Recursive Character Text Splitter

Splitting the textual content into smaller blocks (chunks) to facilitate later processing and improve information retrieval.

7. Embeddings HuggingFace

Transformation of each text chunk into numerical vectors (embeddings) that semantically represent the document content.

8. Pinecone Vector Store

Insertion and storage of the generated embeddings in the vector database.

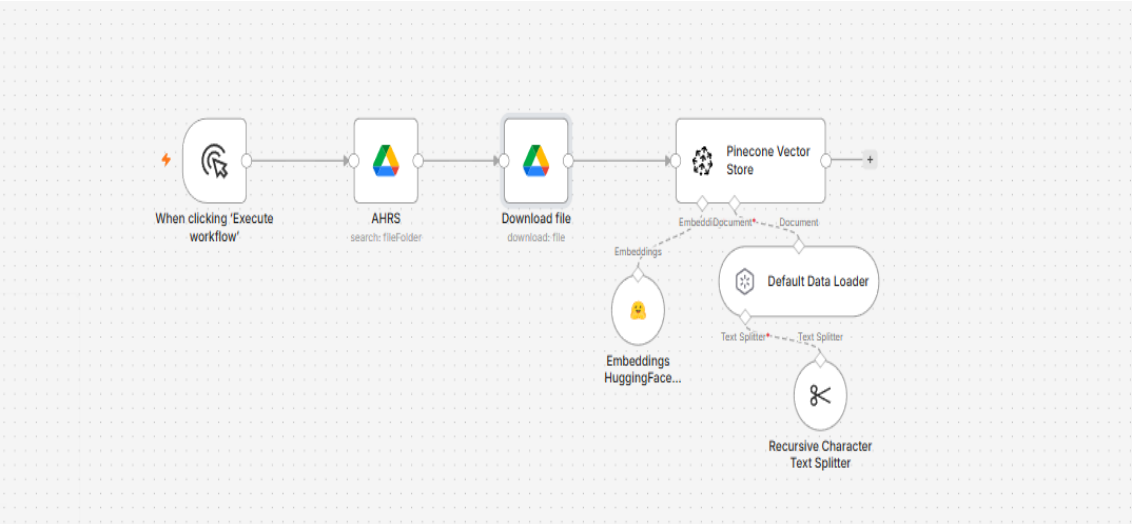


Figure 8-11 Synchronization from Drive to Pinecone (Source: Own Elaboration)

9 FINAL RESULTS

This chapter sets out the methodology used to evaluate the virtual assistant, describes the metrics used and then presents the results obtained in terms of system performance when answering questions related to the functional testing process.

9.1 Evaluation Methodology

Evaluating a technical AI virtual assistant for operator support requires a hybrid approach that measures the AI's technical capability (technical accuracy, operational efficiency...), ease of use (interface, adoption rates…), and practical integration into the industrial environment, including possible scaling to other factory processes.

A complete study would also need to consider development and maintenance costs.

For an effective evaluation, the ideal is to involve end users (technicians) from the start and use a control group to measure real productivity.

However, as there was no access to the assembly line, the test was carried out with university staff (students and lecturers).

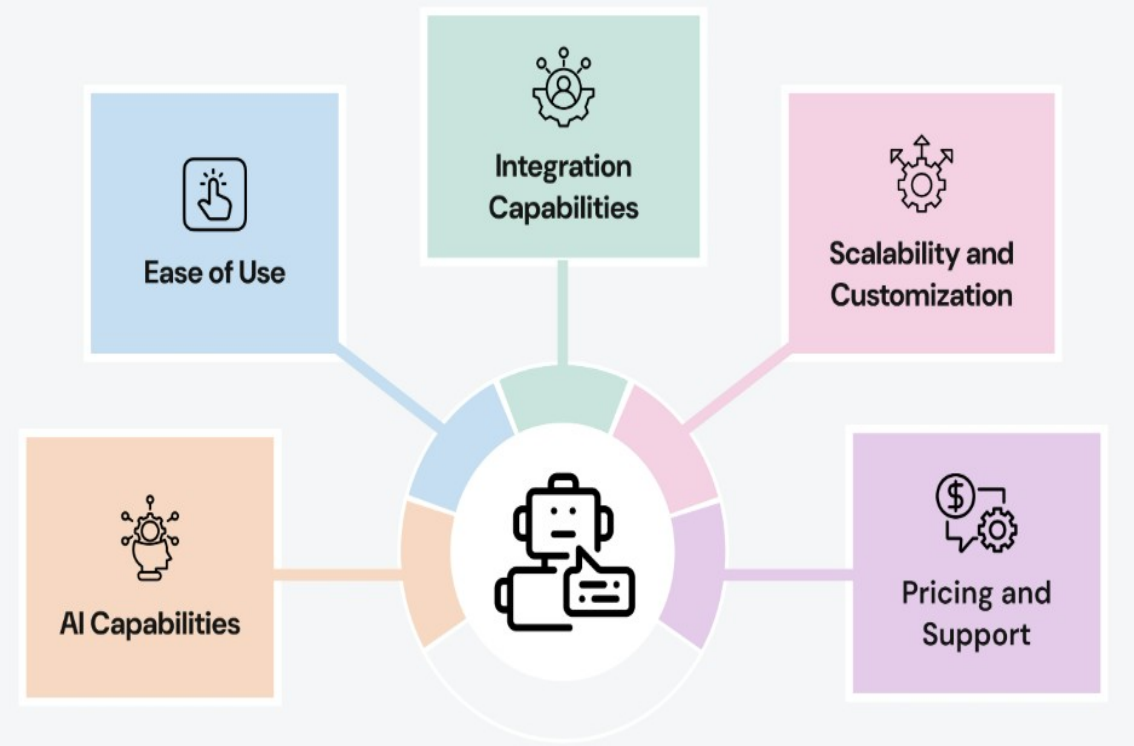


Figure 9-1 Virtual Assistant Evaluation Criteria (Source: Alphabin)

9.2 Metrics for Evaluation

In our case, where there is no access to the industrial environment for which the assistant was designed, we will focus only on evaluating technical accuracy, operational efficiency and ergonomics of use [45-46].

In this way, the evaluation can be divided into three key categories according to the parameters to be measured:

1. Technical Accuracy

- *Truthfulness Rate:* measure whether the answers provided by the AI are accurate and up to date, comparing them with reliable sources.
- *Comprehension Rate:* evaluate whether the AI understands the intent.

2. Operational Efficiency

- *Resolution Rate:* measure what percentage of operators' queries are fully resolved by the AI without the need for a human technician.
- *Mean Resolution Time:* measure how quickly the assistant provides the information and compare it with the time without the assistant.

3. Adoption and Use

- *Accessibility:* evaluate accessibility and the interface in an industrial environment (noise, lighting...)
- *Satisfaction:* gather feedback on whether the help was clear and useful.

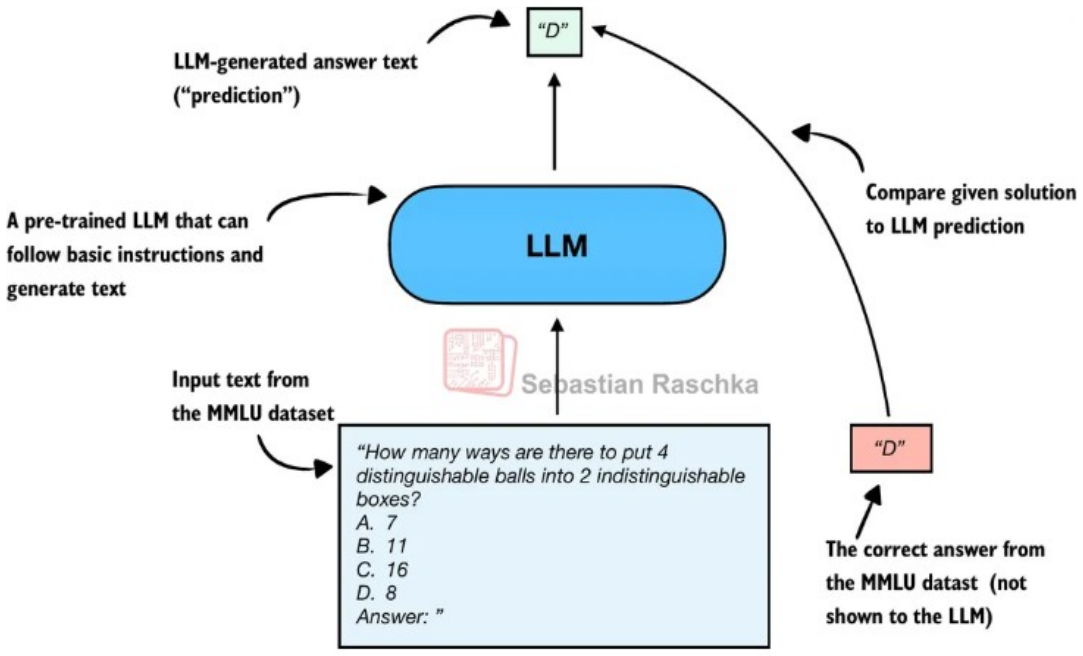


Figure 9-2 Technical Evaluation of the AI Assistant (Source: Magazine.sebastianraschka)

9.3 Survey

Based on the metric above, a control group was chosen consisting of 6 operators with low experience levels (new hires) within the group of avionics test technicians.

Operators with low experience or knowledge were chosen because it is with this type of user that the greatest efficiency is obtained from using virtual assistants.

The working method consists of asking 6 typical questions related to the air navigation testing process, test equipment and system failures, and recording the operators' rating (low, medium, high).

They are scored according to the following criterion: low level 0, medium level 0.5 and high level 1, giving a maximum of 6 points per table.

Accuracy (Technical Truthfulness Rate)

Question/Answer Truthfulness Level	Low	Medium	High	Notes
What is the Radio Altimeter for?			X	16 s
What would I need to test the DME?			X	50 s
How many DMEs does this aircraft have?			X	7 s
How can I do a TCAS self-test?			X	35 s
My TILT panel is broken and I want to replace it with another one; which model should I install?			X	20 s
What is INMARSAT?			X	35 s
Total			**6**	

Accuracy (Comprehension Rate)

Question/Question Comprehension Level	Low	Medium	High	Notes
What is the Radio Altimeter for?			X	None
What label does the Radio Altimeter have on the bus?	X			Manual with ARINC 429 missing
What would I need to test the DME?		X		Too much information
Which ADC model does this aircraft use?			X	None
Tell me the 28 Volt input PIN of the NAV 4000	X			Confuses it with the DME 4000
Tell me the 28 Volt input PIN of the VHF NAV 4000			X	When the VHF context is specified, it answers correctly
Total	**0**	**0,5**	**3**	

Operational Efficiency (Resolution Rate)

Question/Needs Technical Support	No	Partial	Yes	Notes
Can I do the XPDR test with the IFR 4000? It's the only one I have at hand	X			On the first try
How can I test the Radio Altimeter in the hangar without any test equipment?	X			On the first try
Does the TCAS need the XPDR to be operational?	X			On the first try
What is the IDENT button for?	X			On the first try
What is the difference in the RALT Test between the B200 and B300 aircraft models?	X			On the first try
What is the relationship between the weather radar and the AFD, and what is AFD?	X			On the first try
Total	**6**			

Operational Efficiency (Mean Resolution Time)

Question/Resolution Time without/with AI	Without AI	With AI	Notes
What is the IDENT button for?	20 min	18 s	Operator has to find the information in manuals
How can I do a TCAS self-test?	30 min	35 s	Ditto
What is INMARSAT?	10 s	35 s	Expert operator
Tell me the 28 Volt input PIN of the VHF NAV 4000	30 min	20 s	Very specific information
What is the difference in the RALT Test between the B200 and B300 aircraft models?	30 min	18 s	Very specific information
What is the relationship between the weather radar and the AFD, and what is AFD?	35 min	26 s	Very specific information
Total	**8710 s**	**152 s**	**8710/152 = 60**

Adoption and Use (Accessibility Rate)

Operators/Accessibility	Poor	Fair	Good	Notes
Operator 1			X	Fast
Operator 2		X		There is no stop button
Operator 3		X		Cannot pronounce English terminology
Operator 4			X	Easy
Operator 5		X		In a noisy environment it does not understand the question well
Operator 6			X	Useful
Total		**1,5**	**3**	

Adoption and Use (Overall Satisfaction)

Operators/Satisfaction	Poor	Fair	Good	Notes
Operator 1			X	Good
Operator 2			X	Easy to use
Operator 3			X	Easy to use
Operator 4			X	Accessible
Operator 5			X	Fast
Operator 6			X	Fast
Total			**6**	

9.4 Validity Criteria

When determining whether the assistant is valid for use in production, we have to take into account technical and economic criteria.

Due to the demands of the activity (functional tests) and of the product (aircraft safe in flight), errors that could lead to a defective or unsafe product cannot be tolerated.

Therefore, the truthfulness rate of the answers must be the main criterion and the one requiring the highest score. We consider that it must reach 90% accuracy in technical answers. The comprehension rate will normally only affect time, since the question will have to be repeated and rephrased. 60% would be the value we would consider acceptable.

As for the other parameters, such as resolution times or whether additional support is needed, their impact will be on time savings and problem-solving speed. This is more an economic matter and does not affect the functional quality of the product. We believe a 60% score on this point would be sufficient.

Finally, operators' overall satisfaction with the product must be good, otherwise they will tend not to use it. We consider 70% a good starting value, bearing in mind that the more the software is used and improved, the higher this value will be.

The summary table shows the values required to consider the product viable for each parameter and overall.

PARAMETER	VALIDATION LEVEL
Accuracy (Technical Truthfulness Rate)	5,4 (90%)
Accuracy (Comprehension Rate)	3,6 (60%)
Operational Efficiency (Resolution Rate)	3,6 (60%)
Operational Efficiency (Mean Resolution Time)	3,6 (60%)
Adoption and Use (Accessibility Rate)	4,2 (70%)
Adoption and Use (Overall Satisfaction)	4,2 (70%)
Total	**24,6/36**

9.5 Conclusions

From the survey shown in the previous section, the following summary table is obtained.

PARAMETER	OBTAINED	REQUIRED
Accuracy (Technical Truthfulness Rate)	6	5,4 (90%)
Accuracy (Comprehension Rate)	3,5	3,6 (60%)
Operational Efficiency (Resolution Rate)	6	3,6 (60%)
Operational Efficiency (Mean Resolution Time)	6	3,6 (60%)
Adoption and Use (Accessibility Rate)	4,5	4,2 (70%)
Adoption and Use (Overall Satisfaction)	6	4,2 (70%)
Total	**32/36**	24,6/36 (100%)

This table clearly shows the assistant's strengths as well as the opportunities for improvement.

Its main strengths are the truthfulness of the answers provided and the speed of obtaining information. These data give us confidence in the technical validity of the queries and improve operator productivity by obtaining information quickly regardless of their level of experience.

As can be seen, in general the assistant meets the required levels and only falls slightly short on the comprehension rate. However, it is considered that with use, operators will tend to ask questions in a more suitable way and the value will increase.

On the other hand, during development and testing some areas for improvement have been identified, mainly related to the accessibility and usability of the interface.

It is necessary to incorporate new functionalities, add buttons that make certain actions easier and continue correcting the errors detected by operators during real use of the application, with the aim of improving the user experience and increasing the system's adaptability to future needs.

It should be borne in mind that one of the keys to the results lies in the capability of the LLM used as the language model.

It should be noted that the emergence of this type of (generative) AI model occurred barely 3 years ago, so significant improvement is expected in the coming years, which will improve the quality of the answers.

10 ECONOMIC ANALYSIS

The economic analysis includes a breakdown of the investments required for development and the annual operating costs once deployed in production.

It also considers the savings derived from using this system and the payback periods.

10.1 Development Costs

The costs required to develop the assistant before it starts operating in production include:

- **Hardware:** the local server, as specified in the annex, would cost about €10,000. In addition, about €5,000 must be included for installing a power point at the test stand to supply the rack and an Ethernet point for connection to the corporate network.
  No additional investment in tablets for operators is included, since in an I4.0 digital plant model they all already have this tool for consulting the chat.

  The attached table shows the approximate prices of the components, which may vary depending on the manufacturer and the chosen configuration.

Component	Description	Estimated Cost (€)
Rack Cabinet 42U	Rack cabinet with ventilation, trays and cable management	700
Server Dell PowerEdge	Server with Intel Xeon processors, ECC memory, local storage and iDRAC Enterprise	3000
KVM Console	Console with integrated keyboard, monitor and mouse	700
Switch Gigabit/10GbE	Managed switch for rack interconnection	600
NAS QNAP	Enterprise storage system	1000
Enterprise HDDs (4 × 8 TB)	Redundant RAID storage	1000
Enterprise SSD (cache or fast storage)	Performance improvement	800
UPS Online	Uninterruptible power supply (2000-3000	1.500

	VA)	
PCIe HBA/RAID Card	Connection to the storage system	400
Cabling, trays and accessories	Patch cords, organizers, rails, etc	300
TOTAL		10.000€

The proposed architecture mainly uses open-source software or software included with the purchased hardware, which significantly reduces licensing costs.
If enterprise support for Docker or the operating system is required, maintenance contracts costing between €500 and €1,000 per year could be added.

- **Software:** open-source versions at no cost were used for development. Licenses are later requested for the production installation, but their cost is included in the annual operating costs. They would be:
  - Visual Studio Code IDE: €0 (free software).
  - N8N Integration Platform: €0 (free software).
  - Mistral AI LLM: €0 (free software).
  - Pinecone, Qdrant Databases: €0 (free software).
  - Operating Systems, Broadband, Tokens: included in plant IT costs.
- **Programming:** as detailed in chapter 8 of the report, about 1,000 hours of work (6 months) by a mid-level programmer are estimated, at a cost of about €80/hour (U.S. Bureau of Labor Statistics). In total this would amount to about €80,000.
- **Training:** given its ease of use, a 1-hour course for users is considered sufficient, so the training cost is estimated at about €1,000, including the instructor and the cost of operators' hours.

In summary, the initial project costs would be:

Item	Investment Cost (€)
Hardware	15.000
Software	0
Programming	80.000
Training	1.000
Total	**96.000**

10.2 Annual Operating Costs

The recurring expenses to keep the assistant running in production would be:

- **Maintenance:** for updating applications, fixing errors and adapting to new regulations or aeronautical documentation, about 5 hours per month (60 hours per year) are considered, which amounts to about €3,000/year at a technician cost of €50/hour.

- **Infrastructure:** server maintenance by a specialized company is estimated at about $500/year.

- **Licenses:** the required licenses and subscriptions are:

    - N8N (workflow integrator): it is open source and completely free. In addition, it has no execution limits, so no subscription is purchased.
    - Mistral (LLM): its commercial deployment requires a subscription, so Mistral Pro is chosen at a cost of €15/month (€180/year).
    - Pinecone (Database): although there is a free plan for development and experimentation, the standard production plan costs about €50/month (€600/year)

- **Services:** technical support is required, but we consider it included in the plant's own IT support, so it does not represent an extra cost.

  In summary, the operating costs would be:

Item	Annual Cost (€)
Maintenance	3.000
Infrastructure	500
Software Licenses	780
Services	0
Total	**4.280**

10.3 Quantification of Savings

The virtual assistant will reduce costs on the aircraft assembly line through:

- *Reduction of expert operator time:* the assistant guides operators when technical questions arise, so less “expertise” is required and the work can be done by juniors.
  The expert operator costs €50 more per hour and accounts for 50% of the total test execution time (80 h). It is estimated that half of these expert hours can be eliminated and performed by a junior operator at a cost of €100/hour. This means a cost reduction per aircraft of 40 h x (150-100) = €2,000.
- *Reduced preparation time:* the assistant provides information on how the test equipment works, so the preparation/simulation time of the functional test is reduced. A reduction of 1 hour per test is estimated, which would mean about 10 h per aircraft, i.e. about €1,000, considering that preparation is usually done by junior operators whose cost is about €100/hour.
- *Reduced troubleshooting time:* based on manual data and historical records, the assistant makes it possible to find the causes of failure in less time. Although it is difficult to determine the number of failures that occur, it can be considered that on average we can save one hour of the time allocated to each test.
  This gives another saving of about €1,000 per aircraft (10 tests x 1 h x €100/hour).
- *Faster training:* new operators reach the optimal learning curve in less time. Although this means having expert operators sooner, we will not consider any savings in this respect here.

In summary, the savings derived from the project would be:

Item	Saving per Aircraft (€)
Expert Time Reduction	2.000
Preparation Time Reduction	1.000
Troubleshooting Time Reduction	1.000
Training Time Reduction	0
Total	**4.000**

Since the cost of testing per aircraft was about €20,000, this saving of €4,000 represents 20%. Assuming a production of 20 aircraft per year, this would mean an annual saving of €80,000.

10.4 Financial Profitability Metrics

To demonstrate viability we study the typical financial parameters, such as Payback and cumulative ROI.

As this is a high initial investment against a constant cash flow, the best way to present them is with a 3-year projection table.

- **Payback period:** measures the exact time needed to recover the initially invested capital through the cash flows generated by the project.
  It is defined as the ratio between the initial investment cost and the annual net benefit.
  In our case, as the total initial development cost is -€96,000 and the annual recurring cost is -€4,280, with annual savings of about €80,000 for a production of 20 aircraft, the investment can be recovered in 1.27 years (96,000/75,720).

- **Cumulative ROI:** measures the total gain or loss generated by an investment over a specific period of time, adding up all the returns obtained to date.
  It is defined as the ratio between the cumulative cash flow for the year and the initial investment. In our case, for year 3 it would be 139,380/96,000, which gives a good value of 145%.

3-Year Financial Projection (€)	Year 1	Year 2	Year 3
Initial Investment	96.000	0	0
Recurring Cost	4.280	4.280	4.280
Gross Saving	80.000	80.000	80.000
Annual Net Cash Flow	-20.280	75.720	75.720
Cumulative Cash Flow	-20.280	55.400	131.120
Cumulative ROI (%)	-21,13	57,71	136,58

In summary, we have an investment payback period of less than 2 years (a typical criterion in the aeronautical industry) and an excellent ROI.

In short, the economic feasibility analysis shows that the project offers a good cost-benefit ratio, generating a clear return on investment.

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12 ANNEX 1 - SPECIFICATIONS

12.1 General Conditions

12.1.1 Purpose and Scope

These Specifications set out the technical, legal and operational requirements that must be met for the correct installation, commissioning, operation and maintenance of a virtual technical assistance system (THOT) for aircraft assembly line test personnel.

It has been developed within the framework of the Final Degree Project entitled «Implementation of Generative AI in the Air Navigation Test Station of a FAL».

The THOT system consists of a virtual assistant based on Generative Artificial Intelligence (LLM), orchestrated using n8n and equipped with a Retrieval-Augmented Generation (RAG) engine over the technical documentation of the air navigation systems (ADS, AHRS, WRS, RALT, TCAS, GNSS, RNAV VOR/ILS/MKR/ADF, DME, XPDR), with the aim of providing technical support to operators during avionics functional tests at the air navigation station of an aeronautical Final Assembly Line (FAL).

These specifications apply to all phases of the system life cycle: installation, validation, production operation and eventual decommissioning. Any modification to the system not expressly provided for in this document will require written approval from the project's technical manager.

These specifications apply exclusively to the THOT software system (web interface, n8n orchestration, RAG engine) and the associated deployment infrastructure. The avionics test equipment, the aircraft and its navigation systems are outside the scope of this document and are governed by their own technical maintenance manuals (King Air 360 Maintenance Manual, IFR 6000 Operator Manual…).

The THOT system has been designed in compliance with the current regulations applicable to artificial intelligence systems in the aeronautical industrial environment. The reference legal provisions and technical standards are as follows:

Standard	Scope	Application to the system
Reg. (EU) 2024/1689	European AI Act	Transparency towards the operator. Prior training. Human oversight of answers
Reg. (EU) 2016/679 GDPR	Data protection	Operator data in query sessions. Data minimization
LOPDGDD (Organic Law 3/2018)	Workers' digital rights	Right to digital disconnection. Prohibition of abusive workplace monitoring
Royal Decree 1215/1997	Work equipment safety	Tablet/terminal hosting the THOT interface. Usage training

Law 11/2023 Accessibility	Digital accessibility of work tools	THOT interface compatible with accessibility standards
UNE-EN 301549	Requirements for ICT accessibility	Accessibility reference for the THOT web interface
EASA Part-145	Maintenance approved aeronautical maintenance	The assistant acts as advisory support. It does not replace official records

The scope of each provision with respect to the implemented system is set out below.

Regulation (EU) 2024/1689 on Artificial Intelligence: depending on its risk level, the THOT system is subject to obligations of transparency (the operator must be informed that they are interacting with an AI), training (the organization must ensure that staff using the system understand how it works and its limitations) and human oversight (queries relating to safety procedures, fault diagnosis or repair are for guidance only, and the final decision always rests with the qualified operator or technician).

Regulation (EU) 2016/679 (GDPR) and LOPDGDD: during operation, the system records operator interactions with the assistant (query message and session identifier) that may constitute personal data. Their processing must comply with the principle of data minimization, limited to what is strictly necessary to maintain conversation context, and in no case may the query history be used as a mechanism to monitor or evaluate operator performance without informing them in advance.

Royal Decree 1215/1997 and Occupational Risk Prevention: the device used to access the THOT interface (workstation tablet or computer) is considered work equipment for the purposes of these regulations, and its safety and specific operator training in its use must be guaranteed.

Law 11/2023 on Digital Accessibility and UNE-EN 301549: the THOT interface must meet the accessibility criteria required of digital tools for work use, including adequate visual contrast, readable font size and compatibility with assistive technologies, which is especially relevant given its use in hangar environments with variable lighting and noise conditions.

EASA Part-145: air navigation tests assisted by THOT are carried out within aircraft maintenance organizations approved under this regulation. Use of the assistant does not exempt the organization from the traceability and documentary record obligations required by this regulation; THOT acts exclusively as advisory support, with no value as an official maintenance record or as a substitute for certified documentation.

12.1.2 Intellectual Property and Software Licenses

The source code of the THOT interface and of the automation flows developed within this project is the property of the author of the Final Degree Project.

Its use, reproduction, modification or distribution in production environments will require express authorization from the author and, where applicable, from the corresponding academic institution.

The system uses the following third-party software components, whose licenses must be respected in any production deployment:

Component	Function	License	Use in system
n8n	Workflow orchestration	Sustainable Use License (fair-code)	Multi-agent engine and communication with the THOT interface
HTML5 / CSS3 / JavaScript	Frontend THOT	Open standards	Assistant SPA web interface
Marked.js	Markdown rendering	MIT	Display of lists, tables and code in answers
Docker	Containerizatio n	Apache 2.0	Isolation of the n8n and NGINX services
NGINX	Web server	BSD-like (nginx license)	Serving the static THOT interface
Mistral 14B	Language model	Mistral terms of service	Conversational reasoning and answer generation
Pinecone	Vector database	Pinecone terms of service (SaaS)	Storage and indexing of technical manual embeddings
Qdrant	Vector database	Apache 2.0	Deployment alternative on in-house server without SaaS dependency
HuggingFace Embeddings	Embedding generation	Apache 2.0	Transformation of document chunks into numerical vectors
Tailscale	Private mesh VPN network	Tailscale terms of service (free up to a certain number of devices)	Secure remote access to the server without exposure to the public Internet

12.1.3 Responsibilities of the Parties

For the correct application of these specifications, the following roles and their associated responsibilities are defined:

System technical manager (Administrator):

- Keep the RAG knowledge base up to date with the technical manuals of the systems (ADS, AHRS, WRS, RALT, TCAS, GNSS, RNAV, DME, XPDR) and with the test equipment manuals.

- Manage the source document folders and supervise correct synchronization with the vector database.
- Administer access to the Tailscale private network, authorizing or revoking devices as appropriate.
- Keep the n8n automation flows up to date and verify the correct operation of the orchestrator agent.
- Handle operators' requests for information or data erasure in accordance with the GDPR.

Functional Test Operator:

- Use the system exclusively as advisory support during the air navigation functional testing process.
- Always verify the information provided by the assistant against the official documentation before applying it in a safety-critical operation.
- Report to the technical manager any incorrect, inconsistent or potentially unsafe answer generated by the system.
- Do not share access to the Tailscale private network or the THOT interface with unauthorized third parties.

Operating Organization (FAL):

- Ensure that the system is used only in the intended environments (air navigation test station) and with the specified hardware/connectivity.
- Ensure that operators receive the minimum necessary training before using the system autonomously, in accordance with Regulation (EU) 2024/1689.
- Keep the official technical documentation of the aircraft and test equipment available and accessible as the ultimate reference against any answer from the assistant.

12.1.4 Validity and revision of the document

These Specifications are valid from the date of their approval and will be mandatorily reviewed in the following cases: change in the applicable regulations, substantial modification of the system, major version change of any of the software components (n8n, Mistral, Pinecone/Qdrant), or an incident affecting the reliability of the assistant's answers.

The current version of the specifications will be identified by version number and date of last revision. Any previous version is automatically repealed upon publication of a new approved version.

12.2 Technical Conditions Materials/Software

12.2.1 Server hardware requirements

The THOT system's central server is deployed on a centralized rack architecture designed for high availability and continuous 24/7 operation.

The server must be located in an environment with controlled environmental conditions (technical room or rack cabinet with adequate ventilation). If deployed on the shop floor near the hangar, protection against industrial dust and vibration is recommended.

It must meet the following minimum requirements:

Component	Minimum requirement	Recommended
Main server	x86-64, 4 cores	DELL PowerEdge Rack Server, iDRAC Remote Management
RAM	16 GB	16 GB or more Embedding processing
Storage	SSD 50 GB	QNAP NAS system + RAID Data redundancy/integrity
Network connectivity	Ethernet 100 Mbps	Ethernet 1 Gbps
Power supply	Mains power	UPS Uninterruptible Power Supply
Virtualization	Docker Engine	Docker Engine on Ubuntu 24.04 LTS
Physical administration	Direct access	KVM console + dedicated network switch

12.2.2 Network and communications requirements

The network infrastructure must meet the following requirements to guarantee real-time communication between the THOT interface, the orchestration engine and the vector database:

- Access to the system is exclusively through a mesh virtual private network (Tailscale), avoiding exposure of the server to the public Internet.
- The NGINX frontend container exposes the THOT interface on port 80 of the server's private address (e.g. http://100.75.103.85).
- The n8n backend container exposes its Webhook on a mapped host port (e.g. 32769:5678), accessible only through the private network.

- If Pinecone is used as a cloud vector database, stable Internet connectivity is required, with a recommended minimum symmetric bandwidth of 10 Mbps for requests to the embeddings and semantic search API.
- If self-hosted Qdrant on the server itself is chosen, communication between n8n and the vector database takes place on the internal network, with no dependence on external connectivity.
- Each client device (operator's tablet or computer) must have Tailscale installed and belong to the authorized private network in order to access the THOT interface.

12.2.3 Server software requirements

The server runtime environment must have the following software components installed and configured prior to system deployment:

Component	Version / Image	Configuration
Docker Engine	Latest stable	Required for the containerized execution of n8n and NGINX
n8n	n8nio/n8n: latest (Docker)	Environment variables N8N_HOST, N8N_PORT, N8N_PROTOCOL, WEBHOOK_URL, N8N_EDITOR_BASE_URL configured with the server's private IP
NGINX	nginx: stable-alpine (Docker)	Mounted volume with the static files of the THOT interface index.html, script.js estilo.css
Tailscale	Latest stable	Installed on the server and on all authorized client devices
Qdrant (Optional, self-hosted)	Official Docker image	Service deployed on the server itself when not using Pinecone in the cloud

2.4 Orchestration Platform/Language Model Requirements

The multi-agent system logic is implemented entirely in n8n, in accordance with the following minimum required architecture:

- *Webhook Node:* entry point of the workflow; receives the HTTP POST request generated by the THOT interface each time the operator sends a query.
- *AI Agent Node (Orchestrator):* interprets the intent of the query and coordinates the language model, the conversational memory and the connected document retrieval tools.
- *Language model:* Mistral 14B (Mistral Cloud Chat Model), responsible for conversational reasoning and generation of the final answer.
- *Conversational memory (Simple Memory):* maintains the context of the active conversation during the session, associated with a unique session identifier generated by the THOT interface.
- *Retrieval tools (Retriever Tools):* one or more vector tools (e.g. Vector_AHRS, Vector_ADS, etc.) connected to the orchestrator agent, each associated with a document category of the air navigation systems.
- *Respond to Webhook Node:* returns the answer generated by the agent to the THOT interface in JSON format.

The orchestrator node logic must not be modified without documenting the change and validating that the agent's behavior remains consistent with the defined acceptance criteria.

12.2.4 Documentary Knowledge Base (RAG)

The system's knowledge base is built using the Drive → Vector Database synchronization flow, which must meet the following conditions:

- The source documentation (technical manuals of the ADS, AHRS, WRS, RALT, TCAS, GNSS, RNAV, DME and XPDR systems, in PDF format) is organized in folders, where each folder represents an independent documentary agent.
- The synchronization flow (When clicking "Execute Workflow" → Google Drive Search File/Folder → Download File → Default Data Loader → Recursive Character Text Splitter → Embeddings HuggingFace → Pinecone/Qdrant Vector Store) is run manually the first time and every time new documentation is added or existing documentation is updated.
- Each indexed chunk must preserve the traceability of its source document and its document category, so that the assistant can cite the specific source of the information provided.

- No documentation will be added to the knowledge base without first verifying that it corresponds to the current version of the relevant technical manual.

12.2.5 Operator Interface Devices

The operator interacts with the system through the THOT web interface, accessible from any device connected to the Tailscale private network.

Access devices must meet the following minimum requirements, in line with the ergonomic requirements defined in the Project Report:

- Web browser compatible with HTML5, modern JavaScript and Fetch/HTTP requests (a recent version of Google Chrome, Mozilla Firefox or Microsoft Edge).
- Minimum processor equivalent to Intel Core i5/i7 or AMD Ryzen and 16 GB of RAM for fixed workstations.
- Integrated microphone and speaker to support voice recognition, since tests may require hands-free operation.
- Integrated camera, for the future incorporation of visual documentation capabilities (photo/video) during fault diagnosis.
- Easily viewable display, since the testing process involves operations with multiple sequential steps.
- Portability and battery life consistent with the working day (8 h), as checks may be performed both in the cockpit and outside the aircraft.

12.3 Execution and Installation Conditions

12.3.1 Preconditions before start

Before proceeding with the installation of the THOT system, the following preconditions must be verified:

- Availability of the server with Docker Engine installed and operational.
- Access credentials for the Mistral AI API (valid and active API key).
- Access credentials for Pinecone (API key + environment) or, alternatively, deployment of the Qdrant container on the in-house server.
- Access to the Google Drive folder with the technical documentation organized by category (ADS, AHRS, WRS, RALT, TCAS, GNSS, RNAV, DME, XPDR).
- Tailscale account configured for the organization's private network, with the server and authorized operators' devices.

- THOT interface files (index.html, script.js, estilo.css) available for deployment in the NGINX container.

12.3.2 Installation Procedure

The system will be installed following these main steps, with their associated verification criteria:

Step	Action	Command	Success criterion
1	Install Docker Engine on the server	docker --version	Version installed correctly
2	Deploy n8n backend container	Docker run -d --name N8N_contenedor -p 32769:5678 ... n8nio/n8n: latest	Container in running state
3	Configure n8n environment variables	Verification of N8N_HOST, WEBHOOK_URL, N8N_EDITOR_BASE_URL	Editor URL accessible from browser
4	Deploy NGINX frontend container	docker run -d --name THOT_contenedor -p 80:80 -v ".../THOT:/usr/share/nginx/html " nginx:stable-alpine	Container in running state
5	Install and configure Tailscale on the server	Tailscale status	Server visible on the private network with assigned IP
6	Configure Mistral AI and Pinecone/Qdrant credentials in n8n	Manual verification in the n8n credentials panel	Credentials validated without errors
7	Import Orchestrator Agent Flow into n8n	Visual verification of the imported workflow	All nodes (Webhook, AI Agent, Retriever Tools, Respond to Webhook) present and connected
8	Run the Drive → Vector Store synchronization flow	Manual Execute Workflow from n8n	Chunks correctly indexed in Pinecone/Qdrant
9	Verify access to the	Browser access to	Web interface

	THOT interface	http://<Tailscale-private-IP>	loads correctly
10	Perform an end-to-end test query	Sending a technical question from the interface	Answer generated and displayed in the interface within a reasonable time

12.3.3 Phased Commissioning Plan

The transition from the prototype validated in the university environment to production deployment at the FAL test station will take place in three sequential phases.

Moving from one phase to the next requires passing the gate criterion defined for each one.

Phase	Duration	Objective	Gate criterion
Documentary Validation	2–3 weeks	Load and index the complete documentation of the air navigation systems in the vector database.	100% of relevant manuals indexed. Assistant answers correctly cite the documentary source.
Operation Assisted Supervised	3–4 weeks	Deploy the THOT interface on the test station's private network with a small group of operators.	Test sessions without connection errors. Operators manually verify each answer before applying it.
Operation Production	Continuous Deployment	THOT available as a standard support tool during air navigation	Periodic evaluation of technical accuracy. Continuous

		functional tests.	update of the RAG base with new documentation.

12.3.4 Acceptance tests

Before the system is put into service at the test station, the following acceptance tests must be passed satisfactorily, in line with the evaluation methodology described in the Project Report:

- *Test 1 (Start-up):* the n8n and NGINX containers start without errors and the THOT interface is accessible from the Tailscale private network.
- *Test 2 (Conversation):* the assistant answers in Spanish with technical accuracy to queries related to the air navigation systems (ADS, AHRS, WRS, RALT, TCAS, GNSS, RNAV, DME, XPDR).
- *Test 3 (RAG):* the assistant correctly retrieves and cites the technical manual chunk corresponding to the query, without making up information not present in the indexed documentation.
- *Test 4 (Session memory):* the assistant maintains the conversation context across several messages within the same session.
- *Test 5 (Private network access):* only devices connected to the authorized Tailscale network can access the THOT interface.
- *Test 6 (Technical accuracy - Truthfulness Rate):* for a set of control questions, the percentage of answers rated as «high» truthfulness exceeds the minimum threshold set by the organization.
- *Test 7 (Operational efficiency - Resolution Rate):* the percentage of queries fully resolved by the assistant without the intervention of a human technician is recorded and compared with the target threshold.
- *Test 8 (Accessibility):* the interface is readable and operable under the lighting and noise conditions of the test station, in accordance with the ergonomic requirements of the Report.
- *Test 9 (Knowledge base update):* after adding a new document to the corresponding folder and re-running the synchronization flow, the assistant incorporates the new information into its answers without the need to restart the containers.

12.4 Operation and Maintenance Conditions

12.4.1 Normal operating conditions

During normal operation of the THOT system, the following conditions must be respected:

- The system must run continuously during air navigation test shifts, except in installations where start-up and shutdown per shift is chosen.
- The assistant's answers relating to safety procedures, fault diagnosis, repair or use of test equipment are for guidance only. Final responsibility for the operation always lies with the qualified operator or technician, who must verify critical information against the official documentation.
- The operator must not perform any action on the aircraft or test equipment based solely on an answer from the assistant without checking it against the documented procedure.
- In the event of loss of connectivity with the server or the Mistral AI API, the operator will continue the testing process following the usual technical documentation, without relying on the assistant.

12.4.2 Access Control and Role Management

Access to the THOT system is controlled at network level through membership of the Tailscale private network authorized by the organization; access from unauthorized devices or from the public Internet is not allowed.

The n8n administration environment is not part of the operator's normal use of the system and must be restricted exclusively to the technical manager, since it allows the entire logic of the orchestrator agent to be modified.

Access level	Assignment	Capabilities
Operator	Any authorized device on the Tailscale network	Use of the THOT interface and asking technical questions
Technical manager / Administrator	Additional access to the n8n editor and Tailscale network management	Modification of flows, credential management, administration of the knowledge base and authorized devices

12.4.3 System maintenance

The system technical manager must perform the following maintenance tasks at the indicated frequency:

Task	Frequency	Description
Verification of Containers	Weekly	Check that the n8n and NGINX containers remain in running state and restart them in case of failure.
Review of the log of n8n Executions	Weekly	Review the workflow execution history for connection errors with Mistral AI or with the vector database.
Update of Docker Images	Quarterly	Deployment of new versions of n8nio/n8n and nginx : stable-alpine avoiding interruptions during FAL operating hours.
Review of API Credentials	Half-yearly	Verify the validity of the Mistral AI API key and the Pinecone key, renewing them before they expire.
Audit of Tailscale Devices	Monthly	Review the list of authorized devices on the private network and revoke unused access.
Backup of n8n Flows	Monthly	Export of the orchestrator agent workflows and the document synchronization flow.

12.4.4 RAG knowledge base management

The system knowledge base, made up of the technical manuals indexed in Pinecone (or Qdrant), must be kept up to date according to the following criteria:

- When a new revision of a technical manual for any of the air navigation systems is published, the previous document must be replaced in the corresponding folder and the synchronization flow re-run.
- Any technical bulletin, test procedure or specific instruction relevant to the test station may be added to the knowledge base by creating a new document folder or extending an existing one.
- No documentation will be added to the knowledge base without first verifying that it is current and comes from an official source.

- Updating the knowledge base does not require restarting the containers; the new chunks are available to the orchestrator agent's retrieval tools immediately after indexing.

12.4.5 Data protection and GDPR/LOPDGDD compliance

The system processes the operator's personal data (session identifier and query content) in accordance with Regulation (EU) 2016/679 and the LOPDGDD.

The technical manager must ensure compliance with the following obligations:

Data processed and legal basis:

- Session identifier and conversation content during the period of active use. Legal basis: legitimate interest in safe technical support during the testing process (Art. 6.1.f GDPR).

Minimization and retention:

- The system must not use the query history for purposes other than technical support, nor be used as a tool to monitor or evaluate operator performance without informing them in advance, in accordance with the right to digital disconnection recognized by the LOPDGDD.
- The technical manager must configure n8n execution data retention (EXECUTIONS_DATA_PRUNE) to the minimum period necessary to resolve incidents.

Information to the operator:

- Before using the system, the organization must inform operators that they are interacting with an Artificial Intelligence assistant and how their queries are processed, in accordance with Art. 13 GDPR and Regulation (EU) 2024/1689.

12.4.6 Incident procedure

In the event of an incident in the system, the operator must follow the following procedure:

Incident type	Immediate action	Escalation
THOT interface does not load	Verify connection to the Tailscale private network.	Contact the technical manager to check the NGINX container.
The assistant does not	Continue the testing	Contact the technical

respond	process with the official documentation.	manager to check the n8n container and the Mistral AI API.
Incorrect, inconsistent answer or answer without documentary citation	Do not apply the information without verifying it against the official manual. Record the case.	Report to the technical manager for review of the RAG knowledge base or the orchestrator agent flow.
Loss of connectivity with Pinecone/Mistral API	The assistant may stop responding or the quality of its answers may degrade.	Check the server's Internet connectivity. If it persists, consider a temporary migration to self-hosted Qdrant.
Unauthorized access detected on the network Tailscale	Immediately revoke the device from the Tailscale admin panel.	Contact the organization's security manager.

13 ANNEX 2 - FRONTEND

13.1 Frontend Implementation

In order to provide a simple and intuitive way of interacting with the developed RAG multi-agent system, a custom web interface was designed, called **THOT**.

The interface follows this architecture:

- **HTML5** for the structure.
- **CSS3** for visual design and styles.
- **JavaScript** for logic and communication with the backend.

The project is organized as follows:

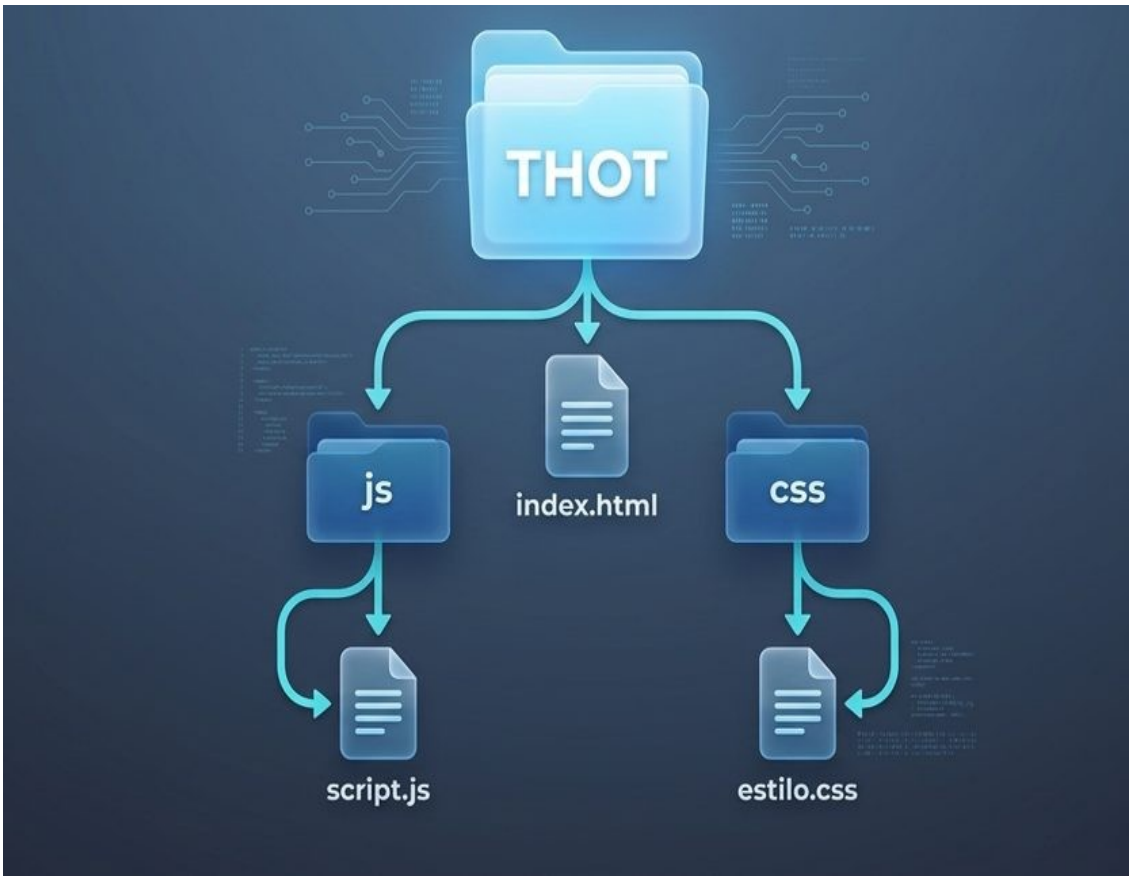


Figure 13-1 THOT Structure (Source: Own Elaboration)

13.1.1 HTML interface structure and organization

The index.html file is the basis of the application. Its main function is to define all the visual elements the user sees:

- Conversation sidebar.
- Welcome screen.
- Message input box.
- Settings pop-up windows.
- Context menus.

The interface was designed following a single-page architecture (SPA, *Single Page Application*), where all interactions take place dynamically through JavaScript without reloading the page.

The HTML document is divided into three main blocks:

- *Header (head):* contains the page configuration, loading of CSS style sheets, fonts and external libraries.
- *Body (body):* includes all the visual elements of the interface, such as the conversation history, the chat area and dialog boxes.
- *Scripts:* responsible for implementing the application's operating logic and communication with N8N.

In addition, the answers generated by the agent are processed using the Marked.js library, which interprets Markdown syntax to display:

- Lists.
- Tables.
- Code blocks.
- Headings.

This significantly improves the readability of the technical information shown to the user.

13.1.2 Operating logic and conversation management

The script.js file implements all the interface's operating logic and manages the interaction between the user and the multi-agent system.

Its main function is to control conversations, dynamically update the interface and establish communication with N8N.

The implemented features include:

- Creating and managing conversations.
- Temporary storage of message history.
- Loading and deleting chats.
- Renaming conversations.
- Sorting and searching conversations.
- Management of pop-up windows and context menus.
- Voice recognition.
- Processing and dynamic display of answers.

One of the most important components is the communication mechanism with N8N. For this, the JavaScript Fetch API is used, which allows asynchronous HTTP requests to be made to a Webhook configured within the automation flow.

When the user sends a query, the application generates an HTTP POST request to the N8N Webhook, sending the information in JSON format. This information contains both the message entered by the user and a unique session identifier associated with the active conversation.

This makes it possible to:

- Keep conversations independent.
- Preserve the context of each session.
- Facilitate memory management within the multi-agent system.
- Allow queries to be processed by the RAG orchestrator.

After receiving the request, N8N triggers the corresponding flow, runs the multi-agent system and returns the generated answer. JavaScript then processes that answer and automatically renders it in the interface.

In addition, while the query is being processed, a visual typing indicator is shown, providing feedback to the user while the system generates the answer.

The overall communication flow implemented can be summarized with the following diagram:

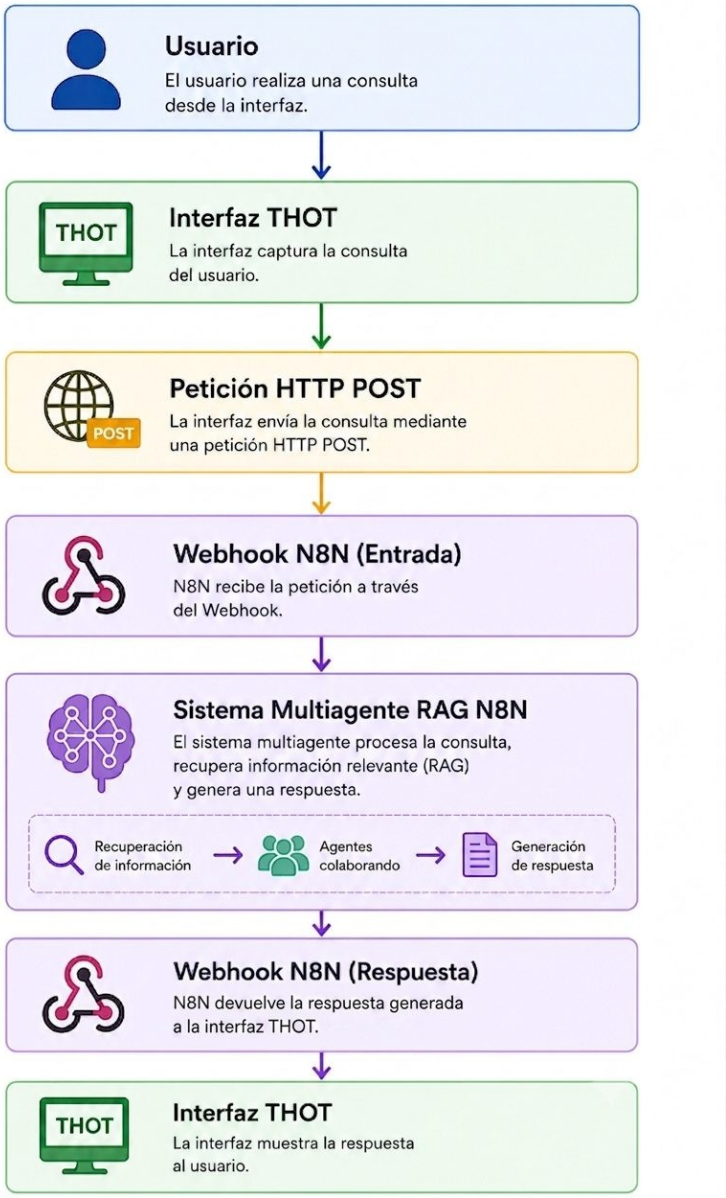


Figure 13-2 Data Flow (Source: Own Elaboration)

This architecture completely decouples the user interface from the multi-agent system logic, allowing N8N flows to be modified or extended without significant changes to the client application.

13.1.3 Visual design and CSS styles

The estilo.css file defines the visual appearance of the application.

Its main objective is to provide an innovative, intuitive and pleasant interface for the user.

In addition, visual effects, smooth animations and a consistent color palette are used to improve the user experience and make the displayed information easier to read.



Figure 13-3 THOT Interface (Source: Own Elaboration)

13.2 Docker as a Deployment Platform

The developed system is designed to be deployed on a single remote server, where the whole architecture runs in containers. The system is accessed through a single public URL, from which the user interacts with the web interface, while the automation flows are processed in the backend.

The infrastructure is based on Docker, which allows the different system components to run in an isolated, reproducible and scalable way within the server.

The deployment consists of three main elements:

- Web user interface served by NGINX.
- Automation and agent orchestration engine using N8N.
- Host server where the Docker containers run.

13.2.1 Docker fundamentals

Docker is a lightweight container-based virtualization technology that allows applications to run in isolation within the same operating system.

Instead of virtualizing complete hardware as in a virtual machine, containers share the host system's kernel, which reduces resource consumption and improves performance.

The main elements used are:

- *Images:* immutable templates containing the minimal operating system, dependencies and application.
- *Containers:* running instances of those images.
- *Volumes:* data persistence mechanisms between the host system and the containers.
- *Docker Networks:* enable internal communication between services.

13.2.2 Docker Images

The system is based on official community-maintained images, which reduces the need for manual configuration and improves deployment stability.

The main ones used are:

- *n8nio/n8n: latest*, which contains the complete workflow execution and automation environment.
- *nginx: stable-alpine*, a lightweight version of the NGINX web server used to serve the user interface.

13.2.3 Docker Containers and Deployment Architecture

The final system deployment is designed to run on a remote server accessible from the Internet.

All services start as independent containers within the same machine, but communicate with each other.

Backend Container (n8n)

This container runs the automation system and the orchestration of RAG agents.

Using the following Windows CMD commands:

```
docker run -d ^
--name N8N_contenedor2 ^
-p 32769:5678 ^
-e N8N_HOST=172.29.39.115 ^
-e N8N_PORT=5678 ^
-e N8N_PROTOCOL=http ^
-e WEBHOOK_URL=http://172.29.39.115:32769/ ^
-e N8N_EDITOR_BASE_URL=http://172.29.39.115:32769/ ^
docker.n8n.io/n8nio/n8n:latest
```

Frontend Container (NGINX)

The user interface is deployed in an NGINX container that serves static HTML, CSS and JavaScript files.

Using the following Windows CMD commands:

```
docker run -d ^
--name THOT_contenedor ^
-p 80:80 ^
-v "C:\Users\Francisco
Sierra\Desktop\THOT_v4:/usr/share/nginx/html" ^
nginx: stable-alpine
```

Deployment architecture

The system is organized under a centralized client-server architecture deployed on a private server, where both the web interface and the backend processing engine run in containers using Docker.

The architecture works as follows:

- The server hosts two independent Docker containers: one based on NGINX, responsible for serving the static web interface (THOT), and another running n8n, responsible for processing all the automation and flow execution logic.
- The system is not accessed through a public Internet address, but through a mesh virtual private network provided by Tailscale, which securely connects authorized devices to the server.
- Once connected to the Tailscale private network, the user can access the THOT web interface by entering the private address assigned to the server in the browser: http://100.75.103.85
- The NGINX web server exposes this interface on port 80, so any device connected to the same private network can load the application directly from the browser.

- The THOT interface sends HTTP requests to the backend implemented in n8n through internal Webhooks, triggering the different processing flows defined in the system.
- n8n runs the corresponding multi-agent flow, processes the query received and returns the generated answer to the frontend interface.
- Finally, the answer is rendered again in the web interface and shown to the user in real time.

Direct access to the internal n8n environment is not part of normal system use, since this service remains deployed only as an internal backend and is accessed remotely only when administration, maintenance or flow modification tasks are needed.

The architecture also allows concurrent multi-user access. For several users to use the system simultaneously, each device only needs to install Tailscale and sign in with the same account or belong to the same authorized private network. Once connected, all devices can access the same web interface via the server's private address, generating independent requests that n8n processes as separate executions.

This approach completely decouples the frontend from the backend, keeps the system isolated from the public Internet, improves security through access restricted by private VPN and facilitates system scalability and maintenance.

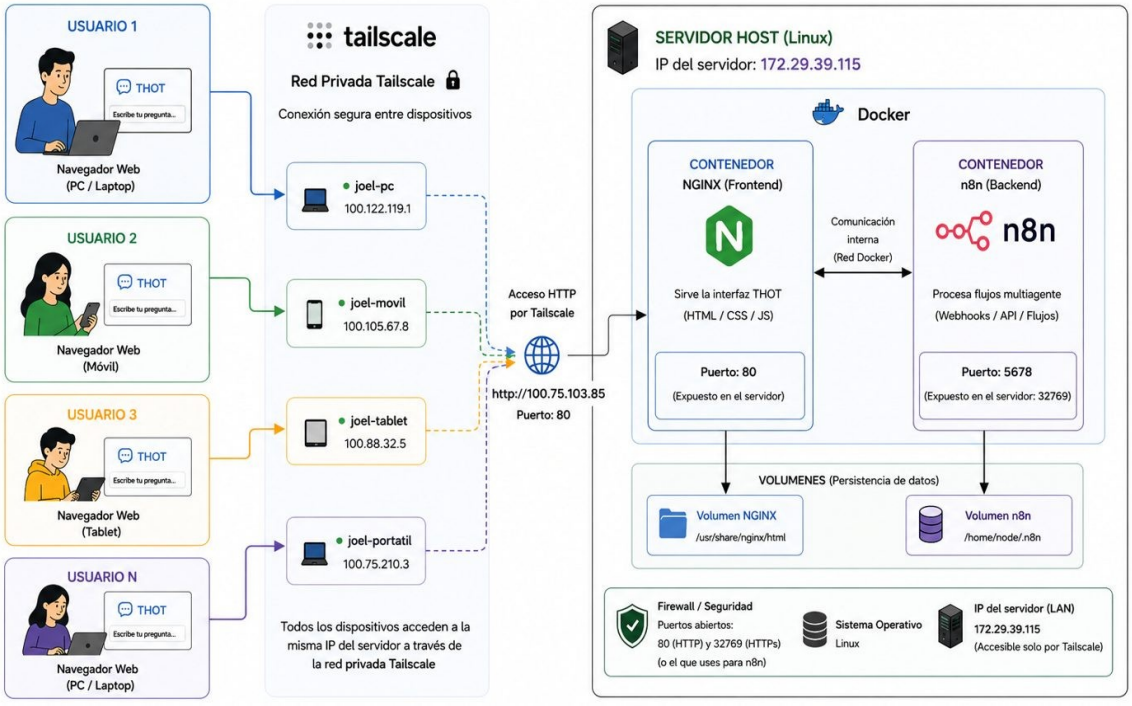


Figure 13-4 Deployment Architecture (Source: Own Elaboration)

13.2.4 Advantages of using Docker

Using Docker in the production environment provides key benefits:

- *Centralized remote access:* the whole system runs on a single server accessible via a URL.
- *Simplified deployment:* services are started with standard commands without complex configurations.
- *Service isolation:* frontend and backend work independently.
- *Future scalability:* possibility of migrating to multiple servers or replicating containers.
- *Simplified maintenance:* services are updated with new images without redesigning the architecture.
- *Controlled security:* access is centralized on the server, reducing exposure points.

In conclusion, Docker allows the developed system to work as a complete platform deployed on a server, where all execution logic is centralized, while the user only interacts through a web interface accessible from anywhere.

13.3 Server Architecture

The system server has been designed following a centralized rack architecture, aimed at high availability, scalability and continuous 24/7 operation.

The Server Rack has the following architecture:

- *Rack Cabinet:* cabinet where all system components are housed and secured, ensuring order, ventilation and physical security.
- *Rack Server (DELL PowerEdge):* main server responsible for centralized processing. It runs continuously (24/7) and has remote management via an integrated iDRAC card, allowing out-of-band administration.
- *KVM Console (Keyboard, Video, Mouse):* system allowing direct physical administration of the server from the laboratory.
- *Network switch:* responsible for interconnecting all rack devices and providing LAN connectivity.
- *Storage system (QNAP NAS + RAID):* storage infrastructure made up of multiple hard drives configured in RAID to guarantee redundancy, integrity and high data availability.

- *HDD/SSD hard drives:* used for mass data storage, with redundant configuration according to the RAID level
- *UPS (Uninterruptible Power Supply):* uninterruptible power supply system that protects the server against power cuts and voltage fluctuations.
- *Storage expansion card (PCIe):* installed in the server to allow direct connection to the external storage system.
- *Vector database (Qdrant):* deployed as a service within the server environment, used for efficient storage and search of embeddings in artificial intelligence applications and semantic information retrieval.
- *Virtualization / container system:* used to deploy services in an isolated and scalable way (Docker).

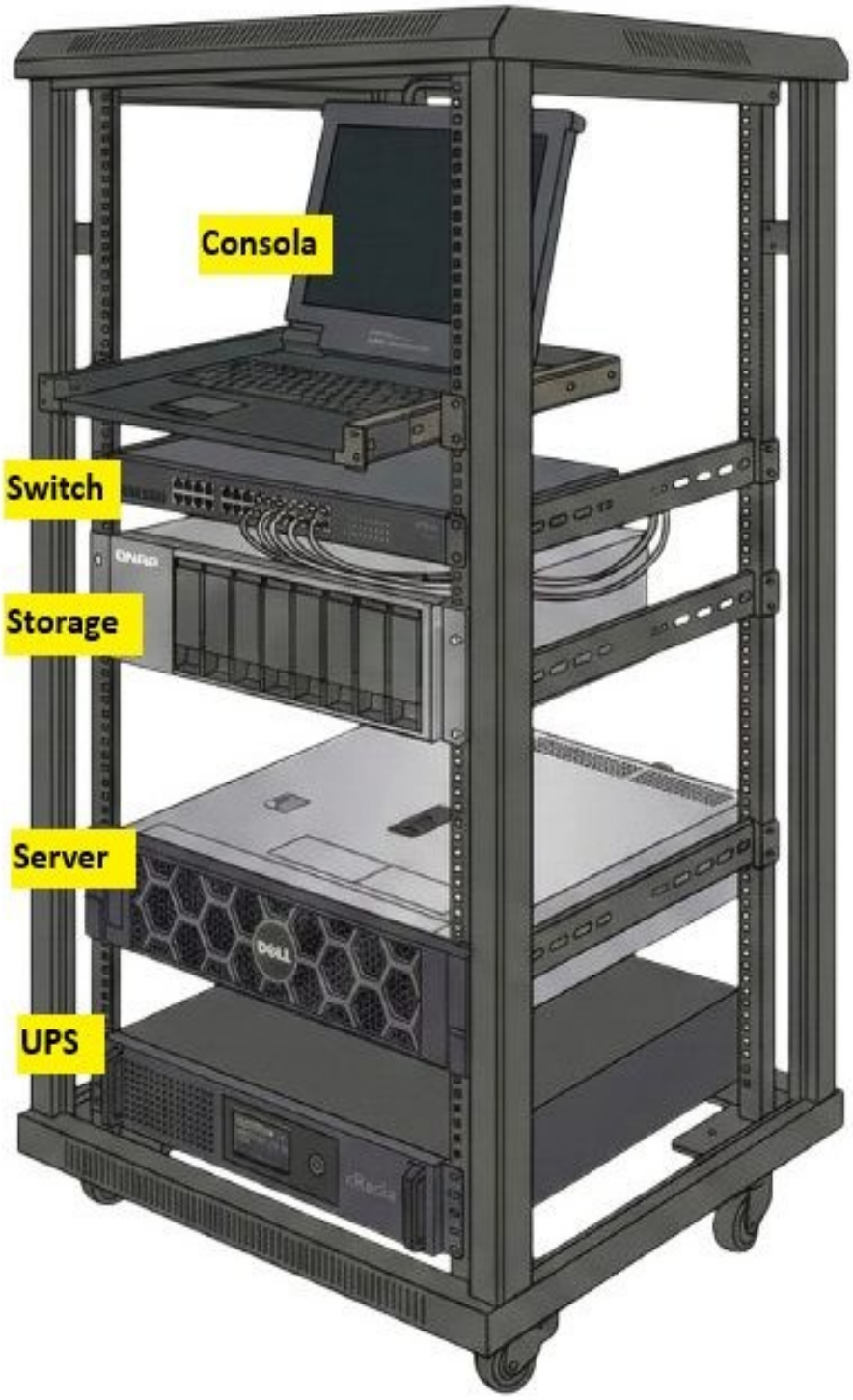


Figure 13-5 Server Architecture (Source: EPS)